

UNIVERSITY OF
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STEP II Past Papers
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Section A: Pure Mathematics

1 Show that, if n is an integer such that

$$(n-3)^3 + n^3 = (n+3)^3, \qquad (*)$$

then n is even and n^2 is a factor of 54. Deduce that there is no integer n which satisfies the equation $(*)$.

Show that, if n is an integer such that

$$(n-6)^3 + n^3 = (n+6)^3, \qquad (**)$$

then n is even. Deduce that there is no integer n which satisfies the equation $(**)$.

2 Use the first four terms of the binomial expansion of $(1-1/50)^{1/2}$, writing $1/50 = 2/100$ to simplify the calculation, to derive the approximation $\sqrt{2} \approx 1.414214$.

Calculate similarly an approximation to the cube root of 2 to six decimal places by considering $(1+N/125)^a$, where a and N are suitable numbers.

[You need not justify the accuracy of your approximations.]

3 Show that the sum S_N of the first N terms of the series

$$\frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \cdots + \frac{2n-1}{n(n+1)(n+2)} + \cdots$$

is

$$\frac{1}{2}\left(\frac{3}{2} + \frac{1}{N+1} - \frac{5}{N+2}\right).$$

What is the limit of S_N as $N \to \infty$?

The numbers a_n are such that

$$\frac{a_n}{a_{n-1}} = \frac{(n-1)(2n-1)}{(n+2)(2n-3)}.$$

Find an expression for a_n/a_1 and hence, or otherwise, evaluate $\sum_{n=1}^{\infty} a_n$ when $a_1 = \frac{2}{9}$.

4 The integral I_n is defined by

$$I_n = \int_0^{\pi} (\pi/2 - x) \sin(nx + x/2) \operatorname{cosec}(x/2)\, \mathrm{d}x,$$

where n is a positive integer. Evaluate $I_n - I_{n-1}$, and hence evaluate I_n leaving your answer in the form of a sum.

5 Define the modulus of a complex number z and give the geometric interpretation of $|\, z_1 - z_2\,|$ for two complex numbers z_1 and z_2. On the basis of this interpretation establish the inequality

$$|\, z_1 + z_2\,| \leqslant |\, z_1\,| + |\, z_2\,|.$$

Use this result to prove, by induction, the corresponding inequality for $|\, z_1 + \cdots + z_n\,|$.

The complex numbers a_1, a_2, …, a_n satisfy $|a_i| \leqslant 3$ $(i = 1, 2, \ldots, n)$. Prove that the equation

$$a_1 z + a_2 z^2 \cdots + a_n z^n = 1$$

has no solution z with $|\, z\,| \leqslant 1/4$.

6 Two curves are given parametrically by

(1) $$x_1 = (\theta + \sin\theta), \quad y_1 = (1 + \cos\theta),$$

and

(2) $$x_2 = (\theta - \sin\theta), \quad y_2 = -(1 + \cos\theta).$$

Find the gradients of the tangents to the curves at the points where $\theta = \pi/2$ and $\theta = 3\pi/2$.

Sketch, using the same axes, the curves for $0 \leqslant \theta \leqslant 2\pi$.

Find the equation of the normal to the curve (1) at the point with parameter θ. Show that this normal is a tangent to the curve (2).

7 Let

$$\begin{aligned} \mathrm{f}(x) &= \tan x - x, \\ \mathrm{g}(x) &= 2 - 2\cos x - x\sin x, \\ \mathrm{h}(x) &= 2x + x\cos 2x - \frac{3}{2}\sin 2x, \\ \mathrm{F}(x) &= \frac{x(\cos x)^{1/3}}{\sin x}. \end{aligned}$$

(i) By considering $\mathrm{f}(0)$ and $\mathrm{f}'(x)$, show that $\mathrm{f}(x) > 0$ for $0 < x < \pi/2$.

(ii) Show similarly that $\mathrm{g}(x) > 0$ for $0 < x < \pi/2$.

(iii) Show that $\mathrm{h}(x) > 0$ for $0 < x < \pi/4$, and hence that

$$x(\sin^2 x + 3\cos^2 x) - 3\sin x \cos x > 0$$

for $0 < x < \pi/4$.

(iv) By considering $\dfrac{\mathrm{F}'(x)}{\mathrm{F}(x)}$, show that $\mathrm{F}'(x) < 0$ for $0 < x < \pi/4$.

8 Points A, B, C in three dimensions have coordinate vectors **a, b, c**, respectively. Show that the lines joining the vertices of the triangle ABC to the mid-points of the opposite sides meet at a point R.

P is a point which is **not** in the plane ABC. Lines are drawn through the mid-points of BC, CA and AB parallel to PA, PB and PC respectively. Write down the vector equations of the lines and show by inspection that these lines meet at a common point Q.

Prove further that the line PQ meets the plane ABC at R.

Section B: Mechanics

9 A light smoothly jointed planar framework in the form of a regular hexagon $ABCDEF$ is suspended smoothly from A and a weight 1kg is suspended from C. The framework is kept rigid by three light rods BD, BE and BF. What is the direction and magnitude of the supporting force which must be exerted on the framework at A?

Indicate on a labelled diagram which rods are in thrust (compression) and which are in tension.

Find the magnitude of the force in BE.

10 A wedge of mass M rests on a smooth horizontal surface. The face of the wedge is a smooth plane inclined at an angle α to the horizontal. A particle of mass m slides down the face of the wedge, starting from rest. At a later time t, the speed V of the wedge, the speed v of the particle and the angle β of the velocity of the particle below the horizontal are as shown in the diagram.

Let y be the vertical distance descended by the particle. Derive the following results, stating in (ii) and (iii) the mechanical principles you use:

(i) $V \sin\alpha = v \sin(\beta - \alpha)$;

(ii) $\tan\beta = (1 + m/M)\tan\alpha$;

(iii) $2gy = v^2(M + m\cos^2\beta)/M$.

Write down a differential equation for y and hence show that

$$y = \frac{gMt^2 \sin^2\beta}{2\,(M + m\cos^2\beta)}.$$

11 A fielder, who is perfectly placed to catch a ball struck by the batsman in a game of cricket, watches the ball in flight. Assuming that the ball is struck at the fielder's eye level and is caught just in front of her eye, show that $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ is constant, where θ is the angle between the horizontal and the fielder's line of sight.

In order to catch the next ball, which is also struck towards her but at a different velocity, the fielder runs at constant speed v towards the batsman. Assuming that the ground is horizontal, show that the fielder should choose v so that $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ remains constant.

Section C: Probability and Statistics

12 The diagnostic test AL has a probability 0.9 of giving a positive result when applied to a person suffering from the rare disease mathematitis. It also has a probability 1/11 of giving a false positive result when applied to a non-sufferer. It is known that only 1% of the population suffer from the disease. Given that the test AL is positive when applied to Frankie, who is chosen at random from the population, what is the probability that Frankie is a sufferer?

In an attempt to identify sufferers more accurately, a second diagnostic test STEP is given to those for whom the test AL gave a positive result. The probablility of STEP giving a positive result on a sufferer is 0.9, and the probability that it gives a false positive result on a non-sufferer is p. Half of those for whom AL was positive and on whom STEP then also gives a positive result are sufferers. Find p.

13 A random variable X has the probability density function

$$\mathrm{f}(x) = \begin{cases} \lambda e^{-\lambda x} & x \geqslant 0, \\ 0 & x < 0. \end{cases}$$

Show that

$$\mathrm{P}(X > s + t \,|X > t) = \mathrm{P}(X > s).$$

The time it takes an assistant to serve a customer in a certain shop is a random variable with the above distribution and the times for different customers are independent. If, when I enter the shop, the only two assistants are serving one customer each, what is the probability that these customers are both still being served at time t after I arrive?

One of the assistants finishes serving his customer and immediately starts serving me. What is the probability that I am still being served when the other customer has finished being served?

14 The staff of Catastrophe College are paid a salary of A pounds per year. With a Teaching Assessment Exercise impending it is decided to try to lower the student failure rate by offering each lecturer an alternative salary of $B/(1 + X)$ pounds, where X is the number of his or her students who fail the end of year examination. Dr Doom has N students, each with independent probability p of failure. Show that she should accept the new salary scheme if

$$A(N + 1)p < B(1 - (1 - p)^{N+1}).$$

Under what circumstances could X, for Dr Doom, be modelled by a Poisson random variable? What would Dr Doom's expected salary be under this model?

Section A: Pure Mathematics

1 Let $x = 10^{100}$, $y = 10^x$, $z = 10^y$, and let

$$a_1 = x!, \quad a_2 = x^y, \quad a_3 = y^x, \quad a_4 = z^x, \quad a_5 = e^{xyz}, \quad a_6 = z^{1/y}, \quad a_7 = y^{z/x}.$$

(i) Use Stirling's approximation $n! \approx \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}$, which is valid for large n, to show that $\log_{10}(\log_{10} a_1) \approx 102$.

(ii) Arrange the seven numbers $a_1, \ldots, a_7$ in ascending order of magnitude, justifying your result.

2 Consider the quadratic equation

$$nx^2 + 2x\sqrt{(pn^2 + q)} + rn + s = 0, \qquad (*)$$

where $p > 0$, $p \neq r$ and $n = 1, 2, 3, \ldots$.

(i) For the case where $p = 3$, $q = 50$, $r = 2$, $s = 15$, find the set of values of n for which equation $(*)$ has no real roots.

(ii) Prove that if $p < r$ and $4q(p - r) > s^2$, then $(*)$ has no real roots for any value of n.

(iii) If $n = 1$, $p - r = 1$ and $q = s^2/8$, show that $(*)$ has real roots if, and only if, $s \leqslant 4 - 2\sqrt{2}$ or $s \geqslant 4 + 2\sqrt{2}$.

3 Let

$$\mathrm{S}_n(x) = e^{x^3} \frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(e^{-x^3}\right).$$

Show that $\mathrm{S}_2(x) = 9x^4 - 6x$ and find $\mathrm{S}_3(x)$.

Prove by induction on n that $\mathrm{S}_n(x)$ is a polynomial. By means of your induction argument, determine the order of this polynomial and the coefficient of the highest power of x.

Show also that if $\dfrac{\mathrm{d}S_n}{\mathrm{d}x} = 0$ for some value a of x, then $S_n(a)S_{n+1}(a) \leqslant 0$.

4 By considering the expansions in powers of x of both sides of the identity

$$(1+x)^n(1+x)^n \equiv (1+x)^{2n},$$

show that

$$\sum_{s=0}^{n} \binom{n}{s}^2 = \binom{2n}{n},$$

where $\displaystyle\binom{n}{s} = \frac{n!}{s!\,(n-s)!}$.

By considering similar identities, or otherwise, show also that:

(i) if n is an even integer, then

$$\sum_{s=0}^{n} (-1)^s \binom{n}{s}^2 = (-1)^{n/2}\binom{n}{n/2};$$

(ii) $$\sum_{t=1}^{n} 2t\binom{n}{t}^2 = n\binom{2n}{n}.$$

5 Show that if α is a solution of the equation

$$5\cos x + 12\sin x = 7,$$

then either

$$\cos\alpha = \frac{35 - 12\sqrt{120}}{169}$$

or $\cos\alpha$ has one other value which you should find.

Prove carefully that if $\pi/2 < \alpha < \pi$, then $\alpha < 3\pi/4$.

6 Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ if

$$y = \frac{ax+b}{cx+d}. \qquad (*)$$

By using changes of variable of the form $(*)$, or otherwise, show that

$$\int_0^1 \frac{1}{(x+3)^2}\,\ln\left(\frac{x+1}{x+3}\right)\mathrm{d}x = \frac{1}{6}\ln 3 - \frac{1}{4}\ln 2 - \frac{1}{12},$$

and evaluate the integrals

$$\int_0^1 \frac{1}{(x+3)^2}\,\ln\left(\frac{x^2+3x+2}{(x+3)^2}\right)\mathrm{d}x \quad \text{and} \quad \int_0^1 \frac{1}{(x+3)^2}\,\ln\left(\frac{x+1}{x+2}\right)\mathrm{d}x.$$

7 The curve C has equation

$$y = \frac{x}{\sqrt{(x^2 - 2x + a)}} ,$$

where the square root is positive. Show that, if $a > 1$, then C has exactly one stationary point.

Sketch C when (i) $a = 2$ and (ii) $a = 1$.

8 Prove that

$$\sum_{k=0}^{n} \sin k\theta = \frac{\cos \frac{1}{2}\theta - \cos(n + \frac{1}{2})\theta}{2 \sin \frac{1}{2}\theta} . \qquad (*)$$

(i) Deduce that, when n is large,

$$\sum_{k=0}^{n} \sin\left(\frac{k\pi}{n}\right) \approx \frac{2n}{\pi} .$$

(ii) By differentiating $(*)$ with respect to θ, or otherwise, show that, when n is large,

$$\sum_{k=0}^{n} k \sin^2\left(\frac{k\pi}{2n}\right) \approx \left(\frac{1}{4} + \frac{1}{\pi^2}\right) n^2 .$$

[The approximations, valid for small θ, $\sin\theta \approx \theta$ and $\cos\theta \approx 1 - \frac{1}{2}\theta^2$ may be assumed.]

Section B: Mechanics

9 In the Z–universe, a star of mass M suddenly blows up, and the fragments, with various initial speeds, start to move away from the centre of mass G which may be regarded as a fixed point. In the subsequent motion the acceleration of each fragment is directed towards G. Moreover, in accordance with the laws of physics of the Z–universe, there are positive constants k_1, k_2 and R such that when a fragment is at a distance x from G, the magnitude of its acceleration is k_1x^3 if $x < R$ and is k_2x^{-4} if $x \geqslant R$. The initial speed of a fragment is denoted by u.

(i) For $x < R$, write down a differential equation for the speed v, and hence determine v in terms of u, k_1 and x for $x < R$.

(ii) Show that if $u < a$, where $2a^2 = k_1R^4$, then the fragment does not reach a distance R from G.

(iii) Show that if $u \geqslant b$, where $6b^2 = 3k_1R^4 + 4k_2/R^3$, then from the moment of the explosion the fragment is always moving away from G.

(iv) If $a < u < b$, determine in terms of k_2, b and u the maximum distance from G attained by the fragment.

10 N particles P_1, P_2, P_3, …, P_N with masses m, qm, q^2m, … , $q^{N-1}m$, respectively, are at rest at distinct points along a straight line in gravity-free space. The particle P_1 is set in motion towards P_2 with velocity V and in every subsequent impact the coefficient of restitution is e, where $0 < e < 1$. Show that after the first impact the velocities of P_1 and P_2 are

$$\left(\frac{1-eq}{1+q}\right)V \quad \text{and} \quad \left(\frac{1+e}{1+q}\right)V,$$

respectively.

Show that if $q \leqslant e$, then there are exactly $N-1$ impacts and that if $q = e$, then the total loss of kinetic energy after all impacts have occurred is equal to

$$\frac{1}{2}me\big(1-e^{N-1}\big)V^2.$$

11 An automated mobile dummy target for gunnery practice is moving anti-clockwise around the circumference of a large circle of radius R in a horizontal plane at a constant angular speed ω. A shell is fired from O, the centre of this circle, with initial speed V and angle of elevation α. Show that if $V^2 < gR$, then no matter what the value of α, or what vertical plane the shell is fired in, the shell cannot hit the target.

Assume now that $V^2 > gR$ and that the shell hits the target, and let β be the angle through which the target rotates between the time at which the shell is fired and the time of impact. Show that β satisfies the equation

$$g^2\beta^4 - 4\omega^2V^2\beta^2 + 4R^2\omega^4 = 0.$$

Deduce that there are exactly two possible values of β.

Let β_1 and β_2 be the possible values of β and let P_1 and P_2 be the corresponding points of impact. By considering the quantities $(\beta_1^2 + \beta_2^2)$ and $\beta_1^2\beta_2^2$, or otherwise, show that the linear distance between P_1 and P_2 is

$$2R\sin\left(\frac{\omega}{g}\sqrt{(V^2 - Rg)}\right).$$

Section C: Probability and Statistics

12 It is known that there are three manufacturers A, B, C, who can produce micro chip MB666. The probability that a randomly selected MB666 is produced by A is $2p$, and the corresponding probabilities for B and C are p and $1-3p$, respectively, where $0 \leqslant p \leqslant \frac{1}{3}$. It is also known that 70% of MB666 micro chips from A are sound and that the corresponding percentages for B and C are 80% and 90%, respectively.

Find in terms of p, the conditional probability, $\mathrm{P}(A|S)$, that if a randomly selected MB666 chip is found to be sound then it came from A, and also the conditional probability, $\mathrm{P}(C|S)$, that if it is sound then it came from C.

A quality inspector took a random sample of one MB666 micro chip and found it to be sound. She then traced its place of manufacture to be A, and so estimated p by calculating the value of p that corresponds to the greatest value of $\mathrm{P}(A|S)$. A second quality inspector also a took random sample of one MB666 chip and found it to be sound. Later he traced its place of manufacture to be C and so estimated p by applying the procedure of his colleague to $\mathrm{P}(C|S)$.

Determine the values of the two estimates and comment briefly on the results obtained.

13 A stick is broken at a point, chosen at random, along its length. Find the probability that the ratio, R, of the length of the shorter piece to the length of the longer piece is less than r.

Find the probability density function for R, and calculate the mean and variance of R.

14 You play the following game. You throw a six-sided fair die repeatedly. You may choose to stop after any throw, except that you must stop if you throw a 1. Your score is the number obtained on your last throw. Determine the strategy that you should adopt in order to maximize your expected score, explaining your reasoning carefully.

Section A: Pure Mathematics

1 A number of the form $1/N$, where N is an integer greater than 1, is called a *unit fraction*.

Noting that

$$\frac{1}{2} = \frac{1}{3} + \frac{1}{6} \quad \text{and} \quad \frac{1}{3} = \frac{1}{4} + \frac{1}{12},$$

guess a general result of the form

$$\frac{1}{N} = \frac{1}{a} + \frac{1}{b} \qquad (*)$$

and hence prove that any unit fraction can be expressed as the sum of two distinct unit fractions.

By writing $(*)$ in the form

$$(a - N)(b - N) = N^2$$

and by considering the factors of N^2, show that if N is prime, then there is only one way of expressing $1/N$ as the sum of two distinct unit fractions.

Prove similarly that any fraction of the form $2/N$, where N is prime number greater than 2, can be expressed uniquely as the sum of two distinct unit fractions.

2 Prove that if $(x - a)^2$ is a factor of the polynomial $\mathrm{p}(x)$, then $\mathrm{p}'(a) = 0$. Prove a corresponding result if $(x - a)^4$ is a factor of $\mathrm{p}(x)$.

Given that the polynomial

$$x^6 + 4x^5 - 5x^4 - 40x^3 - 40x^2 + 32x + k$$

has a factor of the form $(x - a)^4$, find k.

3 The lengths of the sides BC, CA, AB of the triangle ABC are denoted by a, b, c, respectively. Given that

$$b = 8 + \epsilon_1,\ c = 3 + \epsilon_2,\ A = \pi/3 + \epsilon_3,$$

where ϵ_1, ϵ_2, and ϵ_3 are small, show that $a \approx 7 + \eta$, where $\eta = (13\,\epsilon_1 - 2\,\epsilon_2 + 24\sqrt{3}\,\epsilon_3)/14$.

Given now that

$$|\epsilon_1| \leqslant 2 \times 10^{-3}, \quad |\epsilon_2| \leqslant 4 \cdot 9 \times 10^{-2}, \quad |\epsilon_3| \leqslant \sqrt{3} \times 10^{-3},$$

find the range of possible values of η.

4 Prove that

$$(\cos\theta + i\sin\theta)(\cos\phi + i\sin\phi) = \cos(\theta+\phi) + i\sin(\theta+\phi)$$

and that, for every positive integer n,

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta.$$

By considering $(5-i)^2(1+i)$, or otherwise, prove that

$$\arctan(7/17) + 2\arctan(1/5) = \pi/4.$$

Prove also that

$$3\arctan(1/4) + \arctan(1/20) + \arctan(1/1985) = \pi/4.$$

[Note that $\arctan\theta$ is another notation for $\tan^{-1}\theta$.]

5 It is required to approximate a given function $\mathrm{f}(x)$, over the interval $0 \leqslant x \leqslant 1$, by the linear function λx, where λ is chosen to minimise

$$\int_0^1 \Big(\mathrm{f}(x) - \lambda x\Big)^2 \,\mathrm{d}x.$$

Show that

$$\lambda = 3\int_0^1 x\mathrm{f}(x)\,\mathrm{d}x.$$

The residual error, R, of this approximation process is such that

$$R^2 = \int_0^1 \Big(\mathrm{f}(x) - \lambda x\Big)^2 \,\mathrm{d}x.$$

Show that

$$R^2 = \int_0^1 \Big(\mathrm{f}(x)\Big)^2 \,\mathrm{d}x - \frac{1}{3}\lambda^2.$$

Given now that $\mathrm{f}(x) = \sin(\pi x/n)$, show that (i) for large n, $\lambda \approx \pi/n$ and (ii) $\lim_{n\to\infty} R = 0$.

Explain why, prior to any calculation, these results are to be expected.

[You may assume that, when θ is small, $\sin\theta \approx \theta - \theta^3/6$ and $\cos\theta \approx 1 - \theta^2/2$.]

6 Show that

$$\sin\theta = \frac{2t}{1+t^2}, \quad \cos\theta = \frac{1-t^2}{1+t^2}, \quad \frac{1+\cos\theta}{\sin\theta} = \tan(\pi/2 - \theta/2),$$

where $t = \tan(\theta/2)$.

Use the substitution $t = \tan(\theta/2)$ to show that, for $0 < \alpha < \pi/2$,

$$\int_0^{\frac{\pi}{2}} \frac{1}{1+\cos\alpha\sin\theta}\,\mathrm{d}\theta = \frac{\alpha}{\sin\alpha},$$

and deduce a similar result for

$$\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin\alpha\cos\theta}\,\mathrm{d}\theta.$$

7 The line l has vector equation $\mathbf{r} = \lambda\mathbf{s}$, where

$$\mathbf{s} = (\cos\theta + \sqrt{3}\,)\,\mathbf{i} + (\sqrt{2}\,\sin\theta)\,\mathbf{j} + (\cos\theta - \sqrt{3}\,)\,\mathbf{k}$$

and λ is a scalar parameter. Find an expression for the angle between l and the line $\mathbf{r} = \mu(a\,\mathbf{i} + b\,\mathbf{j} + c\,\mathbf{k})$. Show that there is a line m through the origin such that, whatever the value of θ, the acute angle between l and m is $\pi/6$.

A plane has equation $x - z = 4\sqrt{3}$. The line l meets this plane at P. Show that, as θ varies, P describes a circle, with its centre on m. Find the radius of this circle.

8 (i) Let y be the solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 4x\,\mathrm{e}^{-x^2}(y+3)^{\frac{1}{2}} = 0 \qquad (x \geqslant 0),$$

that satisfies the condition $y = 6$ when $x = 0$. Find y in terms of x and show that $y \to 1$ as $x \to \infty$.

(ii) Let y be any solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - x\,\mathrm{e}^{6x^2}(y+3)^{1-k} = 0 \qquad (x \geqslant 0).$$

Find a value of k such that, as $x \to \infty$, $\mathrm{e}^{-3x^2}y$ tends to a finite non-zero limit, which you should determine.

Section B: Mechanics

9 In an aerobatics display, Jane and Karen jump from a great height and go through a period of free fall before opening their parachutes. While in free fall at speed v, Jane experiences air resistance kv per unit mass but Karen, who spread-eagles, experiences air resistance $kv + (2k^2/g)v^2$ per unit mass. Show that Jane's speed can never reach g/k. Obtain the corresponding result for Karen.

Jane opens her parachute when her speed is $g/(3k)$. Show that she has then been in free fall for time $k^{-1}\ln(3/2)$.

Karen also opens her parachute when her speed is $g/(3k)$. Find the time she has then been in free fall.

10 A long light inextensible string passes over a fixed smooth light pulley. A particle of mass 4 kg is attached to one end A of this string and the other end is attached to a second smooth light pulley. A long light inextensible string BC passes over the second pulley and has a particle of mass 2 kg attached at B and a particle of mass of 1 kg attached at C. The system is held in equilibrium in a vertical plane. The string BC is then released from rest. Find the accelerations of the two moving particles.

After T seconds, the end A is released so that all three particles are now moving in a vertical plane. Find the accelerations of A, B and C in this second phase of the motion. Find also, in terms of g and T, the speed of A when B has moved through a total distance of $0.6gT^2$ metres.

11 The string AP has a natural length of 1·5 metres and modulus of elasticity equal to $5g$ newtons. The end A is attached to the ceiling of a room of height 2·5 metres and a particle of mass 0·5 kg is attached to the end P. The end P is released from rest at a point 0·5 metres above the floor and vertically below A. Show that the string becomes slack, but that P does not reach the ceiling.

Show also that while the string is in tension, P executes simple harmonic motion, and that the time in seconds that elapses from the instant when P is released to the instant when P first returns to its original position is

$$\left(\frac{8}{3g}\right)^{\frac{1}{2}} + \left(\frac{3}{5g}\right)^{\frac{1}{2}} \Big(\pi - \arccos(3/7)\Big).$$

[Note that $\arccos x$ is another notation for $\cos^{-1} x$.]

Section C: Probability and Statistics

12 *Tabulated values of $\Phi(\cdot)$, the cumulative distribution function of a standard normal variable, should not be used in this question.*

Henry the commuter lives in Cambridge and his working day starts at his office in London at 0900. He catches the 0715 train to King's Cross with probability p, or the 0720 to Liverpool Street with probability $1-p$. Measured in minutes, journey times for the first train are $N(55, 25)$ and for the second are $N(65, 16)$. Journey times from King's Cross and Liverpool Street to his office are $N(30, 144)$ and $N(25, 9)$, respectively. Show that Henry is more likely to be late for work if he catches the first train.

Henry makes M journeys, where M is large. Writing A for $1-\Phi(20/13)$ and B for $1-\Phi(2)$, find, in terms of A, B, M and p, the expected number, L, of times that Henry will be late and show that for all possible values of p,

$$BM \leqslant L \leqslant AM.$$

Henry noted that in 3/5 of the occasions when he was late, he had caught the King's Cross train. Obtain an estimate of p in terms of A and B.

[A random variable is said to be $N\left(\mu, \sigma^2\right)$ if it has a normal distribution with mean μ and variance σ^2.]

13 A group of biologists attempts to estimate the magnitude, N, of an island population of voles (*Microtus agrestis*). Accordingly, the biologists capture a random sample of 200 voles, mark them and release them. A second random sample of 200 voles is then taken of which 11 are found to be marked. Show that the probability, p_N, of this occurrence is given by

$$p_N = k\frac{\Big((N-200)!\Big)^2}{N!(N-389)!},$$

where k is independent of N.

The biologists then estimate N by calculating the value of N for which p_N is a maximum. Find this estimate.

All unmarked voles in the second sample are marked and then the entire sample is released. Subsequently a third random sample of 200 voles is taken. Write down the probability that this sample contains exactly j marked voles, leaving your answer in terms of binomial coefficients.

Deduce that

$$\sum_{j=0}^{200}\binom{389}{j}\binom{3247}{200-j} = \binom{3636}{200}.$$

14 The random variables X_1, X_2, ... , X_{2n+1} are independently and uniformly distributed on the interval $0 \leqslant x \leqslant 1$. The random variable Y is defined to be the median of X_1, X_2, ... , X_{2n+1}. Given that the probability density function of Y is $\mathrm{g}(y)$, where

$$\mathrm{g}(y) = \begin{cases} ky^n(1-y)^n & \text{if } 0 \leqslant y \leqslant 1 \\ 0 & \text{otherwise,} \end{cases}$$

use the result

$$\int_0^1 y^r(1-y)^s\,dy = \frac{r!s!}{(r+s+1)!}$$

to show that $k = (2n+1)!/(n!)^2$, and evaluate $\mathrm{E}(Y)$ and $\mathrm{Var}\,(Y)$. Hence show that, for any given positive number d, the inequality

$$\mathrm{P}\left(|Y-1/2| < d/\sqrt{n}\right) < \mathrm{P}\left(|\bar{X}-1/2| < d/\sqrt{n}\right)$$

holds provided n is large enough, where $\bar{X}$ is the mean of X_1, X_2, ... , X_{2n+1}.

[You may assume that Y and $\bar{X}$ are normally distributed for large n.]

Section A: Pure Mathematics

1 Use the binomial expansion to obtain a polynomial of degree 2 which is a good approximation to $\sqrt{(1-x)}$ when x is small.

(i) By taking $x = 1/100$, show that $\sqrt{(11)} \approx 79599/24000$, and estimate, correct to 1 significant figure, the error in this approximation. (You may assume that the error is given approximately by the first neglected term in the binomial expansion.)

(ii) Find a rational number which approximates $\sqrt{(1111)}$ with an error of about 2×10^{-12}.

2 Sketch the graph of the function $[x/N]$, for $0 < x < 2N$, where the notation $[y]$ means the integer part of y. (Thus $[2.9] = 2$, $[4] = 4$.)

(i) Prove that

$$\sum_{k=1}^{2N}(-1)^{[k/N]}k = 2N - N^2.$$

(ii) Let

$$S_N = \sum_{k=1}^{2N}(-1)^{[k/N]}2^{-k}.$$

Find S_N in terms of N and determine the limit of S_N as $N \to \infty$.

3 The cuboid $ABCDEFGH$ is such AE, BF, CG, DH are perpendicular to the opposite faces $ABCD$ and $EFGH$, and $AB = 2, BC = 1, AE = \lambda$. Show that if α is the acute angle between the diagonals AG and BH then

$$\cos\alpha = \left|\frac{3-\lambda^2}{5+\lambda^2}\right|$$

Let R be the ratio of the volume of the cuboid to its surface area. Show that $R < 1/3$ for all possible values of λ.

Prove that, if $R \geqslant 1/4$, then $\alpha \leqslant \arccos(1/9)$.

4 Let

$$\mathrm{f}(x) = P\sin x + Q\sin 2x + R\sin 3x\ .$$

Show that if $Q^2 < 4R(P-R)$, then the only values of x for which $\mathrm{f}(x) = 0$ are given by $x = m\pi$, where m is an integer.
[You may assume that $\sin 3x = \sin x(4\cos^2 x - 1)$.]

Now let

$$\mathrm{g}(x) = \sin 2nx + \sin 4nx - \sin 6nx,$$

where n is a positive integer and $0 < x < \pi/2$. Find an expression for the largest root of the equation $\mathrm{g}(x) = 0$, distinguishing between the cases where n is even and n is odd.

5 The curve C_1 passes through the origin in the x–y plane and its gradient is given by

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x(1-x^2)e^{-x^2}.$$

Show that C_1 has a minimum point at the origin and a maximum point at $\left(1, \frac{1}{2}\,e^{-1}\right)$. Find the coordinates of the other stationary point. Give a rough sketch of C_1.

The curve C_2 passes through the origin and its gradient is given by

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x(1-x^2)e^{-x^3}.$$

Show that C_2 has a minimum point at the origin and a maximum point at $(1, k)$, where $k > \frac{1}{2}\,e^{-1}$. (You need not find k.)

6 Show that

$$\int_0^1 \frac{x^4}{1+x^2}\ \mathrm{d}x = \frac{\pi}{4} - \frac{2}{3}\ .$$

Determine the values of

(i) $$\int_0^1 x^3\ \tan^{-1}\left(\frac{1-x}{1+x}\right)\ \mathrm{d}x\ ,$$

(ii) $$\int_0^1 \frac{(1-y)^3}{(1+y)^5}\ \tan^{-1}y\ \mathrm{d}y\ .$$

7 In an Argand diagram, O is the origin and P is the point $2 + 0i$. The points Q, R and S are such that the lengths OP, PQ, QR and RS are all equal, and the angles OPQ, PQR and QRS are all equal to $5\pi/6$, so that the points O, P, Q, R and S are five vertices of a regular 12-sided polygon lying in the upper half of the Argand diagram. Show that Q is the point $2 + \sqrt{3} + i$ and find S.

The point C is the centre of the circle that passes through the points O, P and Q. Show that, if the polygon is rotated anticlockwise about O until C first lies on the real axis, the new position of S is

$$-\frac{1}{2}(3\sqrt{2} + \sqrt{6})(\sqrt{3} - i)\ .$$

8 The function f satisfies $\mathrm{f}(x + 1) = \mathrm{f}(x)$ and $\mathrm{f}(x) > 0$ for all x.

(i) Give an example of such a function.

(ii) The function F satisfies

$$\frac{\mathrm{dF}}{\mathrm{d}x} = \mathrm{f}(x)$$

and $\mathrm{F}(0) = 0$. Show that $\mathrm{F}(n) = n\mathrm{F}(1)$, for any positive integer n.

(iii) Let y be the solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{f}(x)y = 0$$

that satisfies $y = 1$ when $x = 0$. Show that $y(n) \to 0$ as $n \to \infty$, where $n = 1, 2, 3, \ldots$

Section B: Mechanics

9 A particle of unit mass is projected vertically upwards with speed u. At height x, while the particle is moving upwards, it is found to experience a total force F, due to gravity and air resistance, given by $F = \alpha\, \mathrm{e}^{-\beta x}$, where α and β are positive constants. Calculate the energy expended in reaching this height. Show that

$$F = \tfrac{1}{2}\beta v^2 + \alpha - \tfrac{1}{2}\beta u^2 \ ,$$

where v is the speed of the particle, and explain why $\alpha = \frac{1}{2}\beta u^2 + g$, where g is the acceleration due to gravity.

Determine an expression, in terms of y, g and β, for the air resistance experienced by the particle on its downward journey when it is at a distance y below its highest point.

10 Two particles A and B of masses m and km, respectively, are at rest on a smooth horizontal surface. The direction of the line passing through A and B is perpendicular to a vertical wall which is on the other side of B from A. The particle A is now set in motion towards B with speed u. The coefficient of restitution between A and B is e_1 and between B and the wall is e_2. Show that there will be a second collision between A and B provided

$$k < \frac{1 + e_2(1 + e_1)}{e_1} \ .$$

Show that, if $e_1 = \frac{1}{3}$, $e_2 = \frac{1}{2}$ and $k < 5$, then the kinetic energy of A and B immediately after B rebounds from the wall is greater than $mu^2/27$.

11 A two-stage missile is projected from a point A on the ground with horizontal and vertical velocity components u and v, respectively. When it reaches the highest point of its trajectory an internal explosion causes it to break up into two fragments. Immediately after this explosion one of these fragments, P, begins to move vertically upwards with speed v_e, but retains the previous horizontal velocity. Show that P will hit the ground at a distance R from A given by

$$\frac{gR}{u} = v + v_e + \sqrt{(v_e^2 + v^2)}.$$

It is required that the range R should be greater than a certain distance D (where $D > 2uv/g$). Show that this requirement is satisfied if

$$v_e > \frac{gD}{2u}\left(\frac{gD - 2uv}{gD - uv}\right).$$

[*The effect of air resistance is to be neglected.*]

Section C: Probability and Statistics

12 The national lottery of Ruritania is based on the positive integers from 1 to N, where N is very large and fixed. Tickets cost £1 each. For each ticket purchased, the punter (i.e. the purchaser) chooses a number from 1 to N. The winning number is chosen at random, and the jackpot is shared equally amongst those punters who chose the winning number.

A syndicate decides to buy N tickets, choosing every number once to be sure of winning a share of the jackpot. The total number of tickets purchased in this draw is $3.8N$ and the jackpot is £W. Assuming that the non-syndicate punters choose their numbers independently and at random, find the most probable number of winning tickets and show that the expected net loss of the syndicate is approximately

$$N \ - \ \frac{5(1-e^{-2.8})}{14}\, W \ .$$

13 The life times of a large batch of electric light bulbs are independently and identically distributed. The probability that the life time, T hours, of a given light bulb is greater than t hours is given by

$$\mathrm{P}(T > t) \ = \ \frac{1}{(1+kt)^{\alpha}} \ ,$$

where α and k are constants, and $\alpha > 1$. Find the median M and the mean m of T in terms of α and k.

Nine randomly selected bulbs are switched on simultaneously and are left until all have failed. The fifth failure occurs at 1000 hours and the mean life time of all the bulbs is found to be 2400 hours. Show that $\alpha \approx 2$ and find the approximate value of k. Hence estimate the probability that, if a randomly selected bulb is found to last M hours, it will last a further $m - M$ hours.

14 Two coins A and B are tossed together. A has probability p of showing a head, and B has probability $2p$, independent of A, of showing a head, where $0 < p < \frac{1}{2}$. The random variable X takes the value 1 if A shows a head and it takes the value 0 if A shows a tail. The random variable Y takes the value 1 if B shows a head and it takes the value 0 if B shows a tail. The random variable T is defined by

$$T = \lambda X + \tfrac{1}{2}(1-\lambda)Y.$$

Show that $\mathrm{E}(T) = p$ and find an expression for $\mathrm{Var}\,(T)$ in terms of p and λ. Show that as λ varies, the minimum of $\mathrm{Var}\,(T)$ occurs when

$$\lambda = \frac{1-2p}{3-4p}\,.$$

The two coins are tossed n times, where $n > 30$, and $\overline{T}$ is the mean value of T. Let b be a fixed positive number. Show that the maximum value of $\mathrm{P}(|\overline{T} - p| < b)$ as λ varies is approximately $2\Phi(b/s) - 1$, where Φ is the cumulative distribution function of a standard normal variate and

$$s^2 = \frac{p(1-p)(1-2p)}{(3-4p)n}\,.$$

Section A: Pure Mathematics

1 Show that

$$\int_{\pi/6}^{\pi/4} \frac{1}{1-\cos 2\theta}\ \mathrm{d}\theta = \frac{\sqrt{3}}{2} - \frac{1}{2}\,.$$

By using the substitution $x = \sin 2\theta$, or otherwise, show that

$$\int_{\sqrt{3}/2}^{1} \frac{1}{1-\sqrt{(1-x^2)}}\ \mathrm{d}x = \sqrt{3} - 1 - \frac{\pi}{6}\,.$$

Hence evaluate the integral

$$\int_{1}^{2/\sqrt{3}} \frac{1}{y(y-\sqrt{(y^2-1^2)})}\ \mathrm{d}y\,.$$

2 Show that setting $z - z^{-1} = w$ in the quartic equation

$$z^4 + 5z^3 + 4z^2 - 5z + 1 = 0$$

results in the quadratic equation $w^2 + 5w + 6 = 0$. Hence solve the above quartic equation.

Solve similarly the equation

$$2z^8 - 3z^7 - 12z^6 + 12z^5 + 22z^4 - 12z^3 - 12z^2 + 3z + 2 = 0\,.$$

3 The nth Fermat number, F_n, is defined by

$$F_n = 2^{2^n} + 1\,, \qquad n = 0,\ 1,\ 2,\ \ldots\ ,$$

where 2^{2^n} means 2 raised to the power 2^n. Calculate F_0, F_1, F_2 and F_3. Show that, for $k = 1$, $k = 2$ and $k = 3$,

$$F_0 F_1 \ldots F_{k-1} = F_k - 2\,. \qquad (*)$$

Prove, by induction, or otherwise, that $(*)$ holds for all $k \geqslant 1$. Deduce that no two Fermat numbers have a common factor greater than 1.

Hence show that there are infinitely many prime numbers.

4 Give a sketch to show that, if $\mathrm{f}(x) > 0$ for $p < x < q$, then $\int_p^q \mathrm{f}(x)\,\mathrm{d}x > 0$.

(i) By considering $\mathrm{f}(x) = ax^2 - bx + c$ show that, if $a > 0$ and $b^2 < 4ac$, then $3b < 2a + 6c$.

(ii) By considering $\mathrm{f}(x) = a\sin^2 x - b\sin x + c$ show that, if $a > 0$ and $b^2 < 4ac$, then $4b < (a + 2c)\pi$.

(iii) Show that, if $a > 0$, $b^2 < 4ac$ and $q > p > 0$, then

$$b\ln(q/p) < a\left(\frac{1}{p} - \frac{1}{q}\right) + c(q - p)\,.$$

5 The numbers x_n, where $n = 0, 1, 2, \ldots$, satisfy

$$x_{n+1} = kx_n(1 - x_n)\,.$$

(i) Prove that, if $0 < k < 4$ and $0 < x_0 < 1$, then $0 < x_n < 1$ for all n.

(ii) Given that $x_0 = x_1 = x_2 = \cdots = a$, with $a \neq 0$ and $a \neq 1$, find k in terms of a.

(iii) Given instead that $x_0 = x_2 = x_4 = \cdots = a$, with $a \neq 0$ and $a \neq 1$, show that $ab^3 - b^2 + (1 - a) = 0$, where $b = k(1 - a)$. Given, in addition, that $x_1 \neq a$, find the possible values of k in terms of a.

6 The lines l_1, l_2 and l_3 lie in an inclined plane P and pass through a common point A. The line l_2 is a line of greatest slope in P. The line l_1 is perpendicular to l_3 and makes an acute angle α with l_2. The angles between the horizontal and l_1, l_2 and l_3 are $\pi/6$, β and $\pi/4$, respectively. Show that $\cos\alpha\sin\beta = \frac{1}{2}$ and find the value of $\sin\alpha\sin\beta$. Deduce that $\beta = \pi/3$.

The lines l_1 and l_3 are rotated in P about A so that l_1 and l_3 remain perpendicular to each other. The new acute angle between l_1 and l_2 is θ. The new angles which l_1 and l_3 make with the horizontal are ϕ and 2ϕ, respectively. Show that

$$\tan^2\theta = \frac{3 + \sqrt{13}}{2}\,.$$

7 In 3-dimensional space, the lines m_1 and m_2 pass through the origin and have directions $\mathbf{i}+\mathbf{j}$ and $\mathbf{i}+\mathbf{k}$, respectively. Find the directions of the two lines m_3 and m_4 that pass through the origin and make angles of $\pi/4$ with both m_1 and m_2. Find also the cosine of the acute angle between m_3 and m_4.

The points A and B lie on m_1 and m_2 respectively, and are each at distance $\lambda\sqrt{2}$ units from O. The points P and Q lie on m_3 and m_4 respectively, and are each at distance 1 unit from O. If all the coordinates (with respect to axes $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$) of A, B, P and Q are non-negative, prove that:

(i) there are only two values of λ for which AQ is perpendicular to BP;

(ii) there are no non-zero values of λ for which AQ and BP intersect.

8 Find y in terms of x, given that:

$$\text{for } x<0\,, \quad \frac{\mathrm{d}y}{\mathrm{d}x}=-y \quad \text{and} \quad y=a \text{ when } x=-1\,;$$
$$\text{for } x>0\,, \quad \frac{\mathrm{d}y}{\mathrm{d}x}=y \quad \text{and} \quad y=b \text{ when } x=1\,.$$

Sketch a solution curve. Determine the condition on a and b for the solution curve to be continuous (that is, for there to be no 'jump' in the value of y) at $x=0$.

Solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x}=|\,\mathrm{e}^x-1|\,y$$

given that $y=\mathrm{e}^{\,\mathrm{e}}$ when $x=1$ and that y is continuous at $x=0$. Write down the following limits:

(i) $\lim_{x\to+\infty} y\exp(-\mathrm{e}^x)$; (ii) $\lim_{x\to-\infty} y\,\mathrm{e}^{-x}$.

Section B: Mechanics

9 A particle is projected from a point O on a horizontal plane with speed V and at an angle of elevation α. The vertical plane in which the motion takes place is perpendicular to two vertical walls, both of height h, at distances a and b from O. Given that the particle just passes over the walls, find $\tan\alpha$ in terms of a, b and h and show that

$$\frac{2V^2}{g} = \frac{ab}{h} + \frac{(a+b)^2 h}{ab} \, .$$

The heights of the walls are now increased by the same small positive amount δh. A second particle is projected so that it just passes over both walls, and the new angle and speed of projection are $\alpha + \delta\alpha$ and $V + \delta V$, respectively. Show that

$$\sec^2\alpha \, \delta\alpha \approx \frac{a+b}{ab} \, \delta h \, ,$$

and deduce that $\delta\alpha > 0$. Show also that δV is positive if $h > ab/(a+b)$ and negative if $h < ab/(a+b)$.

10 A competitor in a Marathon of $42\frac{3}{8}$ km runs the first t hours of the race at a constant speed of 13 km h^{-1} and the remainder at a constant speed of $14 + 2t/T$ km h^{-1}, where T hours is her time for the race. Show that the minimum possible value of T over all possible values of t is 3.

The speed of another competitor decreases linearly with respect to time from 16 km h^{-1} at the start of the race. If both of these competitors have a run time of 3 hours, find the maximum distance between them at any stage of the race.

11 A rigid straight beam AB has length l and weight W. Its weight per unit length at a distance x from B is $\alpha W l^{-1}(x/l)^{\alpha-1}$, where α is a positive constant. Show that the centre of mass of the beam is at a distance $\alpha l/(\alpha+1)$ from B.

The beam is placed with the end A on a rough horizontal floor and the end B resting against a rough vertical wall. The beam is in a vertical plane at right angles to the plane of the wall and makes an angle of θ with the floor. The coefficient of friction between the floor and the beam is μ and the coefficient of friction between the wall and the beam is also μ. Show that, if the equilibrium is limiting at both A and B, then

$$\tan\theta = \frac{1 - \alpha\mu^2}{(1+\alpha)\mu} \, .$$

Given that $\alpha = 3/2$ and given also that the beam slides for any $\theta < \pi/4$ find the greatest possible value of μ.

Section C: Probability and Statistics

12 On K consecutive days each of L identical coins is thrown M times. For each coin, the probability of throwing a head in any one throw is p (where $0 < p < 1$). Show that the probability that on exactly k of these days more than l of the coins will each produce fewer than m heads can be approximated by

$$\binom{K}{k} q^k (1-q)^{K-k},$$

where

$$q = \Phi\left(\frac{2h - 2l - 1}{2\sqrt{h}}\right), \qquad h = L\Phi\left(\frac{2m - 1 - 2Mp}{2\sqrt{Mp(1-p)}}\right)$$

and $\Phi(.)$ is the cumulative distribution function of a standard normal variate.

Would you expect this approximation to be accurate in the case $K = 7$, $k = 2$, $L = 500$, $l = 4$, $M = 100$, $m = 48$ and $p = 0.6$?

13 Let $\mathrm{F}(x)$ be the cumulative distribution function of a random variable X, which satisfies $\mathrm{F}(a) = 0$ and $\mathrm{F}(b) = 1$, where $a > 0$. Let

$$\mathrm{G}(y) = \frac{\mathrm{F}(y)}{2 - \mathrm{F}(y)} \, .$$

Show that $\mathrm{G}(a) = 0$, $\mathrm{G}(b) = 1$ and that $\mathrm{G}'(y) \geqslant 0$. Show also that

$$\frac{1}{2} \leqslant \frac{2}{(2 - \mathrm{F}(y))^2} \leqslant 2 \, .$$

The random variable Y has cumulative distribution function $\mathrm{G}(y)$. Show that

$$\tfrac{1}{2}\,\mathrm{E}(X) \leqslant \mathrm{E}(Y) \leqslant 2\mathrm{E}(X) \, ,$$

and that

$$\mathrm{Var}\,(Y) \leqslant 2\,\mathrm{Var}\,(X) + \tfrac{7}{4}(\mathrm{E}(X))^2 \, .$$

14 A densely populated circular island is divided into N concentric regions R_1, R_2, ..., R_N, such that the inner and outer radii of R_n are $n-1$ km and n km, respectively. The average number of road accidents that occur in any one day in R_n is $2 - n/N$, independently of the number of accidents in any other region.

Each day an observer selects a region at random, with a probability that is proportional to the area of the region, and records the number of road accidents, X, that occur in it. Show that, in the long term, the average number of recorded accidents per day will be

$$2 - \frac{1}{6}\left(1 + \frac{1}{N}\right)\left(4 - \frac{1}{N}\right) .$$

[Note: $\sum_{n=1}^{N} n^2 = \frac{1}{6}N(N+1)(2N+1)$.]

Show also that

$$\mathrm{P}(X = k) = \frac{\mathrm{e}^{-2}N^{-k-2}}{k!}\sum_{n=1}^{N}(2n-1)(2N-n)^k\,\mathrm{e}^{n/N} .$$

Suppose now that $N = 3$ and that, on a particular day, two accidents were recorded. Show that the probability that R_2 had been selected is

$$\frac{48}{48 + 45\,\mathrm{e}^{1/3} + 25\,\mathrm{e}^{-1/3}} .$$

Section A: Pure Mathematics

1 Consider the equations

$$\begin{aligned} ax - y - z &= 3\,, \\ 2ax - y - 3z &= 7\,, \\ 3ax - y - 5z &= b\,, \end{aligned}$$

where a and b are given constants.

(i) In the case $a = 0$, show that the equations have a solution if and only if $b = 11$.

(ii) In the case $a \neq 0$ and $b = 11$ show that the equations have a solution with $z = \lambda$ for any given number λ.

(iii) In the case $a = 2$ and $b = 11$ find the solution for which $x^2 + y^2 + z^2$ is least.

(iv) Find a value for a for which there is a solution such that $x > 10^6$ and $y^2 + z^2 < 1$.

2 Write down a value of θ in the interval $\pi/4 < \theta < \pi/2$ that satisfies the equation

$$4\cos\theta + 2\sqrt{3}\,\sin\theta = 5\,.$$

Hence, or otherwise, show that

$$\pi = 3\arccos(5/\sqrt{28}) + 3\arctan(\sqrt{3}/2)\,.$$

Show that

$$\pi = 4\arcsin(7\sqrt{2}/10) - 4\arctan(3/4)\,.$$

3 Prove that the cube root of any irrational number is an irrational number.

Let $u_n = 5^{1/(3^n)}$. Given that $\sqrt[3]{5}$ is an irrational number, prove by induction that u_n is an irrational number for every positive integer n.

Hence, or otherwise, give an example of an infinite sequence of irrational numbers which converges to a given integer m.

[An irrational number is a number that cannot be expressed as the ratio of two integers.]

4 The line $y = d$, where $d > 0$, intersects the circle $x^2 + y^2 = R^2$ at G and H. Show that the area of the minor segment GH is equal to

$$R^2 \arccos\left(\frac{d}{R}\right) - d\sqrt{R^2 - d^2}\,. \qquad (*)$$

In the following cases, the given line intersects the given circle. Determine how, in each case, the expression $(*)$ should be modified to give the area of the minor segment.

(i) Line: $y = c$; circle: $(x - a)^2 + (y - b)^2 = R^2$.

(ii) Line: $y = mx + c$; circle: $x^2 + y^2 = R^2$.

(iii) Line: $y = mx + c$; circle: $(x - a)^2 + (y - b)^2 = R^2$.

5 The position vectors of the points A, B and P with respect to an origin O are $a\mathbf{i}$, $b\mathbf{j}$ and $l\mathbf{i} + m\mathbf{j} + n\mathbf{k}$, respectively, where a, b, and n are all non-zero. The points E, F, G and H are the midpoints of OA, BP, OB and AP, respectively. Show that the lines EF and GH intersect.

Let D be the point with position vector $d\mathbf{k}$, where d is non-zero, and let S be the point of intersection of EF and GH. The point T is such that the mid-point of DT is S. Find the position vector of T and hence find d in terms of n if T lies in the plane OAB.

6 The function f is defined by

$$\mathrm{f}(x) = |x - 1|\,,$$

where the domain is $\mathbf{R}$, the set of all real numbers. The function $\mathrm{g}_n = \mathrm{f}^n$, with domain $\mathbf{R}$, so for example $\mathrm{g}_3(x) = \mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$. In separate diagrams, sketch graphs of g_1, g_2, g_3 and g_4.

The function h is defined by

$$\mathrm{h}(x) = \left|\sin\frac{\pi x}{2}\right|\,,$$

where the domain is $\mathbf{R}$. Show that if n is even,

$$\int_0^n \left(\mathrm{h}(x) - \mathrm{g}_n(x)\right)\,\mathrm{d}x = \frac{2n}{\pi} - \frac{n}{2}\,.$$

7 Show that, if $n > 0$, then

$$\int_{e^{1/n}}^{\infty} \frac{\ln x}{x^{n+1}}\,\mathrm{d}x = \frac{2}{n^2 e}\,.$$

You may assume that $\dfrac{\ln x}{x} \to 0$ as $x \to \infty$.

Explain why, if $1 < a < b$, then

$$\int_{b}^{\infty} \frac{\ln x}{x^{n+1}}\,\mathrm{d}x < \int_{a}^{\infty} \frac{\ln x}{x^{n+1}}\,\mathrm{d}x\,.$$

Deduce that

$$\sum_{n=1}^{N} \frac{1}{n^2} < \frac{e}{2}\int_{e^{1/N}}^{\infty} \left(\frac{1-x^{-N}}{x^2-x}\right)\ln x\,\mathrm{d}x\,,$$

where N is any integer greater than 1.

8 It is given that y satisfies

$$\frac{\mathrm{d}y}{\mathrm{d}t} + k\left(\frac{t^2-3t+2}{t+1}\right)y = 0\,,$$

where k is a constant, and $y = A$ when $t = 0$, where A is a positive constant. Find y in terms of t, k and A.

Show that y has two stationary values whose ratio is $(3/2)^{6k}e^{-5k/2}$.

Describe the behaviour of y as $t \to +\infty$ for the case where $k > 0$ and for the case where $k < 0$.

In separate diagrams, sketch the graph of y for $t > 0$ for each of these cases.

Section B: Mechanics

9 AB is a uniform rod of weight W. The point C on AB is such that $AC > CB$. The rod is in contact with a rough horizontal floor at A and with a cylinder at C. The cylinder is fixed to the floor with its axis horizontal. The rod makes an angle α with the horizontal and lies in a vertical plane perpendicular to the axis of the cylinder. The coefficient of friction between the rod and the floor is $\tan\lambda_1$ and the coefficient of friction between the rod and the cylinder is $\tan\lambda_2$.

Show that if friction is limiting both at A and at C, and $\alpha \neq \lambda_2 - \lambda_1$, then the frictional force acting on the rod at A has magnitude

$$\frac{W\sin\lambda_1\,\sin(\alpha-\lambda_2)}{\sin(\alpha+\lambda_1-\lambda_2)}\,.$$

10 A bead B of mass m can slide along a rough horizontal wire. A light inextensible string of length 2ℓ has one end attached to a fixed point A of the wire and the other to B. A particle P of mass $3m$ is attached to the mid-point of the string and B is held at a distance ℓ from A. The bead is released from rest.

Let a_1 and a_2 be the magnitudes of the horizontal and vertical components of the intial acceleration of P. Show by considering the motion of P relative to A, or otherwise, that $a_1 = \sqrt{3}a_2$. Show also that the magnitude of the intial acceleration of B is $2a_1$.

Given that the frictional force opposing the motion of B is equal to $(\sqrt{3}/6)R$, where R is the normal reaction between B and the wire, show that the magnitude of the intial acceleration of P is $g/18$.

11 A particle P_1 is projected with speed V at an angle of elevation α $(> 45^\circ)$, from a point in a horizontal plane. Find T_1, the flight time of P_1, in terms of α, V and g. Show that the time after projection at which the direction of motion of P_1 first makes an angle of 45° with the horizontal is $\frac{1}{2}(1-\cot\alpha)T_1$.

A particle P_2 is projected under the same conditions. When the direction of the motion of P_2 first makes an angle of 45° with the horizontal, the speed of P_2 is instantaneously doubled. If T_2 is the total flight time of P_2, show that

$$\frac{2T_2}{T_1} = 1 + \cot\alpha + \sqrt{1+3\cot^2\alpha}\,.$$

SECTION C: Probability and Statistics

12 The life of a certain species of elementary particles can be described as follows. Each particle has a life time of T seconds, after which it disintegrates into X particles of the same species, where X is a random variable with binomial distribution $\mathrm{B}(2,p)$. A population of these particles starts with the creation of a single such particle at $t=0$. Let X_n be the number of particles in existence in the time interval $nT<t<(n+1)T$, where $n=1,2,\ldots$.

Show that $\mathrm{P}(X_1=2 \text{ and } X_2=2)=6p^4q^2$, where $q=1-p$. Find the possible values of p if it is known that $\mathrm{P}(X_1=2|X_2=2)=9/25$.

Explain briefly why $\mathrm{E}(X_n)=2p\mathrm{E}(X_{n-1})$ and hence determine $\mathrm{E}(X_n)$ in terms of p. Show that for one of the values of p found above $\lim_{n\to\infty}\mathrm{E}(X_n)=0$ and that for the other $\lim_{n\to\infty}\mathrm{E}(X_n)=+\infty$.

13 The random variable X takes the values $k=1,\,2,\,3,\,\ldots$, and has probability distribution

$$\mathrm{P}(X=k)=A\frac{\lambda^k e^{-\lambda}}{k!},$$

where λ is a positive constant. Show that $A=(1-\mathrm{e}^{-\lambda})^{-1}$. Find the mean μ in terms of λ and show that

$$\mathrm{Var}\,(X)=\mu(1-\mu+\lambda)\,.$$

Deduce that $\lambda<\mu<1+\lambda$.

Use a normal approximation to find the value of $P(X=\lambda)$ in the case where $\lambda=100$, giving your answer to 2 decimal places.

14 The probability of throwing a 6 with a biased die is p. It is known that p is equal to one or other of the numbers A and B where $0<A<B<1$. Accordingly the following statistical test of the hypothesis $H_0: p=B$ against the alternative hypothesis $H_1: p=A$ is performed.

The die is thrown repeatedly until a 6 is obtained. Then if X is the total number of throws, H_0 is accepted if $X\leqslant M$, where M is a given positive integer; otherwise H_1 is accepted. Let α be the probability that H_1 is accepted if H_0 is true, and let β be the probability that H_0 is accepted if H_1 is true.

Show that $\beta=1-\alpha^K$, where K is independent of M and is to be determined in terms of A and B. Sketch the graph of β against α.

Section A: Pure Mathematics

1 Find all real values of x that satisfy:

(i) $\sqrt{3x^2+1}+\sqrt{x}-2x-1=0$;

(ii) $\sqrt{3x^2+1}-2\sqrt{x}+x-1=0$;

(iii) $\sqrt{3x^2+1}-2\sqrt{x}-x+1=0$.

2 Prove that, if $|\alpha|<2\sqrt{2}$, then there is no value of x for which

$$x^2-\alpha|x|+2<0\,. \qquad (*)$$

Find the solution set of $(*)$ for $\alpha=3$.

For $\alpha>2\sqrt{2}$, the sum of the lengths of the intervals in which x satisfies $(*)$ is denoted by S. Find S in terms of α and deduce that $S<2\alpha$.

Sketch the graph of S against α .

3 The curve C has equation

$$y=x(x+1)(x-2)^4.$$

Show that the gradient of C is $(x-2)^3(6x^2+x-2)$ and find the coordinates of all the stationary points. Determine the nature of each stationary point and sketch C .

In separate diagrams draw sketches of the curves whose equations are:

(i) $y^2=x(x+1)(x-2)^4$;

(ii) $y=x^2(x^2+1)(x^2-2)^4$.

In each case, you should pay particular attention to the points where the curve meets the x axis.

4

Figure 1

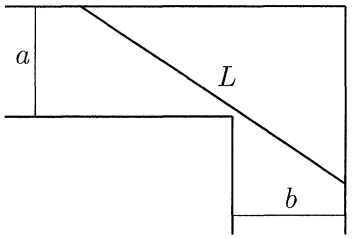
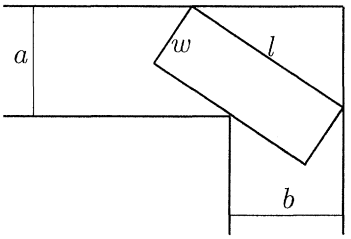


Figure 2



(i) An attempt is made to move a rod of length L from a corridor of width a into a corridor of width b, where $a \neq b$. The corridors meet at right angles, as shown in Figure 1 and the rod remains horizontal. Show that if the attempt is to be successful then

$$L \leqslant a \operatorname{cosec} \alpha + b \sec \alpha \, ,$$

where α satisfies

$$\tan^3 \alpha = \frac{a}{b} \, .$$

(ii) An attempt is made to move a rectangular table-top, of width w and length l, from one corridor to the other, as shown in the Figure 2. The table-top remains horizontal. Show that if the attempt is to be successful then

$$l \leqslant a \operatorname{cosec} \beta + b \sec \beta - 2w \operatorname{cosec} 2\beta,$$

where β satisfies

$$w = \left(\frac{a - b \tan^3 \beta}{1 - \tan^2 \beta} \right) \cos \beta \, .$$

5 Evaluate $\int_0^{\pi} x \sin x \, \mathrm{d}x$ and $\int_0^{\pi} x \cos x \, \mathrm{d}x$.

The function f satisfies the equation

$$\mathrm{f}(t) = t + \int_0^{\pi} \mathrm{f}(x) \sin(x + t) \, \mathrm{d}x \, . \qquad (*)$$

Show that

$$\mathrm{f}(t) = t + A \sin t + B \cos t \, , \qquad (**)$$

where $A = \int_0^{\pi} \mathrm{f}(x) \cos x \, \mathrm{d}x$ and $B = \int_0^{\pi} \mathrm{f}(x) \sin x \, \mathrm{d}x$.

Use the expression $(**)$ to find A and B by substituting for $\mathrm{f}(t)$ and $\mathrm{f}(x)$ in $(*)$ and equating coefficients of $\sin t$ and $\cos t$.

[Turn over

6 The vectors $\mathbf{a}$ and $\mathbf{b}$ lie in the plane Π. Given that $|\mathbf{a}| = 1$ and $\mathbf{a}.\mathbf{b} = 3$, find, in terms of $\mathbf{a}$ and $\mathbf{b}$, a vector $\mathbf{p}$ parallel to $\mathbf{a}$ and a vector $\mathbf{q}$ perpendicular to $\mathbf{a}$, both lying in the plane Π, such that

$$\mathbf{p} + \mathbf{q} = \mathbf{a} + \mathbf{b}\,.$$

The vector $\mathbf{c}$ is not parallel to the plane Π and is such that $\mathbf{a}.\mathbf{c} = -2$ and $\mathbf{b}.\mathbf{c} = 2$. Given that $|\mathbf{b}| = 5$, find, in terms of $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$, vectors $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{R}$ such that $\mathbf{P}$ and $\mathbf{Q}$ are parallel to $\mathbf{p}$ and $\mathbf{q}$, respectively, $\mathbf{R}$ is perpendicular to the plane Π and

$$\mathbf{P} + \mathbf{Q} + \mathbf{R} = \mathbf{a} + \mathbf{b} + \mathbf{c}\,.$$

7 The function f is defined by

$$\mathrm{f}(x) = 2\sin x - x\,.$$

Show graphically that the equation $\mathrm{f}(x) = 0$ has exactly one root in the interval $[\frac{1}{2}\pi, \pi]$. This interval is denoted I_0.

In order to determine the root, a sequence of intervals I_1, I_2, ... is generated in the following way. If the interval $I_n = [a_n, b_n]$, and $c_n = (a_n + b_n)/2$, then

$$I_{n+1} = \begin{cases} [a_n, c_n] & \text{if } \mathrm{f}(a_n)\mathrm{f}(c_n) < 0\,; \\ [c_n, b_n] & \text{if } \mathrm{f}(c_n)\mathrm{f}(b_n) < 0\,. \end{cases}$$

.

By using the approximations $\dfrac{1}{\sqrt{2}} \approx 0.7$ and $\pi \approx \sqrt{10}$, show that $I_2 = [\frac{1}{2}\pi, \frac{5}{8}\pi]$ and find I_3.

8 Let x satisfy the differential equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \left(1 - x^n\right)^{1/n}$$

and the condition $x = 0$ when $t = 0$.

(i) Solve the equation in the case $n = 1$ and sketch the graph of the solution for $t > 0$.

(ii) Prove that $1 - x < (1 - x^2)^{1/2}$ for $0 < x < 1$.

Use this result to sketch the graph of the solution in the case $n = 2$ for $0 < t < \frac{1}{2}\pi$, using the same axes as your previous sketch.

By setting $x = \sin y$, solve the equation in this case.

(iii) Use the result (which you need not prove)

$$(1 - x^2)^{1/2} < (1 - x^3)^{1/3} \quad \text{for} \quad 0 < x < 1\,,$$

to sketch, without solving the equation, the graph of the solution of the equation in the case $n = 3$ using the same axes as your previous sketches. Use your sketch to show that $x = 1$ at a value of t less than $\frac{1}{2}\pi$.

Section B: Mechanics

9 The base of a non-uniform solid hemisphere, of mass M, has radius r. The distance of the centre of gravity, G, of the hemisphere from the base is p and from the centre of the base is $\sqrt{p^2+q^2}$. The hemisphere rests in equilibrium with its curved surface on a horizontal plane.

A particle of mass m, where m is small, is attached to A, the lowest point of the circumference of the base. In the new position of equilibrium, find the angle, α, that the base makes with the horizontal.

The particle is removed and attached to the point B of the base which is at the other end of the diameter through A. In the new position of equilibrium the base makes an angle β with the horizontal. Show that

$$\tan(\alpha-\beta)=\frac{2mMrp}{M^2\left(p^2+q^2\right)-m^2r^2}\,.$$

10 In this question take $g=10ms^{-2}$.

The point A lies on a fixed rough plane inclined at 30° to the horizontal and ℓ is the line of greatest slope through A. A particle P is projected up ℓ from A with initial speed 6ms^{-1}. At time T seconds later, a particle Q is projected from A up ℓ, also with speed 6ms^{-1}. The coefficient of friction between each particle and the plane is $1/(5\sqrt{3})$ and the mass of each particle is 4kg.

(i) Given that $T<1+\sqrt{(3/2)}$, show that the particles collide at a time $(3-\sqrt{6})T+1$ seconds after A is projected.

(ii) In the case $T=1+\sqrt{(2/3)}$, determine the energy lost due to friction from the instant at which P is projected to the time of the collision.

[Turn over

11 The maximum power that can be developed by the engine of train A, of mass m, when travelling at speed v is $Pv^{3/2}$, where P is a constant. The maximum power that can be developed by the engine of train B, of mass $2m$, when travelling at speed v is $2Pv^{3/2}$. For both A and B resistance to motion is equal to kv, where k is a constant.

For $t \leqslant 0$, the engines are crawling along at very low equal speeds. At $t = 0$, both drivers switch on full power and at time t the speeds of A and B are v_A and v_B, respectively.

(i) Show that

$$v_A = \frac{P^2\left(1 - e^{-kt/2m}\right)^2}{k^2}$$

and write down the corresponding result for v_B.

(ii) Find v_A and v_B when $9v_A = 4v_B$. [You may find the substitution $v_A = u^2$ useful.]

(iii) Both engines are switched off when $9v_A = 4v_B$. Show that thereafter $k^2v_B^2 = 4P^2v_A$.

Section C: Probability and Statistics

12 Sketch the graph, for $x \geqslant 0$, of

$$y = kxe^{-ax^2} ,$$

where a and k are positive constants.

The random variable X has probability density function $\mathrm{f}(x)$ given by

$$\mathrm{f}(x) = \begin{cases} kxe^{-ax^2} & \text{for } 0 \leqslant x \leqslant 1 \\ 0 & \text{otherwise.} \end{cases}$$

Show that $k = \dfrac{2a}{1 - e^{-a}}$ and find the mode m in terms of a, distinguishing between the cases $a < \frac{1}{2}$ and $a > \frac{1}{2}$.

Find the median h in terms of a and show that $h > m$ if $a > -\ln\left(2e^{-1/2} - 1\right)$.

Show that $-\ln\left(2e^{-1/2} - 1\right) > \frac{1}{2}$. Show that, if $a > -\ln\left(2e^{-1/2} - 1\right)$, then

$$P(X > m \mid X < h) = \frac{2e^{-1/2} - e^{-a} - 1}{1 - e^{-a}} .$$

13 A bag contains b balls, r of them red and the rest white. In a game the player must remove balls one at a time from the bag (without replacement). She may remove as many balls as she wishes, but if she removes any red ball, she loses and gets no reward at all. If she does not remove a red ball, she is rewarded with £1 for each white ball she has removed.

If she removes n white balls on her first n draws, calculate her expected gain on the next draw and show that it is zero if $n = \dfrac{b - r}{r + 1}$.

Hence, or otherwise, show that she will maximise her expected total reward if she aims to remove n balls, where

$$n = \text{ the integer part of } \frac{b - r}{r + 1} + 1 .$$

With this value of n, show that in the case $r = 1$ and b even, her expected total reward is £$\frac{1}{4}b$, and find her expected total reward in the case $r = 1$ and b odd.

[Turn over

14 Explain why, if A, B and C are three events,

$$\mathrm{P}(\mathrm{A} \cup \mathrm{B} \cup \mathrm{C}) = \mathrm{P}(\mathrm{A}) + \mathrm{P}(\mathrm{B}) + \mathrm{P}(\mathrm{C}) - \mathrm{P}(\mathrm{A} \cap \mathrm{B}) - \mathrm{P}(\mathrm{B} \cap \mathrm{C}) - \mathrm{P}(\mathrm{C} \cap \mathrm{A}) + \mathrm{P}(\mathrm{A} \cap \mathrm{B} \cap \mathrm{C}),$$

where $\mathrm{P}(\mathrm{X})$ denotes the probability of event X.

A cook makes three plum puddings for Christmas. He stirs r silver sixpences thoroughly into the pudding mixture before dividing it into three equal portions. Find an expression for the probability that at least one pudding contains no sixpence. Show that the cook must stir 6 or more sixpences into the mixture if there is to be less than $\frac{1}{3}$ chance that at least one of the puddings contains no sixpence.

Given that the cook stirs 6 sixpences into the mixture and that each pudding contains at least one sixpence, find the probability that there are two sixpences in each pudding.

Section A: Pure Mathematics

1 Find the three values of x for which the derivative of $x^2e^{-x^2}$ is zero.

Given that a and b are distinct positive numbers, find a polynomial $\mathrm{P}(x)$ such that the derivative of $\mathrm{P}(x)e^{-x^2}$ is zero for $x = 0$, $x = \pm a$ and $x = \pm b$, but for no other values of x.

2 For any positive integer N, the function $\mathrm{f}(N)$ is defined by

$$\mathrm{f}(N) = N\Big(1 - \frac{1}{p_1}\Big)\Big(1 - \frac{1}{p_2}\Big)\cdots\Big(1 - \frac{1}{p_k}\Big)$$

where p_1, p_2, ... , p_k are the only prime numbers that are factors of N.
Thus $\mathrm{f}(80) = 80(1 - \frac{1}{2})(1 - \frac{1}{5})$.

(a) (i) Evaluate $\mathrm{f}(12)$ and $\mathrm{f}(180)$.

(ii) Show that $\mathrm{f}(N)$ is an integer for all N.

(b) Prove, or disprove by means of a counterexample, each of the following:

(i) $\mathrm{f}(m)\mathrm{f}(n) = \mathrm{f}(mn)$;

(ii) $\mathrm{f}(p)\mathrm{f}(q) = \mathrm{f}(pq)$ if p and q are distinct prime numbers;

(iii) $\mathrm{f}(p)\mathrm{f}(q) = \mathrm{f}(pq)$ only if p and q are distinct prime numbers.

(c) Find a positive integer m and a prime number p such that $\mathrm{f}(p^m) = 146410$.

3 Give a sketch, for $0 \leqslant x \leqslant \pi/2$, of the curve

$$y = (\sin x - x\cos x)\ ,$$

and show that $0 \leqslant y \leqslant 1$.

Show that:

(i) $\displaystyle\int_0^{\pi/2} y\,\mathrm{d}x = 2 - \frac{\pi}{2}$;

(ii) $\displaystyle\int_0^{\pi/2} y^2\,\mathrm{d}x = \frac{\pi^3}{48} - \frac{\pi}{8}$.

Deduce that $\pi^3 + 18\pi < 96$.

4 The positive numbers a, b and c satisfy $bc = a^2 + 1$. Prove that

$$\arctan\left(\frac{1}{a+b}\right) + \arctan\left(\frac{1}{a+c}\right) = \arctan\left(\frac{1}{a}\right).$$

The positive numbers p, q, r, s, t, u and v satisfy

$$st = (p+q)^2 + 1\,, \qquad uv = (p+r)^2 + 1\,, \qquad qr = p^2 + 1\,.$$

Prove that

$$\arctan\left(\frac{1}{p+q+s}\right) + \arctan\left(\frac{1}{p+q+t}\right) + \arctan\left(\frac{1}{p+r+u}\right) + \arctan\left(\frac{1}{p+r+v}\right) = \arctan\left(\frac{1}{p}\right).$$

Hence show that

$$\arctan\left(\frac{1}{13}\right) + \arctan\left(\frac{1}{21}\right) + \arctan\left(\frac{1}{82}\right) + \arctan\left(\frac{1}{187}\right) = \arctan\left(\frac{1}{7}\right).$$

[Note that $\arctan x$ is another notation for $\tan^{-1} x$.]

5 The angle A of triangle ABC is a right angle and the sides BC, CA and AB are of lengths a, b and c, respectively. Each side of the triangle is tangent to the circle S_1 which is of radius r. Show that $2r = b + c - a$.

Each vertex of the triangle lies on the circle S_2. The ratio of the area of the region between S_1 and the triangle to the area of S_2 is denoted by R. Show that

$$\pi R = -(\pi - 1)q^2 + 2\pi q - (\pi + 1)\,,$$

where $q = \dfrac{b+c}{a}$. Deduce that

$$R \leqslant \frac{1}{\pi(\pi - 1)}\,.$$

6 (i) Write down the general term in the expansion in powers of x of $(1-x)^{-1}$, $(1-x)^{-2}$ and $(1-x)^{-3}$, where $|x| < 1$.

Evaluate $\displaystyle\sum_{n=1}^{\infty} n2^{-n}$ and $\displaystyle\sum_{n=1}^{\infty} n^2 2^{-n}$.

(ii) Show that $(1-x)^{-\frac{1}{2}} = \displaystyle\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2}\,\frac{x^n}{2^{2n}}$, for $|x| < 1$.

Evaluate $\displaystyle\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2 2^{2n} 3^n}$ and $\displaystyle\sum_{n=1}^{\infty} \frac{n(2n)!}{(n!)^2 2^{2n} 3^n}$.

7 The position vectors, relative to an origin O, at time t of the particles P and Q are

$$\cos t\,\mathbf{i} + \sin t\,\mathbf{j} + 0\,\mathbf{k} \qquad \text{and} \qquad \cos(t+\tfrac{1}{4}\pi)\left[\tfrac{3}{2}\mathbf{i} + \tfrac{3\sqrt{3}}{2}\mathbf{k}\right] + 3\sin(t+\tfrac{1}{4}\pi)\,\mathbf{j}\,,$$

respectively, where $0 \leqslant t \leqslant 2\pi$.

(i) Give a geometrical description of the motion of P and Q.

(ii) Let θ be the angle POQ at time t that satisfies $0 \leqslant \theta \leqslant \pi$. Show that

$$\cos\theta = \tfrac{3\sqrt{2}}{8} - \tfrac{1}{4}\cos(2t + \tfrac{1}{4}\pi)\,.$$

(iii) Show that the total time for which $\theta \geqslant \frac{1}{4}\pi$ is $\frac{3}{2}\pi$.

8 For $x \geqslant 0$ the curve C is defined by

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^3y^2}{(1+x^2)^{5/2}}$$

with $y = 1$ when $x = 0$. Show that

$$\frac{1}{y} = \frac{2+3x^2}{3(1+x^2)^{3/2}} + \frac{1}{3}$$

and hence that for large positive x

$$y \approx 3 - \frac{9}{x}\,.$$

Draw a sketch of C.

On a separate diagram draw a sketch of the two curves defined for $x \geqslant 0$ by

$$\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{x^3z^3}{2(1+x^2)^{5/2}}$$

with $z = 1$ at $x = 0$ on one curve, and $z = -1$ at $x = 0$ on the other.

Section B: Mechanics

9 Two particles, A and B, of masses m and $2m$, respectively, are placed on a line of greatest slope, ℓ, of a rough inclined plane which makes an angle of 30° with the horizontal. The coefficient of friction between A and the plane is $\frac{1}{6}\sqrt{3}$ and the coefficient of friction between B and the plane is $\frac{1}{3}\sqrt{3}$. The particles are at rest with B higher up ℓ than A and are connected by a light inextensible string which is taut. A force P is applied to B.

(i) Show that the least magnitude of P for which the two particles move upwards along ℓ is $\frac{11}{8}\sqrt{3}\,mg$ and give, in this case, the direction in which P acts.

(ii) Find the least magnitude of P for which the particles do not slip downwards along ℓ.

10 The points A and B are 180 metres apart and lie on horizontal ground. A missile is launched from A at speed of $100\,\mathrm{m\,s^{-1}}$ and at an acute angle of elevation to the line AB of $\arcsin\frac{3}{5}$. A time T seconds later, an anti-missile missile is launched from B, at speed of $200\,\mathrm{m\,s^{-1}}$ and at an acute angle of elevation to the line BA of $\arcsin\frac{4}{5}$. The motion of both missiles takes place in the vertical plane containing A and B, and the missiles collide.

Taking $g = 10\,\mathrm{m\,s^{-2}}$ and ignoring air resistance, find T.

[Note that $\arcsin\frac{3}{5}$ is another notation for $\sin^{-1}\frac{3}{5}$.]

11 A plane is inclined at an angle $\arctan\frac{3}{4}$ to the horizontal and a small, smooth, light pulley P is fixed to the top of the plane. A string, APB, passes over the pulley. A particle of mass m_1 is attached to the string at A and rests on the inclined plane with AP parallel to a line of greatest slope in the plane. A particle of mass m_2, where $m_2 > m_1$, is attached to the string at B and hangs freely with BP vertical. The coefficient of friction between the particle at A and the plane is $\frac{1}{2}$.

The system is released from rest with the string taut. Show that the acceleration of the particles is $\dfrac{m_2 - m_1}{m_2 + m_1}g$.

At a time T after release, the string breaks. Given that the particle at A does not reach the pulley at any point in its motion, find an expression in terms of T for the time after release at which the particle at A reaches its maximum height. It is found that, regardless of when the string broke, this time is equal to the time taken by the particle at A to descend from its point of maximum height to the point at which it was released. Find the ratio $m_1 : m_2$.

[Note that $\arctan\frac{3}{4}$ is another notation for $\tan^{-1}\frac{3}{4}$.]

Section C: Probability and Statistics

12 The twins Anna and Bella share a computer and never sign their e-mails. When I e-mail them, only the twin currently online responds. The probability that it is Anna who is online is p and she answers each question I ask her truthfully with probability a, independently of all her other answers, even if a question is repeated. The probability that it is Bella who is online is q, where $q = 1 - p$, and she answers each question truthfully with probability b, independently of all her other answers, even if a question is repeated.

(i) I send the twins the e-mail: 'Toss a fair coin and answer the following question. Did the coin come down heads?'. I receive the answer 'yes'. Show that the probability that the coin did come down heads is $\frac{1}{2}$ if and only if $2(ap + bq) = 1$.

(ii) I send the twins the e-mail: 'Toss a fair coin and answer the following question. Did the coin come down heads?'. I receive the answer 'yes'. I then send the e-mail: 'Did the coin come down heads?' and I receive the answer 'no'. Show that the probability (taking into account these answers) that the coin did come down heads is $\frac{1}{2}$.

(iii) I send the twins the e-mail: 'Toss a fair coin and answer the following question. Did the coin come down heads?'. I receive the answer 'yes'. I then send the e-mail: 'Did the coin come down heads?' and I receive the answer 'yes'. Show that, if $2(ap + bq) = 1$, the probability (taking into account these answers) that the coin did come down heads is $\frac{1}{2}$.

13 The number of printing errors on any page of a large book of N pages is modelled by a Poisson variate with parameter λ and is statistically independent of the number of printing errors on any other page. The number of pages in a random sample of n pages (where n is much smaller than N and $n \geqslant 2$) which contain fewer than two errors is denoted by Y. Show that $\mathrm{P}(Y = k) = \binom{n}{k} p^k q^{n-k}$ where $p = (1 + \lambda)e^{-\lambda}$ and $q = 1 - p$.

Show also that, if λ is sufficiently small,

(i) $q \approx \frac{1}{2}\lambda^2$;

(ii) the largest value of n for which $\mathrm{P}(Y = n) \geqslant 1 - \lambda$ is approximately $2/\lambda$;

(iii) $\mathrm{P}(Y > 1 \mid Y > 0) \approx 1 - n(\lambda^2/2)^{n-1}$.

14 The probability density function $\mathrm{f}(x)$ of the random variable X is given by

$$\mathrm{f}(x) = k\left[\phi(x) + \lambda \mathrm{g}(x)\right],$$

where $\phi(x)$ is the probability density function of a normal variate with mean 0 and variance 1, λ is a positive constant, and $\mathrm{g}(x)$ is a probability density function defined by

$$\mathrm{g}(x) = \begin{cases} 1/\lambda & \text{for } 0 \leqslant x \leqslant \lambda\,; \\ 0 & \text{otherwise.} \end{cases}$$

Find μ, the mean of X, in terms of λ, and prove that σ, the standard deviation of X, satisfies.

$$\sigma^2 = \frac{\lambda^4 + 4\lambda^3 + 12\lambda + 12}{12(1+\lambda)^2}\,.$$

In the case $\lambda = 2$:

(i) draw a sketch of the curve $y = \mathrm{f}(x)$;

(ii) express the cumulative distribution function of X in terms of $\Phi(x)$, the cumulative distribution function corresponding to $\phi(x)$;

(iii) evaluate $\mathrm{P}(0 < X < \mu + 2\sigma)$, given that $\Phi(\frac{2}{3} + \frac{2}{3}\sqrt{7}) = 0.9921$.

Section A: Pure Mathematics

1 The sequence of real numbers u_1, u_2, u_3, ... is defined by

$$u_1 = 2\,, \qquad \text{and} \qquad u_{n+1} = k - \frac{36}{u_n} \quad \text{for } n \geqslant 1, \tag{$*$}$$

where k is a constant.

(i) Determine the values of k for which the sequence $(*)$ is:

(a) constant;

(b) periodic with period 2;

(c) periodic with period 4.

(ii) In the case $k = 37$, show that $u_n \geqslant 2$ for all n. Given that in this case the sequence $(*)$ converges to a limit ℓ, find the value of ℓ.

2 Using the series

$$\mathrm{e}^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots\,,$$

show that $\mathrm{e} > \frac{8}{3}$.

Show that $n! > 2^n$ for $n \geqslant 4$ and hence show that $\mathrm{e} < \frac{67}{24}$.

Show that the curve with equation

$$y = 3\mathrm{e}^{2x} + 14\ln(\tfrac{4}{3} - x)\,, \qquad x < \tfrac{4}{3}$$

has a minimum turning point between $x = \frac{1}{2}$ and $x = 1$ and give a sketch to show the shape of the curve.

3 (i) Show that $\left(5+\sqrt{24}\,\right)^4 + \dfrac{1}{\left(5+\sqrt{24}\,\right)^4}$ is an integer.

Show also that

$$0.1 < \frac{1}{5+\sqrt{24}} < \frac{2}{19} < 0.11\,.$$

Hence determine, with clear reasoning, the value of $\left(5+\sqrt{24}\right)^4$ correct to four decimal places.

(ii) If N is an integer greater than 1, show that $\left(N+\sqrt{N^2-1}\right)^k$, where k is a positive integer, differs from the integer nearest to it by less than $\left(2N-\frac{1}{2}\right)^{-k}$.

4 By making the substitution $x = \pi - t$, show that

$$\int_0^\pi x\mathrm{f}(\sin x)\mathrm{d}x = \tfrac{1}{2}\pi\int_0^\pi \mathrm{f}(\sin x)\mathrm{d}x\,,$$

where $\mathrm{f}(\sin x)$ is a given function of $\sin x$.

Evaluate the following integrals:

(i) $\displaystyle\int_0^\pi \frac{x\sin x}{3+\sin^2 x}\,\mathrm{d}x\,;$

(ii) $\displaystyle\int_0^{2\pi} \frac{x\sin x}{3+\sin^2 x}\,\mathrm{d}x\,;$

(iii) $\displaystyle\int_0^\pi \frac{x\left|\sin 2x\right|}{3+\sin^2 x}\,\mathrm{d}x\,.$

5 The notation $\lfloor x\rfloor$ denotes the greatest integer less than or equal to the real number x. Thus, for example, $\lfloor\pi\rfloor = 3$, $\lfloor 18\rfloor = 18$ and $\lfloor -4.2\rfloor = -5$.

(i) Two curves are given by $y = x^2 + 3x - 1$ and $y = x^2 + 3\lfloor x\rfloor - 1$. Sketch the curves, for $1 \leqslant x \leqslant 3$, on the same axes.

Find the area between the two curves for $1 \leqslant x \leqslant n$, where n is a positive integer.

(ii) Two curves are given by $y = x^2 + 3x - 1$ and $y = \lfloor x\rfloor^2 + 3\lfloor x\rfloor - 1$. Sketch the curves, for $1 \leqslant x \leqslant 3$, on the same axes.

Show that the area between the two curves for $1 \leqslant x \leqslant n$, where n is a positive integer, is

$$\tfrac{1}{6}(n-1)(3n+11)\,.$$

6 By considering a suitable scalar product, prove that

$$(ax + by + cz)^2 \leqslant (a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$$

for any real numbers a, b, c, x, y and z. Deduce a necessary and sufficient condition on a, b, c, x, y and z for the following equation to hold:

$$(ax + by + cz)^2 = (a^2 + b^2 + c^2)(x^2 + y^2 + z^2)\,.$$

(i) Show that $(x + 2y + 2z)^2 \leqslant 9(x^2 + y^2 + z^2)$ for all real numbers x, y and z.

(ii) Find real numbers p, q and r that satisfy both

$$p^2 + 4q^2 + 9r^2 = 729 \quad \text{and} \quad 8p + 8q + 3r = 243\,.$$

7 An ellipse has equation $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$. Show that the equation of the tangent at the point $(a\cos\alpha, b\sin\alpha)$ is

$$y = -\frac{b\cot\alpha}{a}\,x + b\,\mathrm{cosec}\,\alpha\,.$$

The point A has coordinates $(-a, -b)$, where a and b are positive. The point E has coordinates $(-a, 0)$ and the point P has coordinates (a, kb), where $0 < k < 1$. The line through E parallel to AP meets the line $y = b$ at the point Q. Show that the line PQ is tangent to the above ellipse at the point given by $\tan(\alpha/2) = k$.

Determine by means of sketches, or otherwise, whether this result holds also for $k = 0$ and $k = 1$.

8 Show that the line through the points with position vectors $\mathbf{x}$ and $\mathbf{y}$ has equation

$$\mathbf{r} = (1 - \alpha)\mathbf{x} + \alpha\mathbf{y}\,,$$

where α is a scalar parameter.

The sides OA and CB of a trapezium $OABC$ are parallel, and $OA > CB$. The point E on OA is such that $OE : EA = 1 : 2$, and F is the midpoint of CB. The point D is the intersection of OC produced and AB produced; the point G is the intersection of OB and EF; and the point H is the intersection of DG produced and OA. Let $\mathbf{a}$ and $\mathbf{c}$ be the position vectors of the points A and C, respectively, with respect to the origin O.

(i) Show that B has position vector $\lambda\mathbf{a} + \mathbf{c}$ for some scalar parameter λ.

(ii) Find, in terms of $\mathbf{a}$, $\mathbf{c}$ and λ only, the position vectors of D, E, F, G and H. Determine the ratio $OH : HA$.

Section B: Mechanics

9 A painter of weight kW uses a ladder to reach the guttering on the outside wall of a house. The wall is vertical and the ground is horizontal. The ladder is modelled as a uniform rod of weight W and length $6a$.

The ladder is not long enough, so the painter stands the ladder on a uniform table. The table has weight $2W$ and a square top of side $\frac{1}{2}a$ with a leg of length a at each corner. The foot of the ladder is at the centre of the table top and the ladder is inclined at an angle $\arctan 2$ to the horizontal. The edge of the table nearest the wall is parallel to the wall.

The coefficient of friction between the foot of the ladder and the table top is $\frac{1}{2}$. The contact between the ladder and the wall is sufficiently smooth for the effects of friction to be ignored.

(i) Show that, if the legs of the table are fixed to the ground, the ladder does not slip on the table however high the painter stands on the ladder.

(ii) It is given that $k = 9$ and that the coefficient of friction between each table leg and the ground is $\frac{1}{3}$. If the legs of the table are not fixed to the ground, so that the table can tilt or slip, determine which occurs first when the painter slowly climbs the ladder.

[Note: $\arctan 2$ is another notation for $\tan^{-1} 2$.]

10 Three particles, A, B and C, of masses m, km and $3m$ respectively, are initially at rest lying in a straight line on a smooth horizontal surface. Then A is projected towards B at speed u. After the collision, B collides with C. The coefficient of restitution between A and B is $\frac{1}{2}$ and the coefficient of restitution between B and C is $\frac{1}{4}$.

(i) Find the range of values of k for which A and B collide for a second time.

(ii) Given that $k = 1$ and that B and C are initially a distance d apart, show that the time that elapses between the two collisions of A and B is $\dfrac{60d}{13u}$.

11 A projectile of unit mass is fired in a northerly direction from a point on a horizontal plain at speed u and an angle θ above the horizontal. It lands at a point A on the plain. In flight, the projectile experiences two forces: gravity, of magnitude g; and a horizontal force of constant magnitude f due to a wind blowing from North to South. Derive an expression, in terms of u, g, f and θ for the distance OA.

(i) Determine the angle α such that, for all $\theta > \alpha$, the wind starts to blow the projectile back towards O before it lands at A.

(ii) An identical projectile, which experiences the same forces, is fired from O in a northerly direction at speed u and angle 45° above the horizontal and lands at a point B on the plain. Given that θ is chosen to maximise OA, show that

$$\frac{OB}{OA} = \frac{g - f}{\sqrt{g^2 + f^2} - f} \ .$$

Describe carefully the motion of the second projectile when $f = g$.

Section C: Probability and Statistics

12 A cricket team has only three bowlers, Arthur, Betty and Cuba, each of whom bowls 30 balls in any match. Past performance reveals that, on average, Arthur takes one wicket for every 36 balls bowled, Betty takes one wicket for every 25 balls bowled, and Cuba takes one wicket for every 41 balls bowled.

(i) In one match, the team took exactly one wicket, but the name of the bowler was not recorded. Using a binomial model, find the probability that Arthur was the bowler.

(ii) Show that the average number of wickets taken by the team in a match is approximately 3. Give with brief justification a suitable model for the number of wickets taken by the team in a match and show that the probability of the team taking at least five wickets in a given match is approximately $\frac{1}{5}$.

[You may use the approximation $e^3 = 20$.]

13 I know that ice-creams come in n different sizes, but I don't know what the sizes are. I am offered one of each in succession, in random order. I am certainly going to choose one — the bigger the better — but I am not allowed more than one. My strategy is to reject the first ice-cream I am offered and choose the first one thereafter that is bigger than the first one I was offered; if the first ice-cream offered is in fact the biggest one, then I have to put up with the last one, however small.

Let $P_n(k)$ be the probability that I choose the kth biggest ice-cream, where $k = 1$ is the biggest and $k = n$ is the smallest.

(i) Show that $P_4(1) = \frac{11}{24}$ and find $P_4(2)$, $P_4(3)$ and $P_4(4)$.

(ii) Find an expression for $P_n(1)$.

14 Sketch the graph of $y = \dfrac{1}{x \ln x}$ for $x > 0$, $x \neq 1$. You may assume that $x \ln x \to 0$ as $x \to 0$.

The continuous random variable X has probability density function

$$\mathrm{f}(x) = \begin{cases} \dfrac{\lambda}{x \ln x} & \text{for } a \leqslant x \leqslant b\,, \\ 0 & \text{otherwise}\,, \end{cases}$$

where a, b and λ are suitably chosen constants.

(i) In the case $a = 1/4$ and $b = 1/2$, find λ.

(ii) In the case $\lambda = 1$ and $a > 1$, show that $b = a^{\mathrm{e}}$.

(iii) In the case $\lambda = 1$ and $a = \mathrm{e}$, show that $\mathrm{P}(\mathrm{e}^{3/2} \leqslant X \leqslant \mathrm{e}^2) \approx \frac{31}{108}$.

(iv) In the case $\lambda = 1$ and $a = \mathrm{e}^{1/2}$, find $\mathrm{P}(\mathrm{e}^{3/2} \leqslant X \leqslant \mathrm{e}^2)$.

Section A: Pure Mathematics

1 *In this question, you are not required to justify the accuracy of the approximations.*

(i) Write down the binomial expansion of $\left(1+\frac{k}{100}\right)^{\frac{1}{2}}$ in ascending powers of k, up to and including the k^3 term.

(a) Use the value $k=8$ to find an approximation to five decimal places for $\sqrt{3}$.

(b) By choosing a suitable integer value of k, find an approximation to five decimal places for $\sqrt{6}$.

(ii) By considering the first two terms of the binomial expansion of $\left(1+\frac{k}{1000}\right)^{\frac{1}{3}}$, show that $\frac{3029}{2100}$ is an approximation to $\sqrt[3]{3}$.

2 A curve has equation $y=2x^3-bx^2+cx$. It has a maximum point at (p,m) and a minimum point at (q,n) where $p>0$ and $n>0$. Let R be the region enclosed by the curve, the line $x=p$ and the line $y=n$.

(i) Express b and c in terms of p and q.

(ii) Sketch the curve. Mark on your sketch the point of inflection and shade the region R. Describe the symmetry of the curve.

(iii) Show that $m-n=(q-p)^3$.

(iv) Show that the area of R is $\frac{1}{2}(q-p)^4$.

3 By writing $x=a\tan\theta$, show that, for $a\neq 0$, $\displaystyle\int\frac{1}{a^2+x^2}\,\mathrm{d}x=\frac{1}{a}\arctan\frac{x}{a}+\text{constant}$.

(i) Let $\displaystyle I=\int_0^{\frac{\pi}{2}}\frac{\cos x}{1+\sin^2 x}\,\mathrm{d}x$.

(a) Evaluate I.

(b) Use the substitution $t=\tan\frac{1}{2}x$ to show that $\displaystyle\int_0^1\frac{1-t^2}{1+6t^2+t^4}\,\mathrm{d}t=\frac{1}{2}I$.

(ii) Evaluate $\displaystyle\int_0^1\frac{1-t^2}{1+14t^2+t^4}\,\mathrm{d}t$.

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4 Given that $\cos A$, $\cos B$ and β are non-zero, show that the equation

$$\alpha \sin(A - B) + \beta \cos(A + B) = \gamma \sin(A + B)$$

reduces to the form

$$(\tan A - m)(\tan B - n) = 0\,,$$

where m and n are independent of A and B, if and only if $\alpha^2 = \beta^2 + \gamma^2$.

Determine all values of x, in the range $0 \leqslant x < 2\pi$, for which:

(i) $2\sin(x - \frac{1}{4}\pi) + \sqrt{3}\cos(x + \frac{1}{4}\pi) = \sin(x + \frac{1}{4}\pi)$;

(ii) $2\sin(x - \frac{1}{6}\pi) + \sqrt{3}\cos(x + \frac{1}{6}\pi) = \sin(x + \frac{1}{6}\pi)$;

(iii) $2\sin(x + \frac{1}{3}\pi) + \sqrt{3}\cos(3x) = \sin(3x)$.

5 In this question, $\mathrm{f}^2(x)$ denotes $\mathrm{f}(\mathrm{f}(x))$, $\mathrm{f}^3(x)$ denotes $\mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$, and so on.

(i) The function f is defined, for $x \neq \pm 1/\sqrt{3}$, by

$$\mathrm{f}(x) = \frac{x + \sqrt{3}}{1 - \sqrt{3}\,x}\,.$$

Find by direct calculation $\mathrm{f}^2(x)$ and $\mathrm{f}^3(x)$, and determine $\mathrm{f}^{2007}(x)$.

(ii) Show that $\mathrm{f}^n(x) = \tan(\theta + \frac{1}{3}n\pi)$, where $x = \tan\theta$ and n is any positive integer.

(iii) The function $\mathrm{g}(t)$ is defined, for $|t| \leqslant 1$ by $\mathrm{g}(t) = \frac{\sqrt{3}}{2}t + \frac{1}{2}\sqrt{1 - t^2}$. Find an expression for $\mathrm{g}^n(t)$ for any positive integer n.

6 **(i)** Differentiate $\ln\left(x + \sqrt{3 + x^2}\right)$ and $x\sqrt{3 + x^2}$ and simplify your answers.
Hence find $\int \sqrt{3 + x^2}\,\mathrm{d}x$.

(ii) Find the two solutions of the differential equation

$$3\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 2x\frac{\mathrm{d}y}{\mathrm{d}x} = 1$$

that satisfy $y = 0$ when $x = 1$.

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[Turn over

7 A function $\mathrm{f}(x)$ is said to be concave on some interval if $\mathrm{f}''(x) < 0$ in that interval. Show that $\sin x$ is concave for $0 < x < \pi$ and that $\ln x$ is concave for $x > 0$.

Let $\mathrm{f}(x)$ be concave on a given interval and let x_1, x_2, ..., x_n lie in the interval. *Jensen's inequality* states that

$$\frac{1}{n}\sum_{k=1}^{n}\mathrm{f}(x_k) \leqslant \mathrm{f}\left(\frac{1}{n}\sum_{k=1}^{n}x_k\right)$$

and that equality holds if and only if $x_1 = x_2 = \cdots = x_n$. You may use this result without proving it.

(i) Given that A, B and C are angles of a triangle, show that

$$\sin A + \sin B + \sin C \leqslant \frac{3\sqrt{3}}{2}.$$

(ii) By choosing a suitable function f, prove that

$$\sqrt[n]{t_1 t_2 \cdots t_n} \leqslant \frac{t_1 + t_2 + \cdots + t_n}{n}$$

for any positive integer n and for any positive numbers t_1, t_2, ..., t_n.

Hence:

(a) show that $x^4 + y^4 + z^4 + 16 \geqslant 8xyz$, where x, y and z are any positive numbers;

(b) find the minimum value of $x^5 + y^5 + z^5 - 5xyz$, where x, y and z are any positive numbers.

8 The points B and C have position vectors $\mathbf{b}$ and $\mathbf{c}$, respectively, relative to the origin A, and A, B and C are not collinear.

(i) The point X has position vector $s\mathbf{b} + t\mathbf{c}$. Describe the locus of X when $s + t = 1$.

(ii) The point P has position vector $\beta\mathbf{b} + \gamma\mathbf{c}$, where β and γ are non-zero, and $\beta + \gamma \neq 1$. The line AP cuts the line BC at D. Show that $BD : DC = \gamma : \beta$.

(iii) The line BP cuts the line CA at E, and the line CP cuts the line AB at F. Show that

$$\frac{AF}{FB} \times \frac{BD}{DC} \times \frac{CE}{EA} = 1.$$

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Section B: Mechanics

9 A solid right circular cone, of mass M, has semi-vertical angle α and smooth surfaces. It stands with its base on a smooth horizontal table. A particle of mass m is projected so that it strikes the curved surface of the cone at speed u. The coefficient of restitution between the particle and the cone is e. The impact has no rotational effect on the cone and the cone has no vertical velocity after the impact.

(i) The particle strikes the cone in the direction of the normal at the point of impact. Explain why the trajectory of the particle immediately after the impact is parallel to the normal to the surface of the cone. Find an expression, in terms of M, m, α, e and u, for the speed at which the cone slides along the table immediately after impact.

(ii) If instead the particle falls vertically onto the cone, show that the speed w at which the cone slides along the table immediately after impact is given by

$$w = \frac{mu(1+e)\sin\alpha\cos\alpha}{M + m\cos^2\alpha}.$$

Show also that the value of α for which w is greatest is given by

$$\cos\alpha = \sqrt{\frac{M}{2M+m}}.$$

10 A solid figure is composed of a uniform solid cylinder of density ρ and a uniform solid hemisphere of density 3ρ. The cylinder has circular cross-section, with radius r, and height $3r$, and the hemisphere has radius r. The flat face of the hemisphere is joined to one end of the cylinder, so that their centres coincide.

The figure is held in equilibrium by a force P so that one point of its flat base is in contact with a rough horizontal plane and its base is inclined at an angle α to the horizontal. The force P is horizontal and acts through the highest point of the base. The coefficient of friction between the solid and the plane is μ. Show that

$$\mu \geqslant \left|\tfrac{9}{8} - \tfrac{1}{2}\cot\alpha\right|.$$

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[Turn over

11 *In this question take the acceleration due to gravity to be* $10\,\mathrm{m\,s^{-2}}$ *and neglect air resistance.*

The point O lies in a horizontal field. The point B lies 50 m east of O. A particle is projected from B at speed $25\,\mathrm{m\,s^{-1}}$ at an angle $\arctan\frac{1}{2}$ above the horizontal and in a direction that makes an angle $60°$ with OB; it passes to the north of O.

(i) Taking unit vectors **i**, **j** and **k** in the directions east, north and vertically upwards, respectively, find the position vector of the particle relative to O at time t seconds after the particle was projected, and show that its distance from O is

$$5(t^2 - \sqrt{5}t + 10)\,\mathrm{m}.$$

When this distance is shortest, the particle is at point P. Find the position vector of P and its horizontal bearing from O.

(ii) Show that the particle reaches its maximum height at P.

(iii) When the particle is at P, a marksman fires a bullet from O directly at P. The initial speed of the bullet is $350\,\mathrm{m\,s^{-1}}$. Ignoring the effect of gravity on the bullet show that, when it passes through P, the distance between P and the particle is approximately 3 m.

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Section C: Probability and Statistics

12 I have two identical dice. When I throw either one of them, the probability of it showing a 6 is p and the probability of it not showing a 6 is q, where $p + q = 1$. As an experiment to determine p, I throw the dice simultaneously until at least one die shows a 6. If both dice show a six on this throw, I stop. If just one die shows a six, I throw the other die until it shows a 6 and then stop.

(i) Show that the probability that I stop after r throws is $pq^{r-1}(2 - q^{r-1} - q^r)$, and find an expression for the expected number of throws.

[**Note:** You may use the result $\sum_{r=0}^{\infty} rx^r = x(1-x)^{-2}$.]

(ii) In a large number of such experiments, the mean number of throws was m. Find an estimate for p in terms of m.

13 Given that $0 < r < n$ and r is much smaller than n, show that $\dfrac{n-r}{n} \approx \mathrm{e}^{-r/n}$.

There are k guests at a party. Assuming that there are exactly 365 days in the year, and that the birthday of any guest is equally likely to fall on any of these days, show that the probability that there are at least two guests with the same birthday is approximately $1 - \mathrm{e}^{-k(k-1)/730}$.

Using the approximation $\frac{253}{365} \approx \ln 2$, find the smallest value of k such that the probability that at least two guests share the same birthday is at least $\frac{1}{2}$.

How many guests must there be at the party for the probability that at least one guest has the same birthday as the host to be at least $\frac{1}{2}$?

14 The random variable X has a continuous probability density function $\mathrm{f}(x)$ given by

$$\mathrm{f}(x) = \begin{cases} 0 & \text{for } x \leqslant 1 \\ \ln x & \text{for } 1 \leqslant x \leqslant k \\ \ln k & \text{for } k \leqslant x \leqslant 2k \\ a - bx & \text{for } 2k \leqslant x \leqslant 4k \\ 0 & \text{for } x \geqslant 4k \end{cases}$$

where k, a and b are constants.

(i) Sketch the graph of $y = \mathrm{f}(x)$.

(ii) Determine a and b in terms of k and find the numerical values of k, a and b.

(iii) Find the median value of X.

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Section A: Pure Mathematics

1 A sequence of points (x_1, y_1), (x_2, y_2), ... in the cartesian plane is generated by first choosing (x_1, y_1) then applying the rule, for $n = 1, 2, \ldots$,

$$(x_{n+1}, y_{n+1}) = (x_n^2 - y_n^2 + a,\ 2x_n y_n + b + 2)\,,$$

where a and b are given real constants.

(i) In the case $a = 1$ and $b = -1$, find the values of (x_1, y_1) for which the sequence is constant.

(ii) Given that $(x_1, y_1) = (-1, 1)$, find the values of a and b for which the sequence has period 2.

2 Let a_n be the coefficient of x^n in the series expansion, in ascending powers of x, of

$$\frac{1+x}{(1-x)^2(1+x^2)}\,,$$

where $|x| < 1$. Show, using partial fractions, that either $a_n = n + 1$ or $a_n = n + 2$ according to the value of n.

Hence find a decimal approximation, to nine significant figures, for the fraction $\dfrac{11\,000}{8181}$. [You are not required to justify the accuracy of your approximation.]

3 **(i)** Find the coordinates of the turning points of the curve $y = 27x^3 - 27x^2 + 4$. Sketch the curve and deduce that $x^2(1-x) \leqslant 4/27$ for all $x \geqslant 0$.

Given that each of the numbers a, b and c lies between 0 and 1, prove by contradiction that at least one of the numbers $bc(1-a)$, $ca(1-b)$ and $ab(1-c)$ is less than or equal to $4/27$.

(ii) Given that each of the numbers p and q lies between 0 and 1, prove that at least one of the numbers $p(1-q)$ and $q(1-p)$ is less than or equal to $1/4$.

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4 A curve is given by

$$x^2 + y^2 + 2axy = 1,$$

where a is a constant satisfying $0 < a < 1$. Show that the gradient of the curve at the point P with coordinates (x, y) is

$$-\frac{x + ay}{ax + y},$$

provided $ax + y \neq 0$. Show that θ, the acute angle between OP and the normal to the curve at P, satisfies

$$\tan\theta = a|y^2 - x^2|\,.$$

Show further that, if $\dfrac{\mathrm{d}\theta}{\mathrm{d}x} = 0$ at P, then:

(i) $a(x^2 + y^2) + 2xy = 0\,;$

(ii) $(1 + a)(x^2 + y^2 + 2xy) = 1\,;$

(iii) $\tan\theta = \dfrac{a}{\sqrt{1 - a^2}}\,.$

5 Evaluate the integrals

$$\int_0^{\pi/2} \frac{\sin 2x}{1 + \sin^2 x}\mathrm{d}x \quad \text{and} \quad \int_0^{\pi/2} \frac{\sin x}{1 + \sin^2 x}\mathrm{d}x\,.$$

Show, using the binomial expansion, that $(1 + \sqrt{2}\,)^5 < 99$. Show also that $\sqrt{2} > 1.4$. Deduce that $2^{\sqrt{2}} > 1 + \sqrt{2}\,$. Use this result to determine which of the above integrals is greater.

6 A curve has the equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = \cos\left(2x + \frac{\pi}{3}\right) + \sin\left(\frac{3x}{2} - \frac{\pi}{4}\right).$$

(i) Find the period of $\mathrm{f}(x)$.

(ii) Determine all values of x in the interval $-\pi \leqslant x \leqslant \pi$ for which $\mathrm{f}(x) = 0$. Find a value of x in this interval at which the curve touches the x-axis without crossing it.

(iii) Find the value or values of x in the interval $0 \leqslant x \leqslant 2\pi$ for which $\mathrm{f}(x) = 2\,$.

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[Turn over

7 **(i)** By writing $y = u(1 + x^2)^{\frac{1}{2}}$, where u is a function of x, find the solution of the equation

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = xy + \frac{x}{1 + x^2}$$

for which $y = 1$ when $x = 0$.

(ii) Find the solution of the equation

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = x^2y + \frac{x^2}{1 + x^3}$$

for which $y = 1$ when $x = 0$.

(iii) Give, without proof, a conjecture for the solution of the equation

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = x^{n-1}y + \frac{x^{n-1}}{1 + x^n}$$

for which $y = 1$ when $x = 0$, where n is an integer greater than 1.

8 The points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$, respectively, relative to the origin O. The points A, B and O are not collinear. The point P lies on AB between A and B such that

$$AP : PB = (1 - \lambda) : \lambda\,.$$

Write down the position vector of P in terms of $\mathbf{a}$, $\mathbf{b}$ and λ. Given that OP bisects $\angle AOB$, determine λ in terms of a and b, where $a = |\mathbf{a}|$ and $b = |\mathbf{b}|$.

The point Q also lies on AB between A and B, and is such that $AP = BQ$. Prove that

$$OQ^2 - OP^2 = (b - a)^2\,.$$

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Section B: Mechanics

9 In this question, use $g = 10\,\mathrm{m\,s^{-2}}$.

In cricket, a fast bowler projects a ball at $40\,\mathrm{m\,s^{-1}}$ from a point $h\,\mathrm{m}$ above the ground, which is horizontal, and at an angle α above the horizontal. The trajectory is such that the ball will strike the stumps at ground level a horizontal distance of 20 m from the point of projection.

(i) Determine, in terms of h, the two possible values of $\tan\alpha$.

Explain which of these two values is the more appropriate one, and deduce that the ball hits the stumps after approximately half a second.

(ii) State the range of values of h for which the bowler projects the ball below the horizontal.

(iii) In the case $h = 2.5$, give an approximate value in degrees, correct to two significant figures, for α. You need not justify the accuracy of your approximation.

[You may use the small-angle approximations $\cos\theta \approx 1$ and $\sin\theta \approx \theta$.]

10 The lengths of the sides of a rectangular billiards table $ABCD$ are given by $AB = DC = a$ and $AD = BC = 2b$. There are small pockets at the midpoints M and N of the sides AD and BC, respectively. The sides of the table may be taken as smooth vertical walls.

A small ball is projected along the table from the corner A. It strikes the side BC at X, then the side DC at Y and then goes directly into the pocket at M. The angles BAX, CXY and DYM are α, β and γ respectively. On each stage of its path, the ball moves with constant speed in a straight line, the speeds being u, v and w respectively. The coefficient of restitution between the ball and the sides is e, where $e > 0$.

(i) Show that $\tan\alpha\tan\beta = e$ and find γ in terms of α.

(ii) Show that $\tan\alpha = \dfrac{(1+2e)b}{(1+e)a}$ and deduce that the shot is possible whatever the value of e.

(iii) Find an expression in terms of e for the fraction of the kinetic energy of the ball that is lost during the motion.

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[Turn over

11 A wedge of mass km has the shape (in cross-section) of a right-angled triangle. It stands on a smooth horizontal surface with one face vertical. The inclined face makes an angle θ with the horizontal surface. A particle P, of mass m, is placed on the inclined face and released from rest. The horizontal face of the wedge is smooth, but the inclined face is rough and the coefficient of friction between P and this face is μ.

(i) When P is released, it slides down the inclined plane at an acceleration a relative to the wedge. Show that the acceleration of the wedge is

$$\frac{a\cos\theta}{k+1}.$$

To a stationary observer, P appears to descend along a straight line inclined at an angle $45°$ to the horizontal. Show that

$$\tan\theta = \frac{k}{k+1}.$$

In the case $k = 3$, find an expression for a in terms of g and μ.

(ii) What happens when P is released if $\tan\theta \leqslant \mu$?

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Section C: Probability and Statistics

12 In the High Court of Farnia, the outcome of each case is determined by three judges: the ass, the beaver and the centaur. Each judge decides its verdict independently. Being simple creatures, they make their decisions entirely at random. Past verdicts show that the ass gives a guilty verdict with probability p, the beaver gives a guilty verdict with probability $p/3$ and the centaur gives a guilty verdict with probability p^2.

Let X be the number of guilty verdicts given by the three judges in a case. Given that $\mathrm{E}(X) = 4/3$, find the value of p.

The probability that a defendant brought to trial is guilty is t. The King pronounces that the defendant is guilty if at least two of the judges give a guilty verdict; otherwise, he pronounces the defendant not guilty. Find the value of t such that the probability that the King pronounces correctly is $1/2$.

13 Bag P and bag Q each contain n counters, where $n \geqslant 2$. The counters are identical in shape and size, but coloured either black or white. First, k counters ($0 \leqslant k \leqslant n$) are drawn at random from bag P and placed in bag Q. Then, k counters are drawn at random from bag Q and placed in bag P.

(i) If initially $n - 1$ counters in bag P are white and one is black, and all n counters in bag Q are white, find the probability in terms of n and k that the black counter ends up in bag P.

Find the value or values of k for which this probability is maximised.

(ii) If initially $n - 1$ counters in bag P are white and one is black, and $n - 1$ counters in bag Q are white and one is black, find the probability in terms of n and k that the black counters end up in the same bag.

Find the value or values of k for which this probability is maximised.

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Section A: Pure Mathematics

1 Two curves have equations $x^4 + y^4 = u$ and $xy = v$, where u and v are positive constants. State the equations of the lines of symmetry of each curve.

The curves intersect at the distinct points A, B, C and D (taken anticlockwise from A). The coordinates of A are (α, β), where $\alpha > \beta > 0$. Write down, in terms of α and β, the coordinates of B, C and D.

Show that the quadrilateral $ABCD$ is a rectangle and find its area in terms of u and v only. Verify that, for the case $u = 81$ and $v = 4$, the area is 14.

2 The curve C has equation

$$y = a^{\sin(\pi \mathrm{e}^x)},$$

where $a > 1$.

(i) Find the coordinates of the stationary points on C.

(ii) Use the approximations $\mathrm{e}^t \approx 1 + t$ and $\sin t \approx t$ (both valid for small values of t) to show that

$$y \approx 1 - \pi x \ln a$$

for small values of x.

(iii) Sketch C.

(iv) By approximating C by means of straight lines joining consecutive stationary points, show that the area between C and the x-axis between the kth and $(k+1)$th maxima is approximately

$$\left(\frac{a^2+1}{2a}\right) \ln\left(1 + \left(k - \tfrac{3}{4}\right)^{-1}\right).$$

3 Prove that

$$\tan\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right) \equiv \sec x - \tan x\,. \qquad (*)$$

(i) Use $(*)$ to find the value of $\tan \tfrac{1}{8}\pi$. Hence show that

$$\tan \tfrac{11}{24}\pi = \frac{\sqrt{3}+\sqrt{2}-1}{\sqrt{3}-\sqrt{6}+1}\,.$$

(ii) Show that

$$\frac{\sqrt{3}+\sqrt{2}-1}{\sqrt{3}-\sqrt{6}+1} = 2+\sqrt{2}+\sqrt{3}+\sqrt{6}\,.$$

(iii) Use $(*)$ to show that

$$\tan \tfrac{1}{48}\pi = \sqrt{16+10\sqrt{2}+8\sqrt{3}+6\sqrt{6}\ } - 2 - \sqrt{2} - \sqrt{3} - \sqrt{6}\,.$$

4 The polynomial $\mathrm{p}(x)$ is of degree 9 and $\mathrm{p}(x) - 1$ is exactly divisible by $(x-1)^5$.

(i) Find the value of $\mathrm{p}(1)$.

(ii) Show that $\mathrm{p}'(x)$ is exactly divisible by $(x-1)^4$.

(iii) Given also that $\mathrm{p}(x) + 1$ is exactly divisible by $(x+1)^5$, find $\mathrm{p}(x)$.

5 Expand and simplify $(\sqrt{x-1}+1)^2$.

(i) Evaluate

$$\int_5^{10} \frac{\sqrt{x+2\sqrt{x-1}\ }+\sqrt{x-2\sqrt{x-1}\ }}{\sqrt{x-1}}\,\mathrm{d}x\,.$$

(ii) Find the total area between the curve

$$y = \frac{\sqrt{x-2\sqrt{x-1}\ }}{\sqrt{x-1}}$$

and the x-axis between the points $x = \frac{5}{4}$ and $x = 10$.

(iii) Evaluate

$$\int_{\frac{5}{4}}^{10} \frac{\sqrt{x+2\sqrt{x-1}\ }+\sqrt{x-2\sqrt{x+1}+2\ }}{\sqrt{x^2-1}}\,\mathrm{d}x\,.$$

6 The Fibonacci sequence F_1, F_2, F_3, ... is defined by $F_1 = 1$, $F_2 = 1$ and

$$F_{n+1} = F_n + F_{n-1} \qquad (n \geqslant 2).$$

Write down the values of F_3, F_4, ..., F_{10}.

Let $S = \displaystyle\sum_{i=1}^{\infty} \frac{1}{F_i}$.

(i) Show that $\dfrac{1}{F_i} > \dfrac{1}{2F_{i-1}}$ for $i \geqslant 4$ and deduce that $S > 3$.

Show also that $S < 3\frac{2}{3}$.

(ii) Show further that $3.2 < S < 3.5$.

7 Let $y = (x-a)^n \mathrm{e}^{bx}\sqrt{1+x^2}$, where n and a are constants and b is a non-zero constant. Show that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(x-a)^{n-1}\mathrm{e}^{bx}\mathrm{q}(x)}{\sqrt{1+x^2}},$$

where $\mathrm{q}(x)$ is a cubic polynomial.

Using this result, determine:

(i) $\displaystyle\int \frac{(x-4)^{14}\mathrm{e}^{4x}(4x^3-1)}{\sqrt{1+x^2}}\,\mathrm{d}x$;

(ii) $\displaystyle\int \frac{(x-1)^{21}\mathrm{e}^{12x}(12x^4-x^2-11)}{\sqrt{1+x^2}}\,\mathrm{d}x$;

(iii) $\displaystyle\int \frac{(x-2)^{6}\mathrm{e}^{4x}(4x^4+x^3-2)}{\sqrt{1+x^2}}\,\mathrm{d}x$.

8 The non-collinear points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, respectively. The points P and Q have position vectors $\mathbf{p}$ and $\mathbf{q}$, respectively, given by

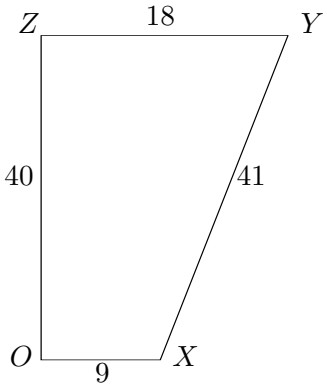
$$\mathbf{p} = \lambda\mathbf{a} + (1-\lambda)\mathbf{b} \quad \text{and} \quad \mathbf{q} = \mu\mathbf{a} + (1-\mu)\mathbf{c}$$

where $0 < \lambda < 1$ and $\mu > 1$. Draw a diagram showing A, B, C, P and Q.

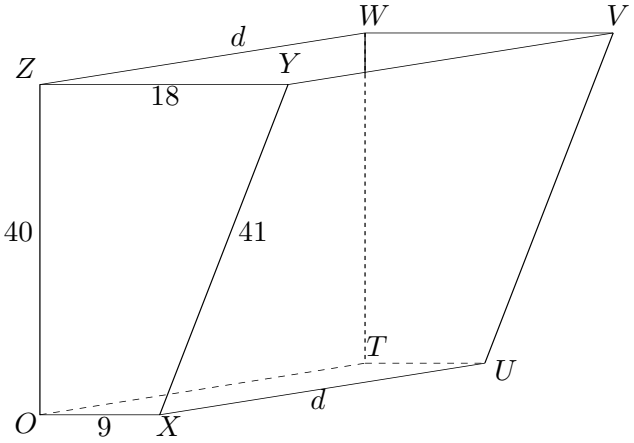
Given that $CQ \times BP = AB \times AC$, find μ in terms of λ, and show that, for all values of λ, the the line PQ passes through the fixed point D, with position vector $\mathbf{d}$ given by $\mathbf{d} = -\mathbf{a} + \mathbf{b} + \mathbf{c}$. What can be said about the quadrilateral $ABDC$?

Section B: Mechanics

9 **(i)** A uniform lamina $OXYZ$ is in the shape of the trapezium shown in the diagram. It is right-angled at O and Z, and OX is parallel to YZ. The lengths of the sides are given by $OX = 9\,\text{cm}$, $XY = 41\,\text{cm}$, $YZ = 18\,\text{cm}$ and $ZO = 40\,\text{cm}$. Show that its centre of mass is a distance $7\,\text{cm}$ from the edge OZ.



(ii) The diagram shows a tank with no lid made of thin sheet metal. The base $OXUT$, the back $OTWZ$ and the front $XUVY$ are rectangular, and each end is a trapezium as in part (i) . The width of the tank is $d\,\text{cm}$.



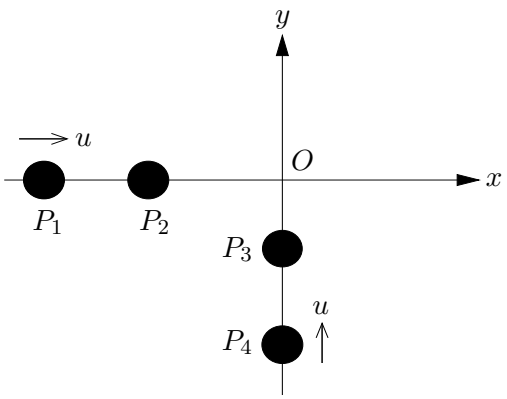
Show that the centre of mass of the tank, when empty, is a distance

$$\frac{3(140 + 11d)}{5(12 + d)}\,\text{cm}$$

from the back of the tank.

The tank is then filled with a liquid. The mass per unit volume of this liquid is k times the mass per unit area of the sheet metal. In the case $d = 20$, find an expression for the distance of the centre of mass of the filled tank from the back of the tank.

10



Four particles P_1, P_2, P_3 and P_4, of masses m_1, m_2, m_3 and m_4, respectively, are arranged on smooth horizontal axes as shown in the diagram.

Initially, P_2 and P_3 are stationary, and both P_1 and P_4 are moving towards O with speed u. Then P_1 and P_2 collide, at the same moment as P_4 and P_3 collide. Subsequently, P_2 and P_3 collide at O, as do P_1 and P_4 some time later. The coefficient of restitution between each pair of particles is e, and $e > 0$.

Show that initially P_2 and P_3 are equidistant from O.

11 A train consists of an engine and n trucks. It is travelling along a straight horizontal section of track. The mass of the engine and of each truck is M. The resistance to motion of the engine and of each truck is R, which is constant. The maximum power at which the engine can work is P.

Obtain an expression for the acceleration of the train when its speed is v and the engine is working at maximum power.

The train starts from rest with the engine working at maximum power. Obtain an expression for the time T taken to reach a given speed V, and show that this speed is only achievable if

$$P > (n+1)RV\,.$$

(i) In the case when $(n+1)RV/P$ is small, use the approximation $\ln(1-x) \approx -x - \frac{1}{2}x^2$ (valid for small x) to obtain the approximation

$$PT \approx \tfrac{1}{2}(n+1)MV^2$$

and interpret this result.

(ii) In the general case, the distance moved from rest in time T is X. *Write down*, with explanation, an equation relating P, T, X, M, V, R and n and hence show that

$$X = \frac{2PT - (n+1)MV^2}{2(n+1)R}\,.$$

Section C: Probability and Statistics

12 A continuous random variable X has probability density function given by

$$\mathrm{f}(x) = \begin{cases} 0 & \text{for } x < 0 \\ k\mathrm{e}^{-2x^2} & \text{for } 0 \leqslant x < \infty \,, \end{cases}$$

where k is a constant.

(i) Sketch the graph of $\mathrm{f}(x)$.

(ii) Find the value of k.

(iii) Determine $\mathrm{E}(X)$ and $\mathrm{Var}\,(X)$.

(iv) Use statistical tables to find, to three significant figures, the median value of X.

13 Satellites are launched using two different types of rocket: the Andover and the Basingstoke. The Andover has four engines and the Basingstoke has six. Each engine has a probability p of failing during any given launch. After the launch, the rockets are retrieved and repaired by replacing some or all of the engines. The cost of replacing each engine is K.

For the Andover, if more than one engine fails, all four engines are replaced. Otherwise, only the failed engine (if there is one) is replaced. Show that the expected repair cost for a single launch using the Andover is

$$4Kp(1 + q + q^2 - 2q^3) \qquad (q = 1 - p) \qquad (*)$$

For the Basingstoke, if more than two engines fail, all six engines are replaced. Otherwise only the failed engines (if there are any) are replaced. Find, in a form similar to $(*)$, the expected repair cost for a single launch using the Basingstoke.

Find the values of p for which the expected repair cost for the Andover is $\frac{2}{3}$ of the expected repair cost for the Basingstoke.

Section A: Pure Mathematics

1 Let P be a given point on a given curve C. The *osculating circle* to C at P is defined to be the circle that satisfies the following two conditions at P: it touches C; and the rate of change of its gradient is equal to the rate of change of the gradient of C.

Find the centre and radius of the osculating circle to the curve $y = 1 - x + \tan x$ at the point on the curve with x-coordinate $\frac{1}{4}\pi$.

2 Prove that

$$\cos 3x = 4\cos^3 x - 3\cos x\,.$$

Find and prove a similar result for $\sin 3x$ in terms of $\sin x$.

(i) Let

$$\mathrm{I}(\alpha) = \int_0^{\alpha} \big(7\sin x - 8\sin^3 x\big)\mathrm{d}x\,.$$

Show that

$$\mathrm{I}(\alpha) = -\tfrac{8}{3}c^3 + c + \tfrac{5}{3}\,,$$

where $c = \cos\alpha$. Write down one value of c for which $\mathrm{I}(\alpha) = 0$.

(ii) Useless Eustace believes that

$$\int \sin^n x\,\mathrm{d}x = \frac{\sin^{n+1} x}{n+1}$$

for $n = 1,\ 2,\ 3, \ldots$. Show that Eustace would obtain the correct value of $\mathrm{I}(\beta)$, where $\cos\beta = -\frac{1}{6}$.

Find all values of α for which he would obtain the correct value of $\mathrm{I}(\alpha)$.

3 The first four terms of a sequence are given by $F_0 = 0$, $F_1 = 1$, $F_2 = 1$ and $F_3 = 2$. The general term is given by

$$F_n = a\lambda^n + b\mu^n\,, \qquad (*)$$

where a, b, λ and μ are independent of n, and a is positive.

(i) Show that $\lambda^2 + \lambda\mu + \mu^2 = 2$, and find the values of λ, μ, a and b.

(ii) Use $(*)$ to evaluate F_6.

(iii) Evaluate $\displaystyle\sum_{n=0}^{\infty} \frac{F_n}{2^{n+1}}$.

4 **(i)** Let

$$I = \int_0^a \frac{\mathrm{f}(x)}{\mathrm{f}(x) + \mathrm{f}(a-x)}\,\mathrm{d}x\,.$$

Use a substitution to show that

$$I = \int_0^a \frac{\mathrm{f}(a-x)}{\mathrm{f}(x) + \mathrm{f}(a-x)}\,\mathrm{d}x$$

and hence evaluate I in terms of a.

Use this result to evaluate the integrals

$$\int_0^1 \frac{\ln(x+1)}{\ln(2+x-x^2)}\,\mathrm{d}x \qquad \text{and} \qquad \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin(x+\frac{\pi}{4})}\,\mathrm{d}x\,.$$

(ii) Evaluate

$$\int_{\frac{1}{2}}^2 \frac{\sin x}{x\big(\sin x + \sin\frac{1}{x}\big)}\,\mathrm{d}x\,.$$

5 The points A and B have position vectors $\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $5\mathbf{i}-\mathbf{j}-\mathbf{k}$, respectively, relative to the origin O. Find $\cos 2\alpha$, where 2α is the angle $\angle AOB$.

(i) The line L_1 has equation $\mathbf{r} = \lambda(m\mathbf{i} + n\mathbf{j} + p\mathbf{k})$. Given that L_1 is inclined equally to OA and to OB, determine a relationship between m, n and p. Find also values of m, n and p for which L_1 is the angle bisector of $\angle AOB$.

(ii) The line L_2 has equation $\mathbf{r} = \mu(u\mathbf{i} + v\mathbf{j} + w\mathbf{k})$. Given that L_2 is inclined at an angle α to OA, where $2\alpha = \angle AOB$, determine a relationship between u, v and w.

Hence describe the surface with Cartesian equation $x^2 + y^2 + z^2 = 2(yz + zx + xy)$.

6 Each edge of the tetrahedron $ABCD$ has unit length. The face ABC is horizontal, and P is the point in ABC that is vertically below D.

(i) Find the length of PD.

(ii) Show that the cosine of the angle between adjacent faces of the tetrahedron is $1/3$.

(iii) Find the radius of the largest sphere that can fit inside the tetrahedron.

7 **(i)** By considering the positions of its turning points, show that the curve with equation

$$y = x^3 - 3qx - q(1+q)\,,$$

where $q > 0$ and $q \neq 1$, crosses the x-axis once only.

(ii) Given that x satisfies the cubic equation

$$x^3 - 3qx - q(1+q) = 0\,,$$

and that

$$x = u + q/u\,,$$

obtain a quadratic equation satisfied by u^3. Hence find the real root of the cubic equation in the case $q > 0$, $q \neq 1$.

(iii) The quadratic equation

$$t^2 - pt + q = 0$$

has roots α and β. Show that

$$\alpha^3 + \beta^3 = p^3 - 3qp\,.$$

It is given that one of these roots is the square of the other. By considering the expression $(\alpha^2 - \beta)(\beta^2 - \alpha)$, find a relationship between p and q. Given further that $q > 0$, $q \neq 1$ and p is real, determine the value of p in terms of q.

8 The curves C_1 and C_2 are defined by

$$y = \mathrm{e}^{-x} \quad (x > 0) \qquad \text{and} \qquad y = \mathrm{e}^{-x}\sin x \quad (x > 0),$$

respectively. Sketch roughly C_1 and C_2 on the same diagram.

Let x_n denote the x-coordinate of the nth point of contact between the two curves, where $0 < x_1 < x_2 < \cdots$, and let A_n denote the area of the region enclosed by the two curves between x_n and x_{n+1}. Show that

$$A_n = \tfrac{1}{2}(\mathrm{e}^{2\pi} - 1)\mathrm{e}^{-(4n+1)\pi/2}$$

and hence find $\displaystyle\sum_{n=1}^{\infty} A_n$.

Section B: Mechanics

9 Two points A and B lie on horizontal ground. A particle P_1 is projected from A towards B at an acute angle of elevation α and simultaneously a particle P_2 is projected from B towards A at an acute angle of elevation β. Given that the two particles collide in the air a horizontal distance b from B, and that the collision occurs after P_1 has attained its maximum height h, show that

$$2h\cot\beta < b < 4h\cot\beta$$

and

$$2h\cot\alpha < a < 4h\cot\alpha\,,$$

where a is the horizontal distance from A to the point of collision.

10 **(i)** In an experiment, a particle A of mass m is at rest on a smooth horizontal table. A particle B of mass bm, where $b > 1$, is projected along the table directly towards A with speed u. The collision is perfectly elastic.

Find an expression for the speed of A after the collision in terms of b and u, and show that, irrespective of the relative masses of the particles, A cannot be made to move at twice the initial speed of B.

(ii) In a second experiment, a particle B_1 is projected along the table directly towards A with speed u. This time, particles B_2, B_3, ..., B_n are at rest in order on the line between B_1 and A. The mass of B_i ($i = 1, 2, \ldots, n$) is $\lambda^{n+1-i}m$, where $\lambda > 1$. All collisions are perfectly elastic. Show that, by choosing n sufficiently large, there is no upper limit on the speed at which A can be made to move.

In the case $\lambda = 4$, determine the least value of n for which A moves at more than $20u$. You may use the approximation $\log_{10} 2 \approx 0.30103$.

11 A uniform rod AB of length $4L$ and weight W is inclined at an angle θ to the horizontal. Its lower end A rests on a fixed support and the rod is held in equilibrium by a string attached to the rod at a point C which is $3L$ from A. The reaction of the support on the rod acts in a direction α to AC and the string is inclined at an angle β to CA. Show that

$$\cot\alpha = 3\tan\theta + 2\cot\beta\,.$$

Given that $\theta = 30°$ and $\beta = 45°$, show that $\alpha = 15°$.

Section C: Probability and Statistics

12 The continuous random variable X has probability density function $\mathrm{f}(x)$, where

$$\mathrm{f}(x) = \begin{cases} a & \text{for } 0 \leqslant x < k \\ b & \text{for } k \leqslant x \leqslant 1 \\ 0 & \text{otherwise,} \end{cases}$$

where $a > b > 0$ and $0 < k < 1$. Show that $a > 1$ and $b < 1$.

(i) Show that

$$\mathrm{E}(X) = \frac{1 - 2b + ab}{2(a - b)}\,.$$

(ii) Show that the median, M, of X is given by $M = \dfrac{1}{2a}$ if $a + b \geqslant 2ab$ and obtain an expression for the median if $a + b \leqslant 2ab$.

(iii) Show that $M < \mathrm{E}(X)$.

13 Rosalind wants to join the Stepney Chess Club. In order to be accepted, she must play a challenge match consisting of several games against Pardeep (the Club champion) and Quentin (the Club secretary), in which she must win at least one game against each of Pardeep and Quentin. From past experience, she knows that the probability of her winning a single game against Pardeep is p and the probability of her winning a single game against Quentin is q, where $0 < p < q < 1$.

(i) The challenge match consists of three games. Before the match begins, Rosalind must choose either to play Pardeep twice and Quentin once or to play Quentin twice and Pardeep once. Show that she should choose to play Pardeep twice.

(ii) In order to ease the entry requirements, it is decided instead that the challenge match will consist of four games. Now, before the match begins, Rosalind must choose whether to play Pardeep three times and Quentin once (strategy 1), or to play Pardeep twice and Quentin twice (strategy 2) or to play Pardeep once and Quentin three times (strategy 3).

Show that, if $q - p > \frac{1}{2}$, Rosalind should choose strategy 1.

If $q - p < \frac{1}{2}$ give examples of values of p and q to show that strategy 2 can be better or worse than strategy 1.

Section A: Pure Mathematics

1 **(i)** Sketch the curve $y = \sqrt{1-x} + \sqrt{3+x}$.

Use your sketch to show that only one real value of x satisfies

$$\sqrt{1-x} + \sqrt{3+x} = x + 1\,,$$

and give this value.

(ii) Determine graphically the number of real values of x that satisfy

$$2\sqrt{1-x} = \sqrt{3+x} + \sqrt{3-x}\,.$$

Solve this equation.

2 Write down the cubes of the integers 1, 2, ... , 10 .

The positive integers x, y and z, where $x < y$, satisfy

$$x^3 + y^3 = kz^3\,, \qquad (*)$$

where k is a given positive integer.

(i) In the case $x + y = k$, show that

$$z^3 = k^2 - 3kx + 3x^2\,.$$

Deduce that $(4z^3 - k^2)/3$ is a perfect square and that $\frac{1}{4}k^2 \leqslant z^3 < k^2$.

Use these results to find a solution of $(*)$ when $k = 20$.

(ii) By considering the case $x + y = z^2$, find two solutions of $(*)$ when $k = 19$.

3 In this question, you may assume without proof that any function f for which $\mathrm{f}'(x) \geqslant 0$ is *increasing*; that is, $\mathrm{f}(x_2) \geqslant \mathrm{f}(x_1)$ if $x_2 \geqslant x_1$.

(i) **(a)** Let $\mathrm{f}(x) = \sin x - x\cos x$. Show that $\mathrm{f}(x)$ is increasing for $0 \leqslant x \leqslant \frac{1}{2}\pi$ and deduce that $\mathrm{f}(x) \geqslant 0$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$.

(b) Given that $\dfrac{\mathrm{d}}{\mathrm{d}x}(\arcsin x) \geqslant 1$ for $0 \leqslant x < 1$, show that

$$\arcsin x \geqslant x \qquad (0 \leqslant x < 1).$$

(c) Let $\mathrm{g}(x) = x\,\mathrm{cosec}\,x$ for $0 < x < \frac{1}{2}\pi$. Show that g is increasing and deduce that

$$(\arcsin x)\,x^{-1} \geqslant x\,\mathrm{cosec}\,x \qquad (0 < x < 1).$$

(ii) Given that $\dfrac{\mathrm{d}}{\mathrm{d}x}(\arctan x) \leqslant 1$ for $x \geqslant 0$, show by considering the function $x^{-1}\tan x$ that

$$(\tan x)(\arctan x) \geqslant x^2 \qquad (0 < x < \tfrac{1}{2}\pi).$$

4 **(i)** Find all the values of θ, in the range $0^\circ < \theta < 180^\circ$, for which $\cos\theta = \sin 4\theta$. Hence show that

$$\sin 18^\circ = \frac{1}{4}\left(\sqrt{5} - 1\right).$$

(ii) Given that

$$4\sin^2 x + 1 = 4\sin^2 2x\,,$$

find all possible values of $\sin x$, giving your answers in the form $p + q\sqrt{5}$ where p and q are rational numbers.

(iii) Hence find two values of α with $0^\circ < \alpha < 90^\circ$ for which

$$\sin^2 3\alpha + \sin^2 5\alpha = \sin^2 6\alpha\,.$$

[Turn over

5 The points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ with respect to an origin O, and O, A and B are non-collinear. The point C, with position vector $\mathbf{c}$, is the reflection of B in the line through O and A. Show that $\mathbf{c}$ can be written in the form

$$\mathbf{c} = \lambda\mathbf{a} - \mathbf{b}$$

where $\lambda = \dfrac{2\,\mathbf{a}.\mathbf{b}}{\mathbf{a}.\mathbf{a}}$.

The point D, with position vector $\mathbf{d}$, is the reflection of C in the line through O and B. Show that $\mathbf{d}$ can be written in the form

$$\mathbf{d} = \mu\mathbf{b} - \lambda\mathbf{a}$$

for some scalar μ to be determined.

Given that A, B and D are collinear, find the relationship between λ and μ. In the case $\lambda = -\frac{1}{2}$, determine the cosine of $\angle AOB$ and describe the relative positions of A, B and D.

6 For any given function f, let

$$I = \int [\mathrm{f}'(x)]^2\,[\mathrm{f}(x)]^n \mathrm{d}x\,, \qquad (*)$$

where n is a positive integer. Show that, if $\mathrm{f}(x)$ satisfies $\mathrm{f}''(x) = k\mathrm{f}(x)\mathrm{f}'(x)$ for some constant k, then $(*)$ can be integrated to obtain an expression for I in terms of $\mathrm{f}(x)$, $\mathrm{f}'(x)$, k and n.

(i) Verify your result in the case $\mathrm{f}(x) = \tan x$. Hence find

$$\int \frac{\sin^4 x}{\cos^8 x}\,\mathrm{d}x\,.$$

(ii) Find

$$\int \sec^2 x\,(\sec x + \tan x)^6\,\mathrm{d}x\,.$$

7 The two sequences a_0, a_1, a_2, ... and b_0, b_1, b_2, ... have general terms

$$a_n = \lambda^n + \mu^n \quad \text{and} \quad b_n = \lambda^n - \mu^n\,,$$

respectively, where $\lambda = 1 + \sqrt{2}$ and $\mu = 1 - \sqrt{2}$.

(i) Show that $\displaystyle\sum_{r=0}^{n} b_r = -\sqrt{2} + \frac{1}{\sqrt{2}}\, a_{n+1}$, and give a corresponding result for $\displaystyle\sum_{r=0}^{n} a_r$.

(ii) Show that, if n is odd,

$$\sum_{m=0}^{2n} \left(\sum_{r=0}^{m} a_r \right) = \tfrac{1}{2} b_{n+1}^2\,,$$

and give a corresponding result when n is even.

(iii) Show that, if n is even,

$$\left(\sum_{r=0}^{n} a_r \right)^2 - \sum_{r=0}^{n} a_{2r+1} = 2\,,$$

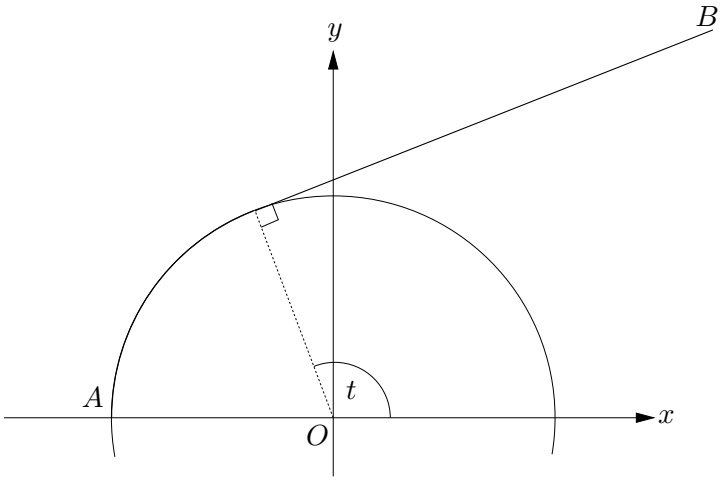
and give a corresponding result when n is odd.

8 The end A of an inextensible string AB of length π is attached to a point on the circumference of a fixed circle of unit radius and centre O. Initially the string is straight and tangent to the circle. The string is then wrapped round the circle until the end B comes into contact with the circle. The string remains taut during the motion, so that a section of the string is in contact with the circumference and the remaining section is straight.

Taking O to be the origin of cartesian coordinates with A at $(-1, 0)$ and B initially at $(-1, \pi)$, show that the curve described by B is given parametrically by

$$x = \cos t + t \sin t\,, \qquad y = \sin t - t \cos t\,,$$

where t is the angle shown in the diagram.



Find the value, t_0, of t for which x takes its maximum value on the curve, and sketch the curve.

Use the area integral $\displaystyle\int y \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t$ to find the area between the curve and the x axis for $\pi \geqslant t \geqslant t_0$.

Find the area swept out by the string (that is, the area between the curve described by B and the semicircle shown in the diagram).

Section B: Mechanics

9 Two particles, A of mass $2m$ and B of mass m, are moving towards each other in a straight line on a smooth horizontal plane, with speeds $2u$ and u respectively. They collide directly. Given that the coefficient of restitution between the particles is e, where $0 < e \leqslant 1$, determine the speeds of the particles after the collision.

After the collision, B collides directly with a smooth vertical wall, rebounding and then colliding directly with A for a second time. The coefficient of restitution between B and the wall is f, where $0 < f \leqslant 1$. Show that the velocity of B after its second collision with A is

$$\tfrac{2}{3}(1 - e^2)u - \tfrac{1}{3}(1 - 4e^2)fu$$

towards the wall and that B moves towards (not away from) the wall for all values of e and f.

10 A particle is projected from a point on a horizontal plane, at speed u and at an angle θ above the horizontal. Let H be the maximum height of the particle above the plane. Derive an expression for H in terms of u, g and θ.

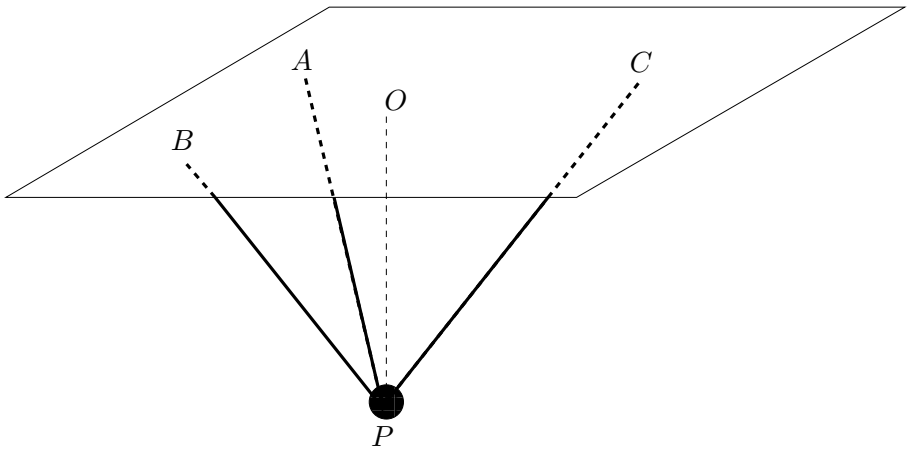
A particle P is projected from a point O on a smooth horizontal plane, at speed u and at an angle θ above the horizontal. At the same instant, a second particle R is projected horizontally from O in such a way that R is vertically below P in the ensuing motion. A light inextensible string of length $\frac{1}{2}H$ connects P and R. Show that the time that elapses before the string becomes taut is

$$(\sqrt{2} - 1)\sqrt{H/g}\,.$$

When the string becomes taut, R leaves the plane, the string remaining taut. Given that P and R have equal masses, determine the total horizontal distance, D, travelled by R from the moment its motion begins to the moment it lands on the plane again, giving your answer in terms of u, g and θ.

Given that $D = H$, find the value of $\tan\theta$.

11 Three non-collinear points A, B and C lie in a horizontal ceiling. A particle P of weight W is suspended from this ceiling by means of three light inextensible strings AP, BP and CP, as shown in the diagram. The point O lies vertically above P in the ceiling.



The angles AOB and AOC are $90° + \theta$ and $90° + \phi$, respectively, where θ and ϕ are acute angles such that $\tan\theta = \sqrt{2}$ and $\tan\phi = \frac{1}{4}\sqrt{2}$.

The strings AP, BP and CP make angles $30°$, $90° - \theta$ and $60°$, respectively, with the vertical, and the tensions in these strings have magnitudes T, U and V respectively.

(i) Show that the unit vector in the direction PB can be written in the form

$$-\frac{1}{3}\mathbf{i} - \frac{\sqrt{2}}{3}\mathbf{j} + \frac{\sqrt{2}}{\sqrt{3}}\mathbf{k},$$

where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the usual mutually perpendicular unit vectors with $\mathbf{j}$ parallel to OA and $\mathbf{k}$ vertically upwards.

(ii) Find expressions in vector form for the forces acting on P.

(iii) Show that $U = \sqrt{6}V$ and find T, U and V in terms of W.

Section C: Probability and Statistics

12 Xavier and Younis are playing a match. The match consists of a series of games and each game consists of three points.

Xavier has probability p and Younis has probability $1 - p$ of winning the first point of any game. In the second and third points of each game, the player who won the previous point has probability p and the player who lost the previous point has probability $1 - p$ of winning the point. If a player wins two consecutive points in a single game, the match ends and that player has won; otherwise the match continues with another game.

(i) Let w be the probability that Younis wins the match. Show that, for $p \neq 0$,

$$w = \frac{1 - p^2}{2 - p}.$$

Show that $w > \frac{1}{2}$ if $p < \frac{1}{2}$, and $w < \frac{1}{2}$ if $p > \frac{1}{2}$. Does w increase whenever p decreases?

(ii) If Xavier wins the match, Younis gives him £1; if Younis wins the match, Xavier gives him £k. Find the value of k for which the game is fair in the case when $p = \frac{2}{3}$.

(iii) What happens when $p = 0$?

13 What property of a distribution is measured by its *skewness*?

(i) One measure of skewness, γ, is given by

$$\gamma = \frac{\mathrm{E}\big((X-\mu)^3\big)}{\sigma^3}\,,$$

where μ and σ^2 are the mean and variance of the random variable X. Show that

$$\gamma = \frac{\mathrm{E}(X^3) - 3\mu\sigma^2 - \mu^3}{\sigma^3}\,.$$

The continuous random variable X has probability density function f where

$$\mathrm{f}(x) = \begin{cases} 2x & \text{for } 0 \leqslant x \leqslant 1\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

Show that for this distribution $\gamma = -\dfrac{2\sqrt{2}}{5}$.

(ii) The *decile skewness*, D, of a distribution is defined by

$$D = \frac{\mathrm{F}^{-1}(\frac{9}{10}) - 2\mathrm{F}^{-1}(\frac{1}{2}) + \mathrm{F}^{-1}(\frac{1}{10})}{\mathrm{F}^{-1}(\frac{9}{10}) - \mathrm{F}^{-1}(\frac{1}{10})}\,,$$

where F^{-1} is the inverse of the cumulative distribution function. Show that, for the above distribution, $D = 2 - \sqrt{5}$.

The *Pearson skewness*, P, of a distribution is defined by

$$P = \frac{3(\mu - M)}{\sigma}\,,$$

where M is the median. Find P for the above distribution and show that $D > P > \gamma$.

Section A: Pure Mathematics

1 Write down the general term in the expansion in powers of x of $(1 - x^6)^{-2}$.

(i) Find the coefficient of x^{24} in the expansion in powers of x of

$$(1 - x^6)^{-2}(1 - x^3)^{-1}.$$

Obtain also, and simplify, formulae for the coefficient of x^n in the different cases that arise.

(ii) Show that the coefficient of x^{24} in the expansion in powers of x of

$$(1 - x^6)^{-2}(1 - x^3)^{-1}(1 - x)^{-1}$$

is 55, and find the coefficients of x^{25} and x^{66}.

2 If $\mathrm{p}(x)$ and $\mathrm{q}(x)$ are polynomials of degree m and n, respectively, what is the degree of $\mathrm{p}(\mathrm{q}(x))$?

(i) The polynomial $\mathrm{p}(x)$ satisfies

$$\mathrm{p}(\mathrm{p}(\mathrm{p}(x))) - 3\mathrm{p}(x) = -2x$$

for all x. Explain carefully why $\mathrm{p}(x)$ must be of degree 1, and find all polynomials that satisfy this equation.

(ii) Find all polynomials that satisfy

$$2\mathrm{p}(\mathrm{p}(x)) + 3[\mathrm{p}(x)]^2 - 4\mathrm{p}(x) = x^4$$

for all x.

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[Turn over

3 Show that, for any function f (for which the integrals exist),

$$\int_0^\infty \mathrm{f}\left(x+\sqrt{1+x^2}\right)\,\mathrm{d}x = \frac{1}{2}\int_1^\infty \left(1+\frac{1}{t^2}\right)\mathrm{f}(t)\,\mathrm{d}t\,.$$

Hence evaluate

$$\int_0^\infty \frac{1}{2x^2+1+2x\sqrt{x^2+1}}\,\mathrm{d}x\,,$$

and, using the substitution $x=\tan\theta$,

$$\int_0^{\frac{1}{2}\pi} \frac{1}{(1+\sin\theta)^3}\,\mathrm{d}\theta\,.$$

4 In this question, you may assume that the infinite series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^{n+1}\frac{x^n}{n} + \cdots$$

is valid for $|x|<1$.

(i) Let n be an integer greater than 1. Show that, for any positive integer k,

$$\frac{1}{(k+1)n^{k+1}} < \frac{1}{kn^k}\,.$$

Hence show that $\ln\left(1+\frac{1}{n}\right) < \frac{1}{n}$. Deduce that

$$\left(1+\frac{1}{n}\right)^n < \mathrm{e}\,.$$

(ii) Show, using an expansion in powers of $\frac{1}{y}$, that $\ln\left(\frac{2y+1}{2y-1}\right) > \frac{1}{y}$ for $y>\frac{1}{2}$.

Deduce that, for any positive integer n,

$$\mathrm{e} < \left(1+\frac{1}{n}\right)^{n+\frac{1}{2}}\,.$$

(iii) Use parts (i) and (ii) to show that as $n\to\infty$

$$\left(1+\frac{1}{n}\right)^n \to \mathrm{e}\,.$$

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5 **(i)** Sketch the curve $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = \frac{1}{(x-a)^2 - 1} \qquad (x \neq a \pm 1),$$

and a is a constant.

(ii) The function $\mathrm{g}(x)$ is defined by

$$\mathrm{g}(x) = \frac{1}{\big((x-a)^2 - 1\big)\big((x-b)^2 - 1\big)} \qquad (x \neq a \pm 1,\ x \neq b \pm 1),$$

where a and b are constants, and $b > a$. Sketch the curves $y = \mathrm{g}(x)$ in the two cases $b > a + 2$ and $b = a + 2$, finding the values of x at the stationary points.

6 A cyclic quadrilateral $ABCD$ has sides AB, BC, CD and DA of lengths a, b, c and d, respectively. The area of the quadrilateral is Q, and angle DAB is θ.

Find an expression for $\cos\theta$ in terms of a, b, c and d, and an expression for $\sin\theta$ in terms of a, b, c, d and Q. Hence show that

$$16Q^2 = 4(ad + bc)^2 - (a^2 + d^2 - b^2 - c^2)^2\,,$$

and deduce that

$$Q^2 = (s-a)(s-b)(s-c)(s-d)\,,$$

where $s = \frac{1}{2}(a + b + c + d)$.

Deduce a formula for the area of a triangle with sides of length a, b and c.

7 Three distinct points, X_1, X_2 and X_3, with position vectors $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ respectively, lie on a circle of radius 1 with its centre at the origin O. The point G has position vector $\frac{1}{3}(\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3)$. The line through X_1 and G meets the circle again at the point Y_1 and the points Y_2 and Y_3 are defined correspondingly.

Given that $\overrightarrow{GY_1} = -\lambda_1 \overrightarrow{GX_1}$, where λ_1 is a positive scalar, show that

$$\overrightarrow{OY_1} = \tfrac{1}{3}\big((1 - 2\lambda_1)\mathbf{x}_1 + (1 + \lambda_1)(\mathbf{x}_2 + \mathbf{x}_3)\big)$$

and hence that

$$\lambda_1 = \frac{3 - \alpha - \beta - \gamma}{3 + \alpha - 2\beta - 2\gamma}\,,$$

where $\alpha = \mathbf{x}_2 \,.\, \mathbf{x}_3$, $\beta = \mathbf{x}_3 \,.\, \mathbf{x}_1$ and $\gamma = \mathbf{x}_1 \,.\, \mathbf{x}_2$.

Deduce that $\dfrac{GX_1}{GY_1} + \dfrac{GX_2}{GY_2} + \dfrac{GX_3}{GY_3} = 3\,.$

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[Turn over

8 The positive numbers α, β and q satisfy $\beta - \alpha > q$. Show that

$$\frac{\alpha^2 + \beta^2 - q^2}{\alpha\beta} - 2 > 0\,.$$

The sequence u_0, u_1, ... is defined by $u_0 = \alpha$, $u_1 = \beta$ and

$$u_{n+1} = \frac{u_n^2 - q^2}{u_{n-1}} \qquad (n \geqslant 1),$$

where α, β and q are given positive numbers (and α and β are such that no term in the sequence is zero). Prove that $u_n(u_n + u_{n+2}) = u_{n+1}(u_{n-1} + u_{n+1})$. Prove also that

$$u_{n+1} - pu_n + u_{n-1} = 0$$

for some number p which you should express in terms of α, β and q.

Hence, or otherwise, show that if $\beta > \alpha + q$, the sequence is strictly increasing (that is, $u_{n+1} - u_n > 0$ for all n). Comment on the case $\beta = \alpha + q$.

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Section B: Mechanics

9 A tennis ball is projected from a height of $2h$ above horizontal ground with speed u and at an angle of α below the horizontal. It travels in a plane perpendicular to a vertical net of height h which is a horizontal distance of a from the point of projection. Given that the ball passes over the net, show that

$$\frac{1}{u^2} < \frac{2(h - a\tan\alpha)}{ga^2\sec^2\alpha}\,.$$

The ball lands before it has travelled a horizontal distance of b from the point of projection. Show that

$$\sqrt{u^2\sin^2\alpha + 4gh} < \frac{bg}{u\cos\alpha} + u\sin\alpha\,.$$

Hence show that

$$\tan\alpha < \frac{h(b^2 - 2a^2)}{ab(b-a)}\,.$$

10 A hollow circular cylinder of internal radius r is held fixed with its axis horizontal. A uniform rod of length $2a$ (where $a < r$) rests in equilibrium inside the cylinder inclined at an angle of θ to the horizontal, where $\theta \neq 0$. The vertical plane containing the rod is perpendicular to the axis of the cylinder. The coefficient of friction between the cylinder and each end of the rod is μ, where $\mu > 0$.

Show that, if the rod is on the point of slipping, then the normal reactions R_1 and R_2 of the lower and higher ends of the rod, respectively, on the cylinder are related by

$$\mu(R_1 + R_2) = (R_1 - R_2)\tan\phi$$

where ϕ is the angle between the rod and the radius to an end of the rod.

Show further that

$$\tan\theta = \frac{\mu r^2}{r^2 - a^2(1+\mu^2)}\,.$$

Deduce that $\lambda < \phi$, where $\tan\lambda = \mu$.

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[Turn over

11 A small block of mass km is initially at rest on a smooth horizontal surface. Particles P_1, P_2, P_3, ... are fired, in order, along the surface from a fixed point towards the block. The mass of the ith particle is im ($i = 1, 2, \ldots$) and the speed at which it is fired is u/i. Each particle that collides with the block is embedded in it. Show that, if the nth particle collides with the block, the speed of the block after the collision is

$$\frac{2nu}{2k + n(n+1)}.$$

In the case $2k = N(N+1)$, where N is a positive integer, determine the number of collisions that occur. Show that the total kinetic energy lost in all the collisions is

$$\tfrac{1}{2}mu^2\left(\sum_{n=2}^{N+1}\frac{1}{n}\right).$$

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Section C: Probability and Statistics

12 A modern villa has complicated lighting controls. In order for the light in the swimming pool to be on, a particular switch in the hallway must be on and a particular switch in the kitchen must be on. There are four identical switches in the hallway and four identical switches in the kitchen. Guests cannot tell whether the switches are on or off, or what they control.

Each Monday morning a guest arrives, and the switches in the hallway are either all on or all off. The probability that they are all on is p and the probability that they are all off is $1 - p$. The switches in the kitchen are each on or off, independently, with probability $\frac{1}{2}$.

(i) On the first Monday, a guest presses one switch in the hallway at random and one switch in the kitchen at random. Find the probability that the swimming pool light is on at the end of this process. Show that the probability that the guest has pressed the swimming pool light switch in the hallway, given that the light is on at the end of the process, is $\dfrac{1-p}{1+2p}$.

(ii) On each of seven Mondays, guests go through the above process independently of each other, and each time the swimming pool light is found to be on at the end of the process. Given that the most likely number of days on which the swimming pool light switch in the hallway was pressed is 3, show that $\frac{1}{4} < p < \frac{5}{14}$.

13 In this question, you may assume that $\displaystyle\int_0^\infty \mathrm{e}^{-x^2/2}\mathrm{d}x = \sqrt{\tfrac{1}{2}\pi}$.

The number of supermarkets situated in any given region can be modelled by a Poisson random variable, where the mean is k times the area of the given region. Find the probability that there are no supermarkets within a circle of radius y.

The random variable Y denotes the distance between a randomly chosen point in the region and the nearest supermarket. Write down $\mathrm{P}(Y < y)$ and hence show that the probability density function of Y is $2\pi yk\mathrm{e}^{-\pi ky^2}$ for $y \geqslant 0$.

Find $\mathrm{E}(Y)$ and show that $\mathrm{Var}\,(Y) = \dfrac{4-\pi}{4\pi k}$.

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Section A: Pure Mathematics

1 **(i)** Find the value of m for which the line $y = mx$ touches the curve $y = \ln x$.

If instead the line intersects the curve when $x = a$ and $x = b$, where $a < b$, show that $a^b = b^a$. Show by means of a sketch that $a < \mathrm{e} < b$.

(ii) The line $y = mx + c$, where $c > 0$, intersects the curve $y = \ln x$ when $x = p$ and $x = q$, where $p < q$. Show by means of a sketch, or otherwise, that $p^q > q^p$.

(iii) Show by means of a sketch that the straight line through the points $(p, \ln p)$ and $(q, \ln q)$, where $\mathrm{e} \leqslant p < q$, intersects the y-axis at a positive value of y. Which is greater, π^{e} or e^{π}?

(iv) Show, using a sketch or otherwise, that if $0 < p < q$ and $\dfrac{\ln q - \ln p}{q - p} = \mathrm{e}^{-1}$, then $q^p > p^q$.

2 For $n \geqslant 0$, let

$$I_n = \int_0^1 x^n (1 - x)^n \mathrm{d}x .$$

(i) For $n \geqslant 1$, show by means of a substitution that

$$\int_0^1 x^{n-1}(1 - x)^n \mathrm{d}x = \int_0^1 x^n (1 - x)^{n-1} \mathrm{d}x$$

and deduce that

$$2\int_0^1 x^{n-1}(1 - x)^n \mathrm{d}x = I_{n-1} .$$

Show also, for $n \geqslant 1$, that

$$I_n = \frac{n}{n+1} \int_0^1 x^{n-1}(1 - x)^{n+1} \mathrm{d}x$$

and hence that $I_n = \dfrac{n}{2(2n+1)} I_{n-1}$.

(ii) When n is a positive integer, show that

$$I_n = \frac{(n!)^2}{(2n+1)!} .$$

(iii) Use the substitution $x = \sin^2 \theta$ to show that $I_{\frac{1}{2}} = \frac{\pi}{8}$, and evaluate $I_{\frac{3}{2}}$.

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3 **(i)** Given that the cubic equation $x^3 + 3ax^2 + 3bx + c = 0$ has three distinct real roots and $c < 0$, show with the help of sketches that either exactly one of the roots is positive or all three of the roots are positive.

(ii) Given that the equation $x^3 + 3ax^2 + 3bx + c = 0$ has three distinct real positive roots show that

$$a^2 > b > 0, \quad a < 0, \quad c < 0\,. \qquad (*)$$

[**Hint**: Consider the turning points.]

(iii) Given that the equation $x^3 + 3ax^2 + 3bx + c = 0$ has three distinct real roots and that

$$ab < 0, \quad c > 0\,,$$

determine, with the help of sketches, the signs of the roots.

(iv) Show by means of an explicit example (giving values for a, b and c) that it is possible for the conditions $(*)$ to be satisfied even though the corresponding cubic equation has only one real root.

4 The line passing through the point $(a, 0)$ with gradient b intersects the circle of unit radius centred at the origin at P and Q, and M is the midpoint of the chord PQ. Find the coordinates of M in terms of a and b.

(i) Suppose b is fixed and positive. As a varies, M traces out a curve (the *locus* of M). Show that $x = -by$ on this curve. Given that a varies with $-1 \leqslant a \leqslant 1$, show that the locus is a line segment of length $2b/(1 + b^2)^{\frac{1}{2}}$. Give a sketch showing the locus and the unit circle.

(ii) Find the locus of M in the following cases, giving in each case its cartesian equation, describing it geometrically and sketching it in relation to the unit circle:

(**a**) a is fixed with $0 < a < 1$, and b varies with $-\infty < b < \infty$;

(**b**) $ab = 1$, and b varies with $0 < b \leqslant 1$.

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5 **(i)** A function $\mathrm{f}(x)$ satisfies $\mathrm{f}(x) = \mathrm{f}(1-x)$ for all x. Show, by differentiating with respect to x, that $\mathrm{f}'(\frac{1}{2}) = 0$. If, in addition, $\mathrm{f}(x) = \mathrm{f}(\frac{1}{x})$ for all (non-zero) x, show that $\mathrm{f}'(-1) = 0$ and that $\mathrm{f}'(2) = 0$.

(ii) The function f is defined, for $x \neq 0$ and $x \neq 1$, by

$$\mathrm{f}(x) = \frac{(x^2 - x + 1)^3}{(x^2 - x)^2}\,.$$

Show that $\mathrm{f}(x) = \mathrm{f}(\frac{1}{x})$ and $\mathrm{f}(x) = \mathrm{f}(1 - x)$.

Given that it has exactly three stationary points, sketch the curve $y = \mathrm{f}(x)$.

(iii) Hence, or otherwise, find all the roots of the equation $\mathrm{f}(x) = \dfrac{27}{4}$ and state the ranges of values of x for which $\mathrm{f}(x) > \dfrac{27}{4}$.

Find also all the roots of the equation $\mathrm{f}(x) = \dfrac{343}{36}$ and state the ranges of values of x for which $\mathrm{f}(x) > \dfrac{343}{36}$.

6 In this question, the following theorem may be used.
Let u_1, u_2, ... be a sequence of (real) numbers. If the sequence is bounded above (that is, $u_n \leqslant b$ for all n, where b is some fixed number) and increasing (that is, $u_n \geqslant u_{n-1}$ for all n), then the sequence tends to a limit (that is, converges).

The sequence u_1, u_2, ... is defined by $u_1 = 1$ and

$$u_{n+1} = 1 + \frac{1}{u_n} \qquad (n \geqslant 1)\,. \qquad (*)$$

(i) Show that, for $n \geqslant 3$,

$$u_{n+2} - u_n = \frac{u_n - u_{n-2}}{(1 + u_n)(1 + u_{n-2})}.$$

(ii) Prove, by induction or otherwise, that $1 \leqslant u_n \leqslant 2$ for all n.

(iii) Show that the sequence u_1, u_3, u_5, ... tends to a limit, and that the sequence u_2, u_4, u_6, ... tends to a limit. Find these limits and deduce that the sequence u_1, u_2, u_3, ... tends to a limit.

Would this conclusion change if the sequence were defined by $(*)$ and $u_1 = 3$?

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7 **(i)** Write down a solution of the equation

$$x^2 - 2y^2 = 1\,, \qquad (*)$$

for which x and y are non-negative integers.

Show that, if $x = p$, $y = q$ is a solution of $(*)$, then so also is $x = 3p + 4q$, $y = 2p + 3q$. Hence find two solutions of $(*)$ for which x is a positive odd integer and y is a positive even integer.

(ii) Show that, if x is an odd integer and y is an even integer, $(*)$ can be written in the form

$$n^2 = \tfrac{1}{2}m(m+1)\,,$$

where m and n are integers.

(iii) The positive integers a, b and c satisfy

$$b^3 = c^4 - a^2\,,$$

where b is a prime number. Express a and c^2 in terms of b in the two cases that arise.

Find a solution of $a^2 + b^3 = c^4$, where a, b and c are positive integers but b is not prime.

8 The function f satisfies $\mathrm{f}(x) > 0$ for $x \geqslant 0$ and is strictly decreasing (which means that $\mathrm{f}(b) < \mathrm{f}(a)$ for $b > a$).

(i) For $t \geqslant 0$, let $A_0(t)$ be the area of the largest rectangle with sides parallel to the coordinate axes that can fit in the region bounded by the curve $y = \mathrm{f}(x)$, the y-axis and the line $y = \mathrm{f}(t)$. Show that $A_0(t)$ can be written in the form

$$A_0(t) = x_0\,(\mathrm{f}(x_0) - \mathrm{f}(t))\,,$$

where x_0 satisfies $x_0\mathrm{f}'(x_0) + \mathrm{f}(x_0) = \mathrm{f}(t)$.

(ii) The function g is defined, for $t > 0$, by

$$\mathrm{g}(t) = \frac{1}{t}\int_0^t \mathrm{f}(x)\mathrm{d}x\,.$$

Show that $t\mathrm{g}'(t) = \mathrm{f}(t) - \mathrm{g}(t)$.

Making use of a sketch show that, for $t > 0$,

$$\int_0^t (\mathrm{f}(x) - \mathrm{f}(t))\,\mathrm{d}x > A_0(t)$$

and deduce that $-t^2\mathrm{g}'(t) > A_0(t)$.

(iii) In the case $\mathrm{f}(x) = \dfrac{1}{1+x}$, use the above to establish the inequality

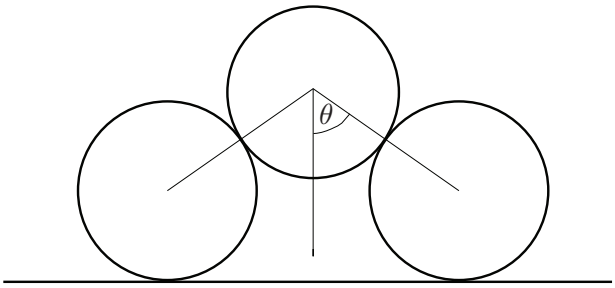
$$\ln\sqrt{1+t} > 1 - \frac{1}{\sqrt{1+t}}\,,$$

for $t > 0$.

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Section B: Mechanics

9 The diagram shows three identical discs in equilibrium in a vertical plane. Two discs rest, not in contact with each other, on a horizontal surface and the third disc rests on the other two. The angle at the upper vertex of the triangle joining the centres of the discs is 2θ.



The weight of each disc is W. The coefficient of friction between a disc and the horizontal surface is μ and the coefficient of friction between the discs is also μ.

(i) Show that the normal reaction between the horizontal surface and a disc in contact with the surface is $\frac{3}{2}W$.

(ii) Find the normal reaction between two discs in contact and show that the magnitude of the frictional force between two discs in contact is $\dfrac{W\sin\theta}{2(1+\cos\theta)}$.

(iii) Show that if $\mu < 2-\sqrt{3}$ there is no value of θ for which equilibrium is possible.

10 A particle is projected at an angle of elevation α (where $\alpha > 0$) from a point A on horizontal ground. At a general point in its trajectory the angle of elevation of the particle from A is θ and its direction of motion is at an angle ϕ above the horizontal (with $\phi \geqslant 0$ for the first half of the trajectory and $\phi \leqslant 0$ for the second half).

Let B denote the point on the trajectory at which $\theta = \frac{1}{2}\alpha$ and let C denote the point on the trajectory at which $\phi = -\frac{1}{2}\alpha$.

(i) Show that, at a general point on the trajectory, $2\tan\theta = \tan\alpha + \tan\phi$.

(ii) Show that, if B and C are the same point, then $\alpha = 60°$.

(iii) Given that $\alpha < 60°$, determine whether the particle reaches the point B first or the point C first.

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11 Three identical particles lie, not touching one another, in a straight line on a smooth horizontal surface. One particle is projected with speed u directly towards the other two which are at rest. The coefficient of restitution in all collisions is e, where $0 < e < 1$.

(i) Show that, after the second collision, the speeds of the particles are $\frac{1}{2}u(1-e)$, $\frac{1}{4}u(1-e^2)$ and $\frac{1}{4}u(1+e)^2$. Deduce that there will be a third collision whatever the value of e.

(ii) Show that there will be a fourth collision if and only if e is less than a particular value which you should determine.

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Section C: Probability and Statistics

12 The random variable U has a Poisson distribution with parameter λ. The random variables X and Y are defined as follows.

$$X = \begin{cases} U & \text{if } U \text{ is } 1, 3, 5, 7, \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$Y = \begin{cases} U & \text{if } U \text{ is } 2, 4, 6, 8, \ldots \\ 0 & \text{otherwise} \end{cases}$$

(i) Find $\mathrm{E}(X)$ and $\mathrm{E}(Y)$ in terms of λ, α and β, where

$$\alpha = 1 + \frac{\lambda^2}{2!} + \frac{\lambda^4}{4!} + \cdots \quad \text{and} \quad \beta = \frac{\lambda}{1!} + \frac{\lambda^3}{3!} + \frac{\lambda^5}{5!} + \cdots .$$

(ii) Show that

$$\mathrm{Var}\,(X) = \frac{\lambda\alpha + \lambda^2\beta}{\alpha + \beta} - \frac{\lambda^2\alpha^2}{(\alpha + \beta)^2}$$

and obtain the corresponding expression for $\mathrm{Var}\,(Y)$. Are there any non-zero values of λ for which $\mathrm{Var}\,(X) + \mathrm{Var}\,(Y) = \mathrm{Var}\,(X + Y)$?

13 A biased coin has probability p of showing a head and probability q of showing a tail, where $p \neq 0$, $q \neq 0$ and $p \neq q$. When the coin is tossed repeatedly, runs occur. A *straight run* of length n is a sequence of n consecutive heads or n consecutive tails. An *alternating run* of length n is a sequence of length n alternating between heads and tails. An alternating run can start with either a head or a tail.

Let S be the length of the longest straight run beginning with the first toss and let A be the length of the longest alternating run beginning with the first toss.

(i) Explain why $\mathrm{P}(A = 1) = p^2 + q^2$ and find $\mathrm{P}(S = 1)$. Show that $\mathrm{P}(S = 1) < \mathrm{P}(A = 1)$.

(ii) Show that $\mathrm{P}(S = 2) = \mathrm{P}(A = 2)$ and determine the relationship between $\mathrm{P}(S = 3)$ and $\mathrm{P}(A = 3)$.

(iii) Show that, for $n > 1$, $\mathrm{P}(S = 2n) > \mathrm{P}(A = 2n)$ and determine the corresponding relationship between $\mathrm{P}(S = 2n + 1)$ and $\mathrm{P}(A = 2n + 1)$. [You are advised *not* to use $p + q = 1$ in this part.]

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Section A: Pure Mathematics

1 In the triangle ABC, the base AB is of length 1 unit and the angles at A and B are α and β respectively, where $0 < \alpha \leqslant \beta$. The points P and Q lie on the sides AC and BC respectively, with $AP = PQ = QB = x$. The line PQ makes an angle of θ with the line through P parallel to AB.

(i) Show that $x\cos\theta = 1 - x\cos\alpha - x\cos\beta$, and obtain an expression for $x\sin\theta$ in terms of x, α and β. Hence show that

$$\big(1 + 2\cos(\alpha+\beta)\big)x^2 - 2(\cos\alpha + \cos\beta)x + 1 = 0\,. \qquad (*)$$

Show that $(*)$ is also satisfied if P and Q lie on AC produced and BC produced, respectively. [By definition, P lies on AC produced if P lies on the line through A and C and the points are in the order A, C, P.]

(ii) State the condition on α and β for $(*)$ to be linear in x. If this condition does not hold (but the condition $0 < \alpha \leqslant \beta$ still holds), show that $(*)$ has distinct real roots.

(iii) Find the possible values of x in the two cases (a) $\alpha = \beta = 45°$ and (b) $\alpha = 30°$, $\beta = 90°$, and illustrate each case with a sketch.

2 This question concerns the inequality

$$\int_0^\pi \big(\mathrm{f}(x)\big)^2\,\mathrm{d}x \leqslant \int_0^\pi \big(\mathrm{f}'(x)\big)^2\,\mathrm{d}x\,. \qquad (*)$$

(i) Show that $(*)$ is satisfied in the case $\mathrm{f}(x) = \sin nx$, where n is a positive integer.

Show by means of counterexamples that $(*)$ is not necessarily satisfied if either $\mathrm{f}(0) \neq 0$ or $\mathrm{f}(\pi) \neq 0$.

(ii) You may now assume that $(*)$ is satisfied for any (differentiable) function f for which $\mathrm{f}(0) = \mathrm{f}(\pi) = 0$.

By setting $\mathrm{f}(x) = ax^2 + bx + c$, where a, b and c are suitably chosen, show that $\pi^2 \leqslant 10$.

By setting $\mathrm{f}(x) = p\sin\frac{1}{2}x + q\cos\frac{1}{2}x + r$, where p, q and r are suitably chosen, obtain another inequality for π.

Which of these inequalities leads to a better estimate for π^2?

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3 **(i)** Show, geometrically or otherwise, that the shortest distance between the origin and the line $y = mx + c$, where $c \geqslant 0$, is $c(m^2 + 1)^{-\frac{1}{2}}$.

(ii) The curve C lies in the x-y plane. Let the line L be tangent to C at a point P on C, and let a be the shortest distance between the origin and L. The curve C has the property that the distance a is the same for all points P on C.

Let P be the point on C with coordinates $(x, y(x))$. Given that the tangent to C at P is not vertical, show that

$$(y - xy')^2 = a^2\big(1 + (y')^2\big)\,. \qquad (*)$$

By first differentiating $(*)$ with respect to x, show that either $y = mx \pm a(1 + m^2)^{\frac{1}{2}}$ for some m or $x^2 + y^2 = a^2$.

(iii) Now suppose that C (as defined above) is a continuous curve for $-\infty < x < \infty$, consisting of the arc of a circle and two straight lines. Sketch an example of such a curve which has a non-vertical tangent at each point.

4 **(i)** By using the substitution $u = 1/x$, show that for $b > 0$

$$\int_{1/b}^{b} \frac{x \ln x}{(a^2 + x^2)(a^2x^2 + 1)}\,\mathrm{d}x = 0\,.$$

(ii) By using the substitution $u = 1/x$, show that for $b > 0$,

$$\int_{1/b}^{b} \frac{\arctan x}{x}\,\mathrm{d}x = \frac{\pi \ln b}{2}\,.$$

(iii) By using the result $\displaystyle\int_0^\infty \frac{1}{a^2 + x^2}\,\mathrm{d}x = \frac{\pi}{2a}$ (where $a > 0$), and a substitution of the form $u = k/x$, for suitable k, show that

$$\int_0^\infty \frac{1}{(a^2 + x^2)^2}\,\mathrm{d}x = \frac{\pi}{4a^3} \qquad (a > 0).$$

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5 Given that $y = xu$, where u is a function of x, write down an expression for $\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

(i) Use the substitution $y = xu$ to solve

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2y + x}{y - 2x}$$

given that the solution curve passes through the point $(1, 1)$.

Give your answer in the form of a quadratic in x and y.

(ii) Using the substitutions $x = X + a$ and $y = Y + b$ for appropriate values of a and b, or otherwise, solve

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x - 2y - 4}{2x + y - 3},$$

given that the solution curve passes through the point $(1, 1)$.

6 By simplifying $\sin(r + \frac{1}{2})x - \sin(r - \frac{1}{2})x$ or otherwise show that, for $\sin \frac{1}{2}x \neq 0$,

$$\cos x + \cos 2x + \cdots + \cos nx = \frac{\sin(n + \frac{1}{2})x - \sin \frac{1}{2}x}{2 \sin \frac{1}{2}x}.$$

The functions S_n, for $n = 1, 2, \ldots$, are defined by

$$\mathrm{S}_n(x) = \sum_{r=1}^{n} \frac{1}{r} \sin rx \qquad (0 \leqslant x \leqslant \pi).$$

(i) Find the stationary points of $\mathrm{S}_2(x)$ for $0 \leqslant x \leqslant \pi$, and sketch this function.

(ii) Show that if $\mathrm{S}_n(x)$ has a stationary point at $x = x_0$, where $0 < x_0 < \pi$, then

$$\sin nx_0 = (1 - \cos nx_0) \tan \tfrac{1}{2}x_0$$

and hence that $\mathrm{S}_n(x_0) \geqslant \mathrm{S}_{n-1}(x_0)$. Deduce that if $\mathrm{S}_{n-1}(x) > 0$ for all x in the interval $0 < x < \pi$, then $\mathrm{S}_n(x) > 0$ for all x in this interval.

(iii) Prove that $\mathrm{S}_n(x) \geqslant 0$ for $n \geqslant 1$ and $0 \leqslant x \leqslant \pi$.

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7 **(i)** The function f is defined by $\mathrm{f}(x) = |x-a| + |x-b|$, where $a < b$. Sketch the graph of $\mathrm{f}(x)$, giving the gradient in each of the regions $x < a$, $a < x < b$ and $x > b$. Sketch on the same diagram the graph of $\mathrm{g}(x)$, where $\mathrm{g}(x) = |2x-a-b|$.

What shape is the quadrilateral with vertices $(a, 0)$, $(b, 0)$, $(b, \mathrm{f}(b))$ and $(a, \mathrm{f}(a))$?

(ii) Show graphically that the equation

$$|x-a| + |x-b| = |x-c|\,,$$

where $a < b$, has 0, 1 or 2 solutions, stating the relationship of c to a and b in each case.

(iii) For the equation

$$|x-a| + |x-b| = |x-c| + |x-d|\,,$$

where $a < b$, $c < d$ and $d-c < b-a$, determine the number of solutions in the various cases that arise, stating the relationship between a, b, c and d in each case.

8 For positive integers n, a and b, the integer c_r $(0 \leqslant r \leqslant n)$ is defined to be the coefficient of x^r in the expansion in powers of x of $(a+bx)^n$. Write down an expression for c_r in terms of r, n, a and b.

For given n, a and b, let m denote a value of r for which c_r is greatest (that is, $c_m \geqslant c_r$ for $0 \leqslant r \leqslant n$).

Show that

$$\frac{b(n+1)}{a+b} - 1 \leqslant m \leqslant \frac{b(n+1)}{a+b}\,.$$

Deduce that m is either a unique integer or one of two consecutive integers.

Let $\mathrm{G}(n, a, b)$ denote the unique value of m (if there is one) or the larger of the two possible values of m.

(i) Evaluate $\mathrm{G}(9, 1, 3)$ and $\mathrm{G}(9, 2, 3)$.

(ii) For any positive integer k, find $\mathrm{G}(2k, a, a)$ and $\mathrm{G}(2k-1, a, a)$ in terms of k.

(iii) For fixed n and b, determine a value of a for which $\mathrm{G}(n, a, b)$ is greatest.

(iv) For fixed n, find the greatest possible value of $\mathrm{G}(n, 1, b)$. For which values of b is this greatest value achieved?

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Section B: Mechanics

9 A uniform rectangular lamina $ABCD$ rests in equilibrium in a vertical plane with the corner A in contact with a rough vertical wall. The plane of the lamina is perpendicular to the wall. It is supported by a light inextensible string attached to the side AB at a distance d from A. The other end of the string is attached to a point on the wall above A where it makes an acute angle θ with the downwards vertical. The side AB makes an acute angle ϕ with the upwards vertical at A. The sides BC and AB have lengths $2a$ and $2b$ respectively. The coefficient of friction between the lamina and the wall is μ.

(i) Show that, when the lamina is in limiting equilibrium with the frictional force acting upwards,

$$d\sin(\theta+\phi) = (\cos\theta + \mu\sin\theta)(a\cos\phi + b\sin\phi)\,. \qquad (*)$$

(ii) How should $(*)$ be modified if the lamina is in limiting equilibrium with the frictional force acting downwards?

(iii) Find a condition on d, in terms of a, b, $\tan\theta$ and $\tan\phi$, which is necessary and sufficient for the frictional force to act upwards. Show that this condition cannot be satisfied if $b(2\tan\theta + \tan\phi) < a$.

10 A particle is projected from a point O on horizontal ground with initial speed u and at an angle of θ above the ground. The motion takes place in the x-y plane, where the x-axis is horizontal, the y-axis is vertical and the origin is O. Obtain the Cartesian equation of the particle's trajectory in terms of u, g and λ, where $\lambda = \tan\theta$.

Now consider the trajectories for different values of θ with u fixed. Show that for a given value of x, the coordinate y can take all values up to a maximum value, Y, which you should determine as a function of x, u and g.

Sketch a graph of Y against x and indicate on your graph the set of points that can be reached by a particle projected from O with speed u.

Hence find the furthest distance from O that can be achieved by such a projectile.

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11 A small smooth ring R of mass m is free to slide on a fixed smooth horizontal rail. A light inextensible string of length L is attached to one end, O, of the rail. The string passes through the ring, and a particle P of mass km (where $k > 0$) is attached to its other end; this part of the string hangs at an acute angle α to the vertical and it is given that α is constant in the motion.

Let x be the distance between O and the ring. Taking the y-axis to be vertically upwards, write down the Cartesian coordinates of P relative to O in terms of x, L and α.

(i) By considering the vertical component of the equation of motion of P, show that

$$km\ddot{x}\cos\alpha = T\cos\alpha - kmg\,,$$

where T is the tension in the string. Obtain two similar equations relating to the horizontal components of the equations of motion of P and R.

(ii) Show that $\dfrac{\sin\alpha}{(1-\sin\alpha)^2} = k$, and deduce, by means of a sketch or otherwise, that motion with α constant is possible for all values of k.

(iii) Show that $\ddot{x} = -g\tan\alpha\,.$

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Section C: Probability and Statistics

12 The lifetime of a fly (measured in hours) is given by the continuous random variable T with probability density function $\mathrm{f}(t)$ and cumulative distribution function $\mathrm{F}(t)$. The *hazard function*, $\mathrm{h}(t)$, is defined, for $\mathrm{F}(t) < 1$, by

$$\mathrm{h}(t) = \frac{\mathrm{f}(t)}{1 - \mathrm{F}(t)}\,.$$

(i) Given that the fly lives to at least time t, show that the probability of its dying within the following δt is approximately $\mathrm{h}(t)\,\delta t$ for small values of δt.

(ii) Find the hazard function in the case $\mathrm{F}(t) = t/a$ for $0 < t < a$. Sketch $\mathrm{f}(t)$ and $\mathrm{h}(t)$ in this case.

(iii) The random variable T is distributed on the interval $t > a$, where $a > 0$, and its hazard function is t^{-1}. Determine the probability density function for T.

(iv) Show that $\mathrm{h}(t)$ is constant for $t > b$ and zero otherwise if and only if $\mathrm{f}(t) = k\mathrm{e}^{-k(t-b)}$ for $t > b$, where k is a positive constant.

(v) The random variable T is distributed on the interval $t > 0$ and its hazard function is given by

$$\mathrm{h}(t) = \left(\frac{\lambda}{\theta^{\lambda}}\right) t^{\lambda-1}\,,$$

where λ and θ are positive constants. Find the probability density function for T.

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13 A random number generator prints out a sequence of integers I_1, I_2, I_3, ... Each integer is independently equally likely to be any one of 1, 2, ..., n, where n is fixed. The random variable X takes the value r, where I_r is the first integer which is a repeat of some earlier integer.

Write down an expression for $\mathrm{P}(X = 4)$.

(i) Find an expression for $\mathrm{P}(X = r)$, where $2 \leqslant r \leqslant n + 1$. Hence show that, for any positive integer n,

$$\frac{1}{n} + \left(1 - \frac{1}{n}\right)\frac{2}{n} + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\frac{3}{n} + \cdots = 1\,.$$

(ii) Write down an expression for $\mathrm{E}(X)$. (You do not need to simplify it.)

(iii) Write down an expression for $\mathrm{P}(X \geqslant k)$.

(iv) Show that, for any discrete random variable Y taking the values 1, 2, ..., N,

$$\mathrm{E}(Y) = \sum_{k=1}^{N} \mathrm{P}(Y \geqslant k)\,.$$

Hence show that, for any positive integer n,

$$\left(1 - \frac{1^2}{n}\right) + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2^2}{n}\right) + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\left(1 - \frac{3^2}{n}\right) + \cdots = 0.$$

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Section A: Pure Mathematics

1 **(i)** By use of calculus, show that $x - \ln(1+x)$ is positive for all positive x. Use this result to show that

$$\sum_{k=1}^{n} \frac{1}{k} > \ln(n+1)\,.$$

(ii) By considering $x + \ln(1-x)$, show that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} < 1 + \ln 2\,.$$

2 In the triangle ABC, angle $BAC = \alpha$ and angle $CBA = 2\alpha$, where 2α is acute, and $BC = x$. Show that $AB = (3 - 4\sin^2\alpha)x$.

The point D is the midpoint of AB and the point E is the foot of the perpendicular from C to AB. Find an expression for DE in terms of x.

The point F lies on the perpendicular bisector of AB and is a distance x from C. The points F and B lie on the same side of the line through A and C. Show that the line FC trisects the angle ACB.

3 Three rods have lengths a, b and c, where $a < b < c$. The three rods can be made into a triangle (possibly of zero area) if $a + b \geqslant c$.

Let T_n be the number of triangles that can be made with three rods chosen from n rods of lengths 1, 2, 3, ... , n (where $n \geqslant 3$). Show that $T_8 - T_7 = 2 + 4 + 6$ and evaluate $T_8 - T_6$. Write down expressions for $T_{2m} - T_{2m-1}$ and $T_{2m} - T_{2m-2}$.

Prove by induction that $T_{2m} = \frac{1}{6}m(m-1)(4m+1)$, and find the corresponding result for an odd number of rods.

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4 **(i)** The continuous function f is defined by

$$\tan \mathrm{f}(x) = x \qquad (-\infty < x < \infty)$$

and $\mathrm{f}(0) = \pi$. Sketch the curve $y = \mathrm{f}(x)$.

(ii) The continuous function g is defined by

$$\tan \mathrm{g}(x) = \frac{x}{1+x^2} \qquad (-\infty < x < \infty)$$

and $\mathrm{g}(0) = \pi$. Sketch the curves $y = \dfrac{x}{1+x^2}$ and $y = \mathrm{g}(x)$.

(iii) The continuous function h is defined by $\mathrm{h}(0) = \pi$ and

$$\tan \mathrm{h}(x) = \frac{x}{1-x^2} \qquad (x \neq \pm 1).$$

(The values of $\mathrm{h}(x)$ at $x = \pm 1$ are such that $\mathrm{h}(x)$ is continuous at these points.) Sketch the curves $y = \dfrac{x}{1-x^2}$ and $y = \mathrm{h}(x)$.

5 In this question, the arctan function satisfies $0 \leqslant \arctan x < \frac{1}{2}\pi$ for $x \geqslant 0$.

(i) Let

$$S_n = \sum_{m=1}^{n} \arctan\left(\frac{1}{2m^2}\right),$$

for $n = 1, 2, 3, \ldots$. Prove by induction that

$$\tan S_n = \frac{n}{n+1}.$$

Prove also that

$$S_n = \arctan \frac{n}{n+1}.$$

(ii) In a triangle ABC, the lengths of the sides AB and BC are $4n^2$ and $4n^4-1$, respectively, and the angle at B is a right angle. Let angle $BCA = 2\alpha_n$. Show that

$$\sum_{n=1}^{\infty} \alpha_n = \tfrac{1}{4}\pi.$$

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6 **(i)** Show that

$$\sec^2\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right) = \frac{2}{1+\sin x}\,.$$

Hence integrate $\dfrac{1}{1+\sin x}$ with respect to x.

(ii) By means of the substitution $y = \pi - x$, show that

$$\int_0^\pi x\mathrm{f}(\sin x)\,\mathrm{d}x = \frac{\pi}{2}\int_0^\pi \mathrm{f}(\sin x)\,\mathrm{d}x,$$

where f is any function for which these integrals exist.

Hence evaluate

$$\int_0^\pi \frac{x}{1+\sin x}\,\mathrm{d}x\,.$$

(iii) Evaluate

$$\int_0^\pi \frac{2x^3 - 3\pi x^2}{(1+\sin x)^2}\,\mathrm{d}x.$$

7 A circle C is said to be *bisected* by a curve X if X meets C in exactly two points and these points are diametrically opposite each other on C.

(i) Let C be the circle of radius a in the x-y plane with centre at the origin. Show, by giving its equation, that it is possible to find a circle of given radius r that bisects C provided $r > a$. Show that no circle of radius r bisects C if $r \leqslant a$.

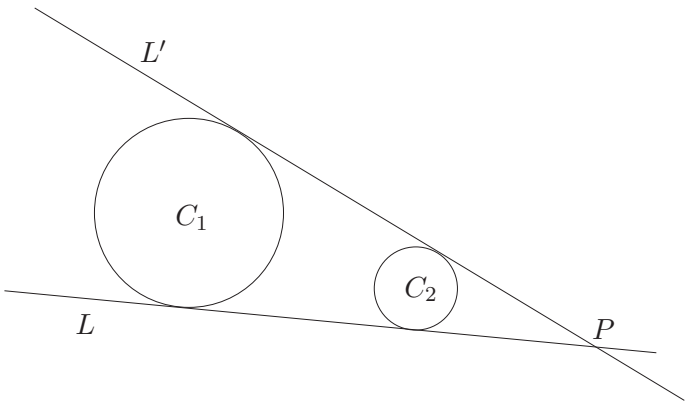
(ii) Let C_1 and C_2 be circles with centres at $(-d, 0)$ and $(d, 0)$ and radii a_1 and a_2, respectively, where $d > a_1$ and $d > a_2$. Let D be a circle of radius r that bisects both C_1 and C_2. Show that the x-coordinate of the centre of D is $\dfrac{a_2^2 - a_1^2}{4d}$.

Obtain an expression in terms of d, r, a_1 and a_2 for the y-coordinate of the centre of D, and deduce that r must satisfy

$$16r^2d^2 \geqslant \left(4d^2 + (a_2 - a_1)^2\right)\left(4d^2 + (a_2 + a_1)^2\right).$$

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8



The diagram above shows two non-overlapping circles C_1 and C_2 of different sizes. The lines L and L' are the two common tangents to C_1 and C_2 such that the two circles lie on the same side of each of the tangents. The lines L and L' intersect at the point P which is called the *focus* of C_1 and C_2.

(i) Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be the position vectors of the centres of C_1 and C_2, respectively. Show that the position vector of P is

$$\frac{r_1\mathbf{x}_2 - r_2\mathbf{x}_1}{r_1 - r_2},$$

where r_1 and r_2 are the radii of C_1 and C_2, respectively.

(ii) The circle C_3 does not overlap either C_1 or C_2 and its radius, r_3, satisfies $r_1 \neq r_3 \neq r_2$. The focus of C_1 and C_3 is Q, and the focus of C_2 and C_3 is R. Show that P, Q and R lie on the same straight line.

(iii) Find a condition on r_1, r_2 and r_3 for Q to lie half-way between P and R.

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Section B: Mechanics

9 An equilateral triangle ABC is made of three light rods each of length a. It is free to rotate in a vertical plane about a horizontal axis through A. Particles of mass $3m$ and $5m$ are attached to B and C respectively. Initially, the system hangs in equilibrium with BC below A.

(i) Show that, initially, the angle θ that BC makes with the horizontal is given by $\sin\theta = \frac{1}{7}$.

(ii) The triangle receives an impulse that imparts a speed v to the particle B. Find the minimum speed v_0 such that the system will perform complete rotations if $v > v_0$.

10 A particle of mass m is pulled along the floor of a room in a straight line by a light string which is pulled at constant speed V through a hole in the ceiling. The floor is smooth and horizontal, and the height of the room is h. Find, in terms of V and θ, the speed of the particle when the string makes an angle of θ with the vertical (and the particle is still in contact with the floor). Find also the acceleration, in terms of V, h and θ.

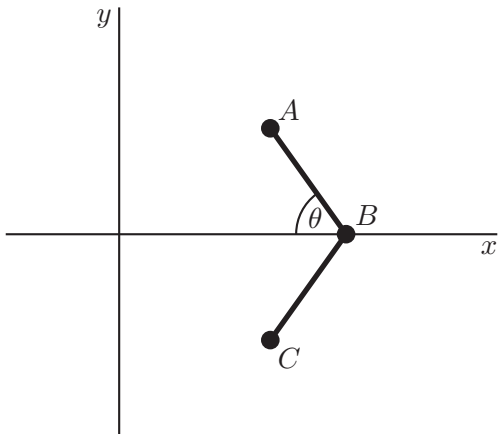
Find the tension in the string and hence show that the particle will leave the floor when

$$\tan^4\theta = \frac{V^2}{gh}\,.$$

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11 Three particles, A, B and C, each of mass m, lie on a smooth horizontal table. Particles A and C are attached to the two ends of a light inextensible string of length $2a$ and particle B is attached to the midpoint of the string. Initially, A, B and C are at rest at points $(0, a)$, $(0, 0)$ and $(0, -a)$, respectively.

An impulse is delivered to B, imparting to it a speed u in the positive x direction. The string remains taut throughout the subsequent motion.



(i) At time t, the angle between the x-axis and the string joining A and B is θ, as shown in the diagram, and B is at $(x, 0)$. Write down the coordinates of A in terms of x, a and θ at this time. Given that the velocity of B is $(v, 0)$, show that the velocity of A is $(v + a\dot{\theta}\sin\theta\,,\, a\dot{\theta}\cos\theta)$, where the dot denotes differentiation with respect to time.

(ii) Show that, before A and C first collide,

$$3v + 2a\dot{\theta}\sin\theta = u \qquad \text{and} \qquad \dot{\theta}^2 = \frac{u^2}{a^2(3 - 2\sin^2\theta)}\,.$$

(iii) When A and C collide, the collision is elastic (no energy is lost). At what value of θ does the second collision between particles A and C occur? (You should justify your answer.)

(iv) When $v = 0$, what are the possible values of θ? Is $v = 0$ whenever θ takes these values?

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Section C: Probability and Statistics

12 Four players A, B, C and D play a coin-tossing game with a fair coin. Each player chooses a sequence of heads and tails, as follows:

Player A: HHT; Player B: THH; Player C: TTH; Player D: HTT.

The coin is then tossed until one of these sequences occurs, in which case the corresponding player is the winner.

(i) Show that, if only A and B play, then A has a probability of $\frac{1}{4}$ of winning.

(ii) If all four players play together, find the probabilities of each one winning.

(iii) Only B and C play. What is the probability of C winning if the first two tosses are TT?

Let the probabilities of C winning if the first two tosses are HT, TH and HH be p, q and r, respectively. Show that $p = \frac{1}{2} + \frac{1}{2}q$.

Find the probability that C wins.

13 The maximum height X of flood water each year on a certain river is a random variable with probability density function f given by

$$\mathrm{f}(x) = \begin{cases} \lambda \mathrm{e}^{-\lambda x} & \text{for } x \geqslant 0\,, \\ 0 & \text{otherwise,} \end{cases}$$

where λ is a positive constant.

It costs ky pounds each year to prepare for flood water of height y or less, where k is a positive constant and $y \geqslant 0$. If $X \leqslant y$ no further costs are incurred but if $X > y$ the additional cost of flood damage is $a(X - y)$ pounds where a is a positive constant.

(i) Let C be the total cost of dealing with the floods in the year. Show that the expectation of C is given by

$$\mathrm{E}(C) = ky + \frac{a}{\lambda}\mathrm{e}^{-\lambda y}\,.$$

How should y be chosen in order to minimise $\mathrm{E}(C)$, in the different cases that arise according to the value of a/k?

(ii) Find the variance of C, and show that the more that is spent on preparing for flood water in advance the smaller this variance.

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Section A: Pure Mathematics

1 The curve C_1 has parametric equations $x = t^2$, $y = t^3$, where $-\infty < t < \infty$. Let O denote the point $(0, 0)$. The points P and Q on C_1 are such that $\angle POQ$ is a right angle. Show that the tangents to C_1 at P and Q intersect on the curve C_2 with equation $4y^2 = 3x - 1$.

Determine whether C_1 and C_2 meet, and sketch the two curves on the same axes.

2 Use the factor theorem to show that $a + b - c$ is a factor of

$$(a + b + c)^3 - 6(a + b + c)(a^2 + b^2 + c^2) + 8(a^3 + b^3 + c^3)\,. \qquad (*)$$

Hence factorise $(*)$ completely.

(i) Use the result above to solve the equation

$$(x + 1)^3 - 3(x + 1)(2x^2 + 5) + 2(4x^3 + 13) = 0\,.$$

(ii) By setting $d + e = c$, or otherwise, show that $(a + b - d - e)$ is a factor of

$$(a + b + d + e)^3 - 6(a + b + d + e)(a^2 + b^2 + d^2 + e^2) + 8(a^3 + b^3 + d^3 + e^3)$$

and factorise this expression completely.

Hence solve the equation

$$(x + 6)^3 - 6(x + 6)(x^2 + 14) + 8(x^3 + 36) = 0\,.$$

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3 For each non-negative integer n, the polynomial f_n is defined by

$$\mathrm{f}_n(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}\,.$$

(i) Show that $\mathrm{f}_n'(x) = \mathrm{f}_{n-1}(x)$ (for $n \geqslant 1$).

(ii) Show that, if a is a real root of the equation

$$\mathrm{f}_n(x) = 0\,, \qquad (*)$$

then $a < 0$.

(iii) Let a and b be distinct real roots of $(*)$, for $n \geqslant 2$. Show that $\mathrm{f}_n'(a)\,\mathrm{f}_n'(b) > 0$ and use a sketch to deduce that $\mathrm{f}_n(c) = 0$ for some number c between a and b.

Deduce that $(*)$ has at most one real root. How many real roots does $(*)$ have if n is odd? How many real roots does $(*)$ have if n is even?

4 Let

$$y = \frac{x^2 + x\sin\theta + 1}{x^2 + x\cos\theta + 1}\,.$$

(i) Given that x is real, show that

$$(y\cos\theta - \sin\theta)^2 \geqslant 4(y-1)^2\,.$$

Deduce that

$$y^2 + 1 \geqslant 4(y-1)^2\,,$$

and hence that

$$\frac{4-\sqrt{7}}{3} \leqslant y \leqslant \frac{4+\sqrt{7}}{3}\,.$$

(ii) In the case $y = \dfrac{4+\sqrt{7}}{3}$, show that

$$\sqrt{y^2+1} = 2(y-1)$$

and find the corresponding values of x and $\tan\theta$.

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5 In this question, the definition of $\binom{p}{q}$ is taken to be

$$\binom{p}{q} = \begin{cases} \dfrac{p!}{q!(p-q)!} & \text{if } p \geqslant q \geqslant 0\,, \\ 0 & \text{otherwise .} \end{cases}$$

(i) Write down the coefficient of x^n in the binomial expansion for $(1-x)^{-N}$, where N is a positive integer, and write down the expansion using the Σ summation notation.

By considering $(1-x)^{-1}(1-x)^{-N}$, where N is a positive integer, show that

$$\sum_{j=0}^{n} \binom{N+j-1}{j} = \binom{N+n}{n}.$$

(ii) Show that, for any positive integers m, n and r with $r \leqslant m+n$,

$$\binom{m+n}{r} = \sum_{j=0}^{r} \binom{m}{j}\binom{n}{r-j}.$$

(iii) Show that, for any positive integers m and N,

$$\sum_{j=0}^{n} (-1)^j \binom{N+m}{n-j}\binom{m+j-1}{j} = \binom{N}{n}.$$

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6 This question concerns solutions of the differential equation

$$(1-x^2)\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + k^2y^2 = k^2 \qquad (*)$$

where k is a positive integer.

For each value of k, let $y_k(x)$ be the solution of $(*)$ that satisfies $y_k(1) = 1$; you may assume that there is only one such solution for each value of k.

(i) Write down the differential equation satisfied by $y_1(x)$ and verify that $y_1(x) = x$.

(ii) Write down the differential equation satisfied by $y_2(x)$ and verify that $y_2(x) = 2x^2 - 1$.

(iii) Let $z(x) = 2\big(y_n(x)\big)^2 - 1$. Show that

$$(1-x^2)\left(\frac{\mathrm{d}z}{\mathrm{d}x}\right)^2 + 4n^2z^2 = 4n^2$$

and hence obtain an expression for $y_{2n}(x)$ in terms of $y_n(x)$.

(iv) Let $v(x) = y_n\big(y_m(x)\big)$. Show that $v(x) = y_{mn}(x)$.

7 Show that

$$\int_0^a \mathrm{f}(x)\mathrm{d}x = \int_0^a \mathrm{f}(a-x)\mathrm{d}x\,, \qquad (*)$$

where f is any function for which the integrals exist.

(i) Use $(*)$ to evaluate

$$\int_0^{\frac{1}{2}\pi} \frac{\sin x}{\cos x + \sin x}\,\mathrm{d}x\,.$$

(ii) Evaluate

$$\int_0^{\frac{1}{4}\pi} \frac{\sin x}{\cos x + \sin x}\,\mathrm{d}x\,.$$

(iii) Evaluate

$$\int_0^{\frac{1}{4}\pi} \ln(1+\tan x)\,\mathrm{d}x\,.$$

(iv) Evaluate

$$\int_0^{\frac{1}{4}\pi} \frac{x}{\cos x\,(\cos x + \sin x)}\,\mathrm{d}x\,.$$

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8 Evaluate the integral

$$\int_{m-\frac{1}{2}}^{\infty} \frac{1}{x^2}\,\mathrm{d}x \qquad (m > \tfrac{1}{2})\,.$$

Show by means of a sketch that

$$\sum_{r=m}^{n} \frac{1}{r^2} \approx \int_{m-\frac{1}{2}}^{n+\frac{1}{2}} \frac{1}{x^2}\,\mathrm{d}x\,, \qquad (*)$$

where m and n are positive integers with $m < n$.

(i) You are given that the infinite series $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r^2}$ converges to a value denoted by E. Use $(*)$ to obtain the following approximations for E:

$$E \approx 2\,; \quad E \approx \frac{5}{3}\,; \quad E \approx \frac{33}{20}\,.$$

(ii) Show that, when r is large, the error in approximating $\dfrac{1}{r^2}$ by $\displaystyle\int_{r-\frac{1}{2}}^{r+\frac{1}{2}} \frac{1}{x^2}\,\mathrm{d}x$ is approximately $\dfrac{1}{4r^4}$.

Given that $E \approx 1.645$, show that $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r^4} \approx 1.08$.

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Section B: Mechanics

9 A small bullet of mass m is fired into a block of wood of mass M which is at rest. The speed of the bullet on entering the block is u. Its trajectory within the block is a horizontal straight line and the resistance to the bullet's motion is R, which is constant.

(i) The block is fixed. The bullet travels a distance a inside the block before coming to rest. Find an expression for a in terms of m, u and R.

(ii) Instead, the block is free to move on a smooth horizontal table. The bullet travels a distance b inside the block before coming to rest relative to the block, at which time the block has moved a distance c on the table. Find expressions for b and c in terms of M, m and a.

10 A thin uniform wire is bent into the shape of an isosceles triangle ABC, where AB and AC are of equal length and the angle at A is 2θ. The triangle ABC hangs on a small rough horizontal peg with the side BC resting on the peg. The coefficient of friction between the wire and the peg is μ. The plane containing ABC is vertical. Show that the triangle can rest in equilibrium with the peg in contact with any point on BC provided

$$\mu \geqslant 2\tan\theta(1+\sin\theta)\,.$$

11 **(i)** Two particles move on a smooth horizontal surface. The positions, in Cartesian coordinates, of the particles at time t are $(a+ut\cos\alpha\,,\ ut\sin\alpha)$ and $(vt\cos\beta\,,\ b+vt\sin\beta)$, where a, b, u and v are positive constants, α and β are constant acute angles, and $t \geqslant 0$.

Given that the two particles collide, show that

$$u\sin(\theta+\alpha) = v\sin(\theta+\beta)\,,$$

where θ is the acute angle satisfying $\tan\theta = \dfrac{b}{a}$.

(ii) A gun is placed on the top of a vertical tower of height b which stands on horizontal ground. The gun fires a bullet with speed v and (acute) angle of elevation β. Simultaneously, a target is projected from a point on the ground a horizontal distance a from the foot of the tower. The target is projected with speed u and (acute) angle of elevation α, in a direction directly away from the tower.

Given that the target is hit before it reaches the ground, show that

$$2u\sin\alpha(u\sin\alpha - v\sin\beta) > bg\,.$$

Explain, with reference to part (i), why the target can only be hit if $\alpha > \beta$.

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Section C: Probability and Statistics

12 Starting with the result $\mathrm{P}(A \cup B) = \mathrm{P}(A) + P(B) - \mathrm{P}(A \cap B)$, prove that

$$\mathrm{P}(A \cup B \cup C) = \mathrm{P}(A) + \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(A \cap B) - \mathrm{P}(B \cap C) - \mathrm{P}(C \cap A) + \mathrm{P}(A \cap B \cap C)\,.$$

Write down, without proof, the corresponding result for four events A, B, C and D.

A pack of n cards, numbered 1, 2, ..., n, is shuffled and laid out in a row. The result of the shuffle is that each card is equally likely to be in any position in the row. Let E_i be the event that the card bearing the number i is in the ith position in the row. Write down the following probabilities:

(i) $\mathrm{P}(E_i)$;

(ii) $\mathrm{P}(E_i \cap E_j)$, where $i \neq j$;

(iii) $\mathrm{P}(E_i \cap E_j \cap E_k)$, where $i \neq j$, $j \neq k$ and $k \neq i$.

Hence show that the probability that at least one card is in the same position as the number it bears is

$$1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + (-1)^{n+1}\frac{1}{n!}\,.$$

Find the probability that exactly one card is in the same position as the number it bears.

13 **(i)** The random variable X has a binomial distribution with parameters n and p, where $n = 16$ and $p = \frac{1}{2}$. Show, using an approximation in terms of the standard normal density function $\frac{1}{\sqrt{2\pi}}\,\mathrm{e}^{-\frac{1}{2}x^2}$, that

$$\mathrm{P}(X = 8) \approx \frac{1}{2\sqrt{2\pi}}\,.$$

(ii) By considering a binomial distribution with parameters $2n$ and $\frac{1}{2}$, show that

$$(2n)! \approx \frac{2^{2n}(n!)^2}{\sqrt{n\pi}}\,.$$

(iii) By considering a Poisson distribution with parameter n, show that

$$n! \approx \sqrt{2\pi n}\,\mathrm{e}^{-n}\,n^n\,.$$

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Section A: Pure Mathematics

1 **Note:** In this question you may use without proof the result $\dfrac{\mathrm{d}}{\mathrm{d}x}\big(\arctan x\big) = \dfrac{1}{1+x^2}$.

Let

$$I_n = \int_0^1 x^n \arctan x \,\mathrm{d}x \,,$$

where $n = 0, 1, 2, 3, \ldots$.

(i) Show that, for $n \geqslant 0$,

$$(n+1)I_n = \frac{\pi}{4} - \int_0^1 \frac{x^{n+1}}{1+x^2}\,\mathrm{d}x$$

and evaluate I_0.

(ii) Find an expression, in terms of n, for $(n+3)I_{n+2} + (n+1)I_n$.

Use this result to evaluate I_4.

(iii) Prove by induction that, for $n \geqslant 1$,

$$(4n+1)I_{4n} = A - \frac{1}{2}\sum_{r=1}^{2n}(-1)^r\frac{1}{r}\,,$$

where A is a constant to be determined.

2 The sequence of numbers $x_0, x_1, x_2, \ldots$ satisfies

$$x_{n+1} = \frac{ax_n - 1}{x_n + b}\,.$$

(You may assume that a, b and x_0 are such that $x_n + b \neq 0$.)

Find an expression for x_{n+2} in terms of a, b and x_n.

(i) Show that $a + b = 0$ is a necessary condition for the sequence to be periodic with period 2.

Note: The sequence is said to be periodic with period k if $x_{n+k} = x_n$ for all n, and there is no integer m with $0 < m < k$ such that $x_{n+m} = x_n$ for all n.

(ii) Find necessary and sufficient conditions for the sequence to have period 4.

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3 **(i)** Sketch, on x-y axes, the set of all points satisfying $\sin y = \sin x$, for $-\pi \leqslant x \leqslant \pi$ and $-\pi \leqslant y \leqslant \pi$. You should give the equations of all the lines on your sketch.

(ii) Given that

$$\sin y = \tfrac{1}{2} \sin x$$

obtain an expression, in terms of x, for y' when $0 \leqslant x \leqslant \frac{1}{2}\pi$ and $0 \leqslant y \leqslant \frac{1}{2}\pi$, and show that

$$y'' = -\frac{3 \sin x}{(4 - \sin^2 x)^{\frac{3}{2}}} \,.$$

Use these results to sketch the set of all points satisfying $\sin y = \frac{1}{2} \sin x$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$ and $0 \leqslant y \leqslant \frac{1}{2}\pi$.

Hence sketch the set of all points satisfying $\sin y = \frac{1}{2} \sin x$ for $-\pi \leqslant x \leqslant \pi$ and $-\pi \leqslant y \leqslant \pi$.

(iii) Without further calculation, sketch the set of all points satisfying $\cos y = \frac{1}{2} \sin x$ for $-\pi \leqslant x \leqslant \pi$ and $-\pi \leqslant y \leqslant \pi$.

4 The Schwarz inequality is

$$\left(\int_a^b \mathrm{f}(x)\, \mathrm{g}(x)\, \mathrm{d}x \right)^2 \leqslant \left(\int_a^b \big(\mathrm{f}(x)\big)^2 \mathrm{d}x \right) \left(\int_a^b \big(\mathrm{g}(x)\big)^2 \mathrm{d}x \right). \qquad (*)$$

(i) By setting $\mathrm{f}(x) = 1$ in $(*)$, and choosing $\mathrm{g}(x)$, a and b suitably, show that for $t > 0$,

$$\frac{\mathrm{e}^t - 1}{\mathrm{e}^t + 1} \leqslant \frac{t}{2} \,.$$

(ii) By setting $\mathrm{f}(x) = x$ in $(*)$, and choosing $\mathrm{g}(x)$ suitably, show that

$$\int_0^1 \mathrm{e}^{-\frac{1}{2}x^2} \mathrm{d}x \geqslant 12\big(1 - \mathrm{e}^{-\frac{1}{4}}\big)^2 \,.$$

(iii) Use $(*)$ to show that

$$\frac{64}{25\pi} \leqslant \int_0^{\frac{1}{2}\pi} \sqrt{\sin x}\, \mathrm{d}x \leqslant \sqrt{\frac{\pi}{2}} \,.$$

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5 A curve C is determined by the parametric equations

$$x = at^2\,,\ y = 2at\,,$$

where $a > 0$.

(i) Show that the normal to C at a point P, with non-zero parameter p, meets C again at a point N, with parameter n, where

$$n = -\left(p + \frac{2}{p}\right).$$

(ii) Show that the distance $|PN|$ is given by

$$|PN|^2 = 16a^2\frac{(p^2+1)^3}{p^4}$$

and that this is minimised when $p^2 = 2$.

(iii) The point Q, with parameter q, is the point at which the circle with diameter PN cuts C again. By considering the gradients of QP and QN, show that

$$2 = p^2 - q^2 + \frac{2q}{p}\,.$$

Deduce that $|PN|$ is at its minimum when Q is at the origin.

6 Let

$$S_n = \sum_{r=1}^{n}\frac{1}{\sqrt{r}}\,,$$

where n is a positive integer.

(i) Prove by induction that

$$S_n \leqslant 2\sqrt{n} - 1\,.$$

(ii) Show that $(4k+1)\sqrt{k+1} > (4k+3)\sqrt{k}$ for $k \geqslant 0$.

Determine the smallest number C such that

$$S_n \geqslant 2\sqrt{n} + \frac{1}{2\sqrt{n}} - C\,.$$

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[Turn over

7 The functions f and g are defined, for $x > 0$, by

$$\mathrm{f}(x) = x^x\,, \qquad \mathrm{g}(x) = x^{\mathrm{f}(x)}\,.$$

(i) By taking logarithms, or otherwise, show that $\mathrm{f}(x) > x$ for $0 < x < 1$. Show further that $x < \mathrm{g}(x) < \mathrm{f}(x)$ for $0 < x < 1$.

Write down the corresponding results for $x > 1$.

(ii) Find the value of x for which $\mathrm{f}'(x) = 0$.

(iii) Use the result $x \ln x \to 0$ as $x \to 0$ to find $\lim\limits_{x\to 0} \mathrm{f}(x)$, and write down $\lim\limits_{x\to 0} \mathrm{g}(x)$.

(iv) Show that $x^{-1} + \ln x \geqslant 1$ for $x > 0$.

Using this result, or otherwise, show that $\mathrm{g}'(x) > 0$.

Sketch the graphs, for $x > 0$, of $y = x$, $y = \mathrm{f}(x)$ and $y = \mathrm{g}(x)$ on the same axes.

8 All vectors in this question lie in the same plane.

The vertices of the non-right-angled triangle ABC have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, respectively. The non-zero vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular to BC and CA, respectively.

Write down the vector equation of the line through A perpendicular to BC, in terms of $\mathbf{u}$, $\mathbf{a}$ and a parameter λ.

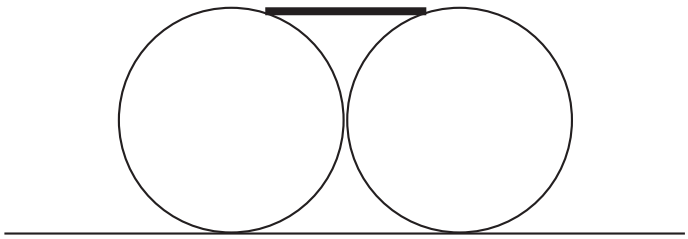
The line through A perpendicular to BC intersects the line through B perpendicular to CA at P. Find the position vector of P in terms of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{u}$.

Hence show that the line CP is perpendicular to the line AB.

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Section B: Mechanics

9 Two identical rough cylinders of radius r and weight W rest, not touching each other but a negligible distance apart, on a horizontal floor. A thin flat rough plank of width $2a$, where $a < r$, and weight kW rests symmetrically and horizontally on the cylinders, with its length parallel to the axes of the cylinders and its faces horizontal. A vertical cross-section is shown in the diagram below.



The coefficient of friction at all four contacts is $\frac{1}{2}$. The system is in equilibrium.

(i) Let F be the frictional force between one cylinder and the floor, and let R be the normal reaction between the plank and one cylinder. Show that

$$R\sin\theta = F(1+\cos\theta)\,,$$

where θ is the acute angle between the plank and the tangent to the cylinder at the point of contact.

Deduce that $2\sin\theta \leqslant 1+\cos\theta$.

(ii) Show that

$$N = \left(1+\frac{2}{k}\right)\left(\frac{1+\cos\theta}{\sin\theta}\right)F\,,$$

where N is the normal reaction between the floor and one cylinder.

Write down the condition that the cylinder does not slip on the floor and show that it is satisfied with no extra restrictions on θ.

(iii) Show that $\sin\theta \leqslant \frac{4}{5}$ and hence that $r \leqslant 5a$.

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[Turn over

10 A car of mass m makes a journey of distance $2d$ in a straight line. It experiences air resistance and rolling resistance so that the total resistance to motion when it is moving with speed v is $Av^2 + R$, where A and R are constants.

The car starts from rest and moves with constant acceleration a for a distance d. Show that the work done by the engine for this half of the journey is

$$\int_0^d (ma + R + Av^2)\,\mathrm{d}x$$

and that it can be written in the form

$$\int_0^w \frac{(ma + R + Av^2)v}{a}\,\mathrm{d}v\,,$$

where $w = \sqrt{2ad}$.

For the second half of the journey, the acceleration of the car is $-a$.

(i) In the case $R > ma$, show that the work done by the engine for the whole journey is

$$2Aad^2 + 2Rd\,.$$

(ii) In the case $ma - 2Aad < R < ma$, show that at a certain speed the driving force required to maintain the constant acceleration falls to zero.

Thereafter, the engine does no work (and the driver applies the brakes to maintain the constant acceleration). Show that the work done by the engine for the whole journey is

$$2Aad^2 + 2Rd + \frac{(ma - R)^2}{4Aa}\,.$$

11 Two thin vertical parallel walls, each of height $2a$, stand a distance a apart on horizontal ground. The projectiles in this question move in a plane perpendicular to the walls.

(i) A particle is projected with speed $\sqrt{5ag}$ towards the two walls from a point A at ground level. It just clears the first wall. By considering the energy of the particle, find its speed when it passes over the first wall.

Given that it just clears the second wall, show that the angle its trajectory makes with the horizontal when it passes over the first wall is $45°$.

Find the distance of A from the foot of the first wall.

(ii) A second particle is projected with speed $\sqrt{5ag}$ from a point B at ground level towards the two walls. It passes a distance h above the first wall, where $h > 0$. Show that it does not clear the second wall.

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Section C: Probability and Statistics

12 Adam and Eve are catching fish. The number of fish, X, that Adam catches in a fixed time interval T has a Poisson distribution with parameter λ. The number of fish, Y, that Eve catches in the same time interval has a Poisson distribution with parameter μ. The two Poisson variables are independent.

(i) By considering $\mathrm{P}(X+Y=r)$, show that the total number of fish caught by Adam and Eve in time T also has a Poisson distribution.

(ii) Given that Adam and Eve catch a total of k fish in time T, where k is fixed, show that the number caught by Adam has a binomial distribution.

(iii) Given that Adam and Eve start fishing at the same time, find the probability that the first fish is caught by Adam.

(iv) You are now given that, for a Poisson distribution with parameter θ, the expected time from any starting point until the next event is θ^{-1}.

Find the expected time from the moment Adam and Eve start fishing until they have each caught at least one fish.

13 In a television game show, a contestant has to open a door using a key. The contestant is given a bag containing n keys, where $n \geqslant 2$. Only one key in the bag will open the door. There are three versions of the game. In each version, the contestant starts by choosing a key at random from the bag.

(i) In version 1, after each failed attempt at opening the door the key that has been tried is put back into the bag and the contestant again selects a key at random from the bag. By considering the binomial expansion of $(1-q)^{-2}$, or otherwise, find the expected number of attempts required to open the door.

(ii) In version 2, after each failed attempt at opening the door the key that has been tried is put aside and the contestant selects another key at random from the bag. Find the expected number of attempts required to open the door.

(iii) In version 3, after each failed attempt at opening the door the key that has been tried is put back into the bag and another incorrect key is added to the bag. The contestant then selects a key at random from the bag. Show that the probability that the contestant draws the correct key at the kth attempt is

$$\frac{n-1}{(n+k-1)(n+k-2)}\,.$$

Show also, using partial fractions, that the expected number of attempts required to open the door is infinite.

You may use without proof the result that $\displaystyle\sum_{m=1}^{N}\frac{1}{m}\to\infty$ as $N\to\infty$.

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Section A: Pure Mathematics

1 Show that, if k is a root of the quartic equation

$$x^4 + ax^3 + bx^2 + ax + 1 = 0\,, \qquad (*)$$

then k^{-1} is a root.

You are now given that a and b in $(*)$ are both real and are such that the roots are all real.

(i) Write down all the values of a and b for which $(*)$ has only one distinct root.

(ii) Given that $(*)$ has exactly three distinct roots, show that either $b = 2a - 2$ or $b = -2a - 2$.

(iii) Solve $(*)$ in the case $b = 2a - 2$, giving your solutions in terms of a.

Given that a and b are both real and that the roots of $(*)$ are all real, find necessary and sufficient conditions, in terms of a and b, for $(*)$ to have exactly three distinct real roots.

2 A function $\mathrm{f}(x)$ is said to be *concave* for $a < x < b$ if

$$t\,\mathrm{f}(x_1) + (1-t)\,\mathrm{f}(x_2) \leqslant \mathrm{f}\big(tx_1 + (1-t)x_2\big)\,,$$

for $a < x_1 < b$, $a < x_2 < b$ and $0 \leqslant t \leqslant 1$.

Illustrate this definition by means of a sketch, showing the chord joining the points $\big(x_1, \mathrm{f}(x_1)\big)$ and $\big(x_2, \mathrm{f}(x_2)\big)$, in the case $x_1 < x_2$ and $\mathrm{f}(x_1) < \mathrm{f}(x_2)$.

Explain why a function $\mathrm{f}(x)$ satisfying $\mathrm{f}''(x) < 0$ for $a < x < b$ is concave for $a < x < b$.

(i) By choosing t, x_1 and x_2 suitably, show that, if $\mathrm{f}(x)$ is concave for $a < x < b$, then

$$\mathrm{f}\Big(\frac{u+v+w}{3}\Big) \geqslant \frac{\mathrm{f}(u) + \mathrm{f}(v) + \mathrm{f}(w)}{3}\,,$$

for $a < u < b$, $a < v < b$ and $a < w < b$.

(ii) Show that, if A, B and C are the angles of a triangle, then

$$\sin A + \sin B + \sin C \leqslant \frac{3\sqrt{3}}{2}\,.$$

(iii) By considering $\ln(\sin x)$, show that, if A, B and C are the angles of a triangle, then

$$\sin A \times \sin B \times \sin C \leqslant \frac{3\sqrt{3}}{8}\,.$$

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[Turn over

3 **(i)** Let

$$\mathrm{f}(x) = \frac{1}{1+\tan x}$$

for $0 \leqslant x < \frac{1}{2}\pi$.

Show that $\mathrm{f}'(x) = -\dfrac{1}{1+\sin 2x}$ and hence find the range of $\mathrm{f}'(x)$.

Sketch the curve $y = \mathrm{f}(x)$.

(ii) The function $\mathrm{g}(x)$ is continuous for $-1 \leqslant x \leqslant 1$.

Show that the curve $y = \mathrm{g}(x)$ has rotational symmetry of order 2 about the point (a, b) on the curve if and only if

$$\mathrm{g}(x) + \mathrm{g}(2a - x) = 2b\,.$$

Given that the curve $y = \mathrm{g}(x)$ passes through the origin and has rotational symmetry of order 2 about the origin, write down the value of

$$\int_{-1}^{1} \mathrm{g}(x)\,\mathrm{d}x\,.$$

(iii) Show that the curve $y = \dfrac{1}{1+\tan^k x}$, where k is a positive constant and $0 < x < \frac{1}{2}\pi$,

has rotational symmetry of order 2 about a certain point (which you should specify) and evaluate

$$\int_{\frac{1}{6}\pi}^{\frac{1}{3}\pi} \frac{1}{1+\tan^k x}\,\mathrm{d}x\,.$$

4 In this question, you may use the following identity without proof:

$$\cos A + \cos B = 2\cos\tfrac{1}{2}(A+B)\,\cos\tfrac{1}{2}(A-B)\,.$$

(i) Given that $0 \leqslant x \leqslant 2\pi$, find all the values of x that satisfy the equation

$$\cos x + 3\cos 2x + 3\cos 3x + \cos 4x = 0\,.$$

(ii) Given that $0 \leqslant x \leqslant \pi$ and $0 \leqslant y \leqslant \pi$ and that

$$\cos(x+y) + \cos(x-y) - \cos 2x = 1\,,$$

show that either $x = y$ or x takes one specific value which you should find.

(iii) Given that $0 \leqslant x \leqslant \pi$ and $0 \leqslant y \leqslant \pi$, find the values of x and y that satisfy the equation

$$\cos x + \cos y - \cos(x+y) = \tfrac{3}{2}\,.$$

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5 In this question, you should ignore issues of convergence.

(i) Write down the binomial expansion, for $|x| < 1$, of $\dfrac{1}{1+x}$ and deduce that

$$\ln(1+x) = -\sum_{n=1}^{\infty} \frac{(-x)^n}{n}$$

for $|x| < 1$.

(ii) Write down the series expansion in powers of x of e^{-ax}. Use this expansion to show that

$$\int_0^{\infty} \frac{(1-\mathrm{e}^{-ax})\,\mathrm{e}^{-x}}{x}\,\mathrm{d}x = \ln(1+a) \qquad (|a| < 1)\,.$$

(iii) Deduce the value of

$$\int_0^1 \frac{x^p - x^q}{\ln x}\,\mathrm{d}x \qquad (|p| < 1,\ |q| < 1)\,.$$

6 **(i)** Find all pairs of positive integers (n, p), where p is a prime number, that satisfy

$$n! + 5 = p\,.$$

(ii) In this part of the question you may use the following two theorems:

1. For $n \geqslant 7$, $1! \times 3! \times \cdots \times (2n-1)! > (4n)!$.
2. For every positive integer n, there is a prime number between $2n$ and $4n$.

Find all pairs of positive integers (n, m) that satisfy

$$1! \times 3! \times \cdots \times (2n-1)! = m!\,.$$

7 The points O, A and B are the vertices of an acute-angled triangle. The points M and N lie on the sides OA and OB respectively, and the lines AN and BM intersect at Q. The position vector of A with respect to O is $\mathbf{a}$, and the position vectors of the other points are labelled similarly.

Given that $|MQ| = \mu|QB|$, and that $|NQ| = \nu|QA|$, where μ and ν are positive and $\mu\nu < 1$, show that

$$\mathbf{m} = \frac{(1+\mu)\nu}{1+\nu}\,\mathbf{a}\,.$$

The point L lies on the side OB, and $|OL| = \lambda|OB|$. Given that ML is parallel to AN, express λ in terms of μ and ν.

What is the geometrical significance of the condition $\mu\nu < 1$?

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[Turn over

8 **(i)** Use the substitution $v = \sqrt{y}$ to solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \alpha y^{\frac{1}{2}} - \beta y \qquad (y \geqslant 0, \quad t \geqslant 0)\,,$$

where α and β are positive constants. Find the non-constant solution $y_1(x)$ that satisfies $y_1(0) = 0$.

(ii) Solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \alpha y^{\frac{2}{3}} - \beta y \qquad (y \geqslant 0, \quad t \geqslant 0)\,,$$

where α and β are positive constants. Find the non-constant solution $y_2(x)$ that satisfies $y_2(0) = 0$.

(iii) In the case $\alpha = \beta$, sketch $y_1(x)$ and $y_2(x)$ on the same axes, indicating clearly which is $y_1(x)$ and which is $y_2(x)$. You should explain how you determined the positions of the curves relative to each other.

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Section B: Mechanics

9 Two small beads, A and B, of the same mass, are threaded onto a vertical wire on which they slide without friction, and which is fixed to the ground at P. They are released simultaneously from rest, A from a height of $8h$ above P and B from a height of $17h$ above P.

When A reaches the ground for the first time, it is moving with speed V. It then rebounds with coefficient of restitution $\frac{1}{2}$ and subsequently collides with B at height H above P.

Show that $H = \frac{15}{8}h$ and find, in terms of g and h, the speeds u_A and u_B of the two beads just before the collision.

When A reaches the ground for the second time, it is again moving with speed V. Determine the coefficient of restitution between the two beads.

10 A uniform elastic string lies on a smooth horizontal table. One end of the string is attached to a fixed peg, and the other end is pulled at constant speed u. At time $t = 0$, the string is taut and its length is a. Obtain an expression for the speed, at time t, of the point on the string which is a distance x from the peg at time t.

An ant walks along the string starting at $t = 0$ at the peg. The ant walks at constant speed v along the string (so that its speed relative to the peg is the sum of v and the speed of the point on the string beneath the ant). At time t, the ant is a distance x from the peg. Write down a first order differential equation for x, and verify that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{x}{a+ut}\right) = \frac{v}{a+ut}\,.$$

Show that the time T taken for the ant to reach the end of the string is given by

$$uT = a(\mathrm{e}^k - 1)\,,$$

where $k = u/v$.

On reaching the end of the string, the ant turns round and walks back to the peg. Find in terms of T and k the time taken for the journey back.

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[Turn over

11 The axles of the wheels of a motorbike of mass m are a distance b apart. Its centre of mass is a horizontal distance of d from the front axle, where $d < b$, and a vertical distance h above the road, which is horizontal and straight. The engine is connected to the rear wheel. The coefficient of friction between the ground and the rear wheel is μ, where $\mu < b/h$, and the front wheel is smooth.

You may assume that the sum of the moments of the forces acting on the motorbike about the centre of mass is zero. By taking moments about the centre of mass show that, as the acceleration of the motorbike increases from zero, the rear wheel will slip before the front wheel loses contact with the road if

$$\mu < \frac{b-d}{h}\,. \qquad (*)$$

If the inequality $(*)$ holds and the rear wheel does not slip, show that the maximum acceleration is

$$\frac{\mu d g}{b - \mu h}\,.$$

If the inequality $(*)$ does not hold, find the maximum acceleration given that the front wheel remains in contact with the road.

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Section C: Probability and Statistics

12 In a game, I toss a coin repeatedly. The probability, p, that the coin shows Heads on any given toss is given by

$$p = \frac{N}{N+1},$$

where N is a positive integer. The outcomes of any two tosses are independent.

The game has two versions. In each version, I can choose to stop playing after any number of tosses, in which case I win £H, where H is the number of Heads I have tossed. However, the game may end before that, in which case I win nothing.

(i) In version 1, the game ends when the coin first shows Tails (if I haven't stopped playing before that).

I decide from the start to toss the coin until a total of h Heads have been shown, unless the game ends before then. Find, in terms of h and p, an expression for my expected winnings and show that I can maximise my expected winnings by choosing $h = N$.

(ii) In version 2, the game ends when the coin shows Tails on two *consecutive* tosses (if I haven't stopped playing before that).

I decide from the start to toss the coin until a total of h Heads have been shown, unless the game ends before then. Show that my expected winnings are

$$\frac{hN^h(N+2)^h}{(N+1)^{2h}}.$$

In the case $N = 2$, use the approximation $\log_3 2 \approx 0.63$ to show that the maximum value of my expected winnings is approximately £3.

13 Four children, A, B, C and D, are playing a version of the game 'pass the parcel'. They stand in a circle, so that $ABCDA$ is the clockwise order. Each time a whistle is blown, the child holding the parcel is supposed to pass the parcel immediately exactly one place clockwise. In fact each child, independently of any other past event, passes the parcel clockwise with probability $\frac{1}{4}$, passes it anticlockwise with probability $\frac{1}{4}$ and fails to pass it at all with probability $\frac{1}{2}$. At the start of the game, child A is holding the parcel.

The probability that child A is holding the parcel just after the whistle has been blown for the nth time is A_n, and B_n, C_n and D_n are defined similarly.

(i) Find A_1, B_1, C_1 and D_1. Find also A_2, B_2, C_2 and D_2.

(ii) By first considering $B_{n+1} + D_{n+1}$, or otherwise, find B_n and D_n.

Find also expressions for A_n and C_n in terms of n.

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Section A: Pure Mathematics

1 Let $\mathrm{f}(x) = (x-p)\mathrm{g}(x)$, where g is a polynomial. Show that the tangent to the curve $y = \mathrm{f}(x)$ at the point with $x = a$, where $a \neq p$, passes through the point $(p, 0)$ if and only if $\mathrm{g}'(a) = 0$.

The curve C has equation

$$y = A(x-p)(x-q)(x-r),$$

where p, q and r are constants with $p < q < r$, and A is a non-zero constant.

(i) The tangent to C at the point with $x = a$, where $a \neq p$, passes through the point $(p, 0)$. Show that $2a = q + r$ and find an expression for the gradient of this tangent in terms of A, q and r.

(ii) The tangent to C at the point with $x = c$, where $c \neq r$, passes through the point $(r, 0)$. Show that this tangent is parallel to the tangent in part (i) if and only if the tangent to C at the point with $x = q$ does not meet the curve again.

2 The function f satisfies $\mathrm{f}(0) = 0$ and $\mathrm{f}'(t) > 0$ for $t > 0$. Show by means of a sketch that, for $x > 0$,

$$\int_0^x \mathrm{f}(t)\,\mathrm{d}t + \int_0^{\mathrm{f}(x)} \mathrm{f}^{-1}(y)\,\mathrm{d}y = x\mathrm{f}(x).$$

(i) The (real) function g is defined, for all t, by

$$\big(\mathrm{g}(t)\big)^3 + \mathrm{g}(t) = t.$$

Prove that $\mathrm{g}(0) = 0$, and that $\mathrm{g}'(t) > 0$ for all t.

Evaluate $\displaystyle\int_0^2 \mathrm{g}(t)\,\mathrm{d}t$.

(ii) The (real) function h is defined, for all t, by

$$\big(\mathrm{h}(t)\big)^3 + \mathrm{h}(t) = t + 2.$$

Evaluate $\displaystyle\int_0^8 \mathrm{h}(t)\,\mathrm{d}t$.

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[Turn over

3 For any two real numbers x_1 and x_2, show that

$$|x_1 + x_2| \leqslant |x_1| + |x_2|.$$

Show further that, for any real numbers x_1, x_2, ..., x_n,

$$|x_1 + x_2 + \cdots + x_n| \leqslant |x_1| + |x_2| + \cdots + |x_n|.$$

(i) The polynomial f is defined by

$$\mathrm{f}(x) = 1 + a_1 x + a_2 x^2 + \cdots + a_{n-1} x^{n-1} + x^n$$

where the coefficients are real and satisfy $|a_i| \leqslant A$ for $i = 1, 2, \ldots, n-1$, where $A \geqslant 1$.

(a) If $|x| < 1$, show that

$$|\mathrm{f}(x) - 1| \leqslant \frac{A|x|}{1 - |x|}.$$

(b) Let ω be a real root of f, so that $\mathrm{f}(\omega) = 0$. In the case $|\omega| < 1$, show that

$$\frac{1}{1 + A} \leqslant |\omega| \leqslant 1 + A. \qquad (*)$$

(c) Show further that the inequalities $(*)$ also hold if $|\omega| \geqslant 1$.

(ii) Find the integer root or roots of the quintic equation

$$135x^5 - 135x^4 - 100x^3 - 91x^2 - 126x + 135 = 0.$$

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4 You are not required to consider issues of convergence in this question.

For any sequence of numbers $a_1,\ a_2,\ \ldots\ ,\ a_m,\ \ldots\ ,\ a_n$, the notation $\prod_{i=m}^{n} a_i$ denotes the product $a_m a_{m+1} \cdots a_n$.

(i) Use the identity $2\cos x \sin x = \sin(2x)$ to evaluate the product $\cos(\frac{\pi}{9})\cos(\frac{2\pi}{9})\cos(\frac{4\pi}{9})$.

(ii) Simplify the expression

$$\prod_{k=0}^{n} \cos\left(\frac{x}{2^k}\right) \qquad\qquad (0 < x < \tfrac{1}{2}\pi).$$

Using differentiation, or otherwise, show that, for $0 < x < \frac{1}{2}\pi$,

$$\sum_{k=0}^{n} \frac{1}{2^k} \tan\left(\frac{x}{2^k}\right) = \frac{1}{2^n} \cot\left(\frac{x}{2^n}\right) - 2\cot(2x).$$

(iii) Using the results $\lim_{\theta\to 0} \dfrac{\sin\theta}{\theta} = 1$ and $\lim_{\theta\to 0} \dfrac{\tan\theta}{\theta} = 1$, show that

$$\prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right) = \frac{\sin x}{x}$$

and evaluate

$$\sum_{j=2}^{\infty} \frac{1}{2^{j-2}} \tan\left(\frac{\pi}{2^j}\right).$$

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[Turn over

5 The sequence u_0, u_1, ... is said to be a *constant sequence* if $u_n = u_{n+1}$ for $n = 0, 1, 2, \ldots$. The sequence is said to be a *sequence of period* 2 if $u_n = u_{n+2}$ for $n = 0, 1, 2, \ldots$ and the sequence is not constant.

(i) A sequence of real numbers is defined by $u_0 = a$ and $u_{n+1} = \mathrm{f}(u_n)$ for $n = 0, 1, 2, \ldots$, where

$$\mathrm{f}(x) = p + (x - p)x,$$

and p is a given real number.

Find the values of a for which the sequence is constant.

Show that the sequence has period 2 for some value of a if and only if $p > 3$ or $p < -1$.

(ii) A sequence of real numbers is defined by $u_0 = a$ and $u_{n+1} = \mathrm{f}(u_n)$ for $n = 0, 1, 2, \ldots$, where

$$\mathrm{f}(x) = q + (x - p)x,$$

and p and q are given real numbers.

Show that there is no value of a for which the sequence is constant if and only if $\mathrm{f}(x) > x$ for all x.

Deduce that, if there is no value of a for which the sequence is constant, then there is no value of a for which the sequence has period 2.

Is it true that, if there is no value of a for which the sequence has period 2, then there is no value of a for which the sequence is constant?

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6 **Note**: You may assume that if the functions $y_1(x)$ and $y_2(x)$ both satisfy one of the differential equations in this question, then the curves $y = y_1(x)$ and $y = y_2(x)$ do not intersect.

(i) Find the solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y + x + 1$$

that has the form $y = mx + c$, where m and c are constants.

Let $y_3(x)$ be the solution of this differential equation with $y_3(0) = k$. Show that any stationary point on the curve $y = y_3(x)$ lies on the line $y = -x - 1$. Deduce that solution curves with $k < -2$ cannot have any stationary points.

Show further that any stationary point on the solution curve is a local minimum.

Use the substitution $Y = y + x$ to solve the differential equation, and sketch, on the same axes, the solutions with $k = 0$, $k = -2$ and $k = -3$.

(ii) Find the two solutions of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x^2 + y^2 - 2xy - 4x + 4y + 3$$

that have the form $y = mx + c$.

Let $y_4(x)$ be the solution of this differential equation with $y_4(0) = -2$. (Do not attempt to find this solution.)

Show that any stationary point on the curve $y = y_4(x)$ lies on one of two lines that you should identify. What can be said about the gradient of the curve at points between these lines?

Sketch the curve $y = y_4(x)$. You should include on your sketch the two straight line solutions and the two lines of stationary points.

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[Turn over

7 **(i)** The points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, respectively. Each of these vectors is a unit vector (so $\mathbf{a}\,.\,\mathbf{a} = 1$, for example) and

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}.$$

Show that $\mathbf{a}\,.\,\mathbf{b} = -\frac{1}{2}$. What can be said about the triangle ABC? You should justify your answer.

(ii) The four distinct points A_i ($i = 1, 2, 3, 4$) have unit position vectors $\mathbf{a}_i$ and

$$\sum_{i=1}^{4} \mathbf{a}_i = \mathbf{0}.$$

Show that $\mathbf{a}_1\,.\,\mathbf{a}_2 = \mathbf{a}_3\,.\,\mathbf{a}_4$.

(a) Given that the four points lie in a plane, determine the shape of the quadrilateral with vertices A_1, A_2, A_3 and A_4.

(b) Given instead that the four points are the vertices of a regular tetrahedron, find the length of the sides of this tetrahedron.

8 The domain of the function f is the set of all 2×2 matrices and its range is the set of real numbers. Thus, if $\mathbf{M}$ is a 2×2 matrix, then $\mathrm{f}(\mathbf{M}) \in \mathbb{R}$.

The function f has the property that $\mathrm{f}(\mathbf{MN}) = \mathrm{f}(\mathbf{M})\mathrm{f}(\mathbf{N})$ for any 2×2 matrices $\mathbf{M}$ and $\mathbf{N}$.

(i) You are given that there is a matrix $\mathbf{M}$ such that $\mathrm{f}(\mathbf{M}) \neq 0$. Let $\mathbf{I}$ be the 2×2 identity matrix. By considering $\mathrm{f}(\mathbf{MI})$, show that $\mathrm{f}(\mathbf{I}) = 1$.

(ii) Let $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. You are given that $\mathrm{f}(\mathbf{J}) \neq 1$. By considering $\mathbf{J}^2$, evaluate $\mathrm{f}(\mathbf{J})$.

Using $\mathbf{J}$, show that, for any real numbers a, b, c and d,

$$\mathrm{f}\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = -\mathrm{f}\left(\begin{pmatrix} c & d \\ a & b \end{pmatrix}\right) = \mathrm{f}\left(\begin{pmatrix} d & c \\ b & a \end{pmatrix}\right).$$

(iii) Let $\mathbf{K} = \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$ where $k \in \mathbb{R}$. Use $\mathbf{K}$ to show that, if the second row of the matrix $\mathbf{A}$ is a multiple of the first row, then $\mathrm{f}(\mathbf{A}) = 0$.

(iv) Let $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. By considering the matrices $\mathbf{P}^2$, $\mathbf{P}^{-1}$, and $\mathbf{K}^{-1}\mathbf{PK}$ for suitable values of k, evaluate $\mathrm{f}(\mathbf{P})$.

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Section B: Mechanics

9 A particle P is projected from a point O on horizontal ground with speed u and angle of projection α, where $0 < \alpha < \frac{1}{2}\pi$.

(i) Show that if $\sin\alpha < \dfrac{2\sqrt{2}}{3}$, then the distance OP is increasing throughout the flight.

Show also that if $\sin\alpha > \dfrac{2\sqrt{2}}{3}$, then OP will be decreasing at some time before the particle lands.

(ii) At the same time as P is projected, a particle Q is projected horizontally from O with speed v along the ground in the opposite direction from the trajectory of P. The ground is smooth. Show that if

$$2\sqrt{2}v > (\sin\alpha - 2\sqrt{2}\cos\alpha)u,$$

then QP is increasing throughout the flight of P.

10 A small light ring is attached to the end A of a uniform rod AB of weight W and length $2a$. The ring can slide on a rough horizontal rail.

One end of a light inextensible string of length $2a$ is attached to the rod at B and the other end is attached to a point C on the rail so that the rod makes an angle of θ with the rail, where $0 < \theta < 90°$. The rod hangs in the same vertical plane as the rail.

A force of kW acts vertically downwards on the rod at B and the rod is in equilibrium.

(i) You are given that the string will break if the tension T is greater than W. Show that (assuming that the ring does not slip) the string will break if

$$2k + 1 > 4\sin\theta.$$

(ii) Show that (assuming that the string does not break) the ring will slip if

$$2k + 1 > (2k + 3)\mu\tan\theta,$$

where μ is the coefficient of friction between the rail and the ring.

(iii) You are now given that $\mu\tan\theta < 1$.

Show that, when k is increased gradually from zero, the ring will slip before the string breaks if

$$\mu < \frac{2\cos\theta}{1 + 2\sin\theta}.$$

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[Turn over

Section C: Probability and Statistics

11 **(i)** The three integers n_1, n_2 and n_3 satisfy $0 < n_1 < n_2 < n_3$ and $n_1 + n_2 > n_3$. Find the number of ways of choosing the pair of numbers n_1 and n_2 in the cases $n_3 = 9$ and $n_3 = 10$.

Given that $n_3 = 2n+1$, where n is a positive integer, write down an expression (which you need not prove is correct) for the number of ways of choosing the pair of numbers n_1 and n_2. Simplify your expression.

Write down and simplify the corresponding expression when $n_3 = 2n$, where n is a positive integer.

(ii) You have N rods, of lengths 1, 2, 3, ... , N (one rod of each length). You take the rod of length N, and choose two more rods at random from the remainder, each choice of two being equally likely. Show that, in the case $N = 2n + 1$ where n is a positive integer, the probability that these three rods can form a triangle (of non-zero area) is

$$\frac{n-1}{2n-1}.$$

Find the corresponding probability in the case $N = 2n$, where n is a positive integer.

(iii) You have $2M+1$ rods, of lengths 1, 2, 3, ... , $2M+1$ (one rod of each length), where M is a positive integer. You choose three at random, each choice of three being equally likely. Show that the probability that the rods can form a triangle (of non-zero area) is

$$\frac{(4M+1)(M-1)}{2(2M+1)(2M-1)}.$$

Note: $\sum_{k=1}^{K} k^2 = \frac{1}{6}K(K+1)(2K+1)$.

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12 The random variable X has the probability density function on the interval $[0, 1]$:

$$\mathrm{f}(x) = \begin{cases} nx^{n-1} & 0 \leqslant x \leqslant 1, \\ 0 & \text{elsewhere,} \end{cases}$$

where n is an integer greater than 1.

(i) Let $\mu = \mathrm{E}(X)$. Find an expression for μ in terms of n, and show that the variance, σ^2, of X is given by

$$\sigma^2 = \frac{n}{(n+1)^2(n+2)}.$$

(ii) In the case $n = 2$, show without using decimal approximations that the interquartile range is less than 2σ.

(iii) Write down the first three terms and the $(k+1)$th term (where $0 \leqslant k \leqslant n$) of the binomial expansion of $(1+x)^n$ in ascending powers of x.

By setting $x = \dfrac{1}{n}$, show that μ is less than the median and greater than the lower quartile.

Note: You may assume that

$$1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots < 4.$$

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